# An open curtailment dataset and ~~a diagnostic for~~ an evaluation of adaptive conformal calibration under regime change

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**Abstract**

Curtailment of variable renewable generation is zero-inflated, heavy-tailed and drifts as the system changes, a demanding target for calibrated uncertainty. We study it in Chile's National Electricity System, where ~~roughly~~ about 19,000 GWh were curtailed between 2022 and mid-2026~~and the distribution of magnitudes shifted. We contribute~~. We release an open, ~~plant-level,~~ errata-corrected, plant-level dataset of daily and hourly curtailment ~~covering~~ (53 ~~continuous months(hydropower from June 2024) and~~ months, 300 plants~~across four technologies, reconciling~~) that reconciles exactly with the operator's ~~official annual totals; and a diagnostic. Paired with a zero-inflated lognormal~~ annual totals, and use it to evaluate conformal calibration of seven-day-ahead upper limits under a documented distributional change. With a lognormal hurdle base model, ~~a scheme combining a probability-integral-transform score , score~~ an adaptive scheme combining score transport and online recalibration narrows the ~~intervals~~ limits of the most divergent window by ~~about~~ a third at near-nominal coverage. ~~That~~ Across nine base models from two families the gain does not ~~survive a change of base model. On a multi-quantile gradient-boosting predictive fitted under the same protocol, the static conformal split is already at nominal coverage and~~ transfer, and it is largely mechanical: a constant-threshold oracle that matches the adaptive coverage without adapting in time explains 84% of the variation in width change across 45 model-window cells, as a controlled simulation confirms. The gain measures the drift of the base model's calibration, which here depends more on the base model than on the window. Improving the base model dominates: early stopping alone lowers the hurdle's interval score by 35%. On high-curtailment days defined by the outcome every method loses coverage, and the ~~adaptive scheme returns slightly wider intervals. Across eight base models and five windows the width reduction grows with how much the static split over-covers ($r = -0.873$, bootstrap interval $[-0.912, -0.645]$; slope $-4.89$ per cent of width per percentage point), so the layer recovers base-model over-dispersion rather than distribution shift. Ablations agree: online adaptation earns its cost, score transport does not. Panel dependence is severe: intraclass correlation by date 0.238. We separate finite-sample marginal validity under exchangeability, long-run adaptive control, and empirical coverage in a dependent panel. The practical implication is that fixing the base predictive dominates fixing the calibration layer~~transport-based scheme, which leaves more curtailed energy uncovered than a static split, is not the method we recommend for operation. A static conformal wrapper around a multi-quantile predictive gives the lowest operational cost among methods with near-nominal coverage. Dataset and code are public.

*Keywords:* renewable curtailment, probabilistic forecasting, conformal prediction, distribution shift, ~~model-agnostic~~ forecast evaluation, power system operations

**Marked-up version (revision 2).** Deleted text is struck through in red and new text is underlined in blue. Tables, figures, the algorithm, displayed equations and the bibliography are marked as whole blocks. The main text was re-read and edited throughout, so many paragraphs appear as replaced; the response letter lists every substantive change.

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## 1. Introduction

The share of variable renewable energy has grown faster than the transmission and demand able to absorb it, and curtailment, the deliberate reduction of available solar, wind or hydropower output, has become a routine consequence of that imbalance~~rather than an exceptional event~~. O'Shaughnessy et al. [1] document the pattern across systems worldwide and show that solar curtailment rises sharply once penetration passes a threshold that ~~varies by system but~~ is reached quickly where network expansion lags. In Chile's National Electricity System (Sistema Eléctrico Nacional, SEN), recorded curtailment rose from about 1.5 TWh in 2022 to more than 6 TWh per year by 2024, and the cumulative figure over 2022 to mid-2026 reaches roughly 19 TWh. For ~~the owners of affected plants this is generated~~ plant owners this is energy that is ~~never remunerated; for the system it is a growing signal of the mismatch between where and when renewable energy is produced and where and when it can be used~~generated but never remunerated.

Anticipating curtailment is therefore ~~of direct operational and commercial value~~commercially relevant, and it is ~~considerably~~ harder than forecasting generation. Generation depends ~~primarily~~ mainly on weather and installed capacity~~, which are well understood and increasingly predictable~~. Curtailment depends on the state of the whole system: net load, transmission congestion, the marginal cost of energy and the ~~behaviour~~ operation of other assets. It is zero-inflated and heavy-tailed: on many days a plant curtails nothing, and on the days it does the amounts vary widely. A single predicted number conveys neither the probability that curtailment occurs nor the range it may take, so what a plant owner needs is a ~~downstream, system-level quantity, and this makes it both more difficult and less studied than generation forecasting.~~ calibrated statement of uncertainty, for instance an upper prediction limit that is exceeded with a known probability.

~~Two features of the case at hand make it an informative test bed rather than merely an application. First, curtailment in the SEN is highly concentrated: a small number of northern regions, where solar capacity has outpaced the north-to-south transmission corridor, account for the large majority of national curtailment. Second, and central to this paper, the statistical~~ The Chilean case adds a further difficulty. The distribution of curtailment changed ~~materially over~~ during the study period~~. The change is gradual rather than abrupt: a formal~~ : a change-point analysis ~~of the monthly mean log-magnitude, described in Section 3, places the shift in~~ (Section 3.4) places a ramp of roughly eighteen months ~~running from~~ between mid-2023 ~~to~~ and late 2024, ~~with no detected break in 2025 and none in October 2024. A model calibrated on one part of that trajectory is miscalibrated on a later part, not because the model is wrong but because the system it describes has changed. Two properties make the change a demanding case rather than a textbook one. It~~ during which the positive magnitudes roughly tripled. The change has no single identifiable cause~~: the growth of solar capacity~~ , since solar growth behind a saturated north–south corridor, the entry of battery storage, transmission reinforcements and changes in ~~northern mining demand all act over the same months, and the data released here cannot separate them. And~~ mining demand overlap in time, and it is superimposed on a strong annual cycle~~, so that separating seasonal from secular movement is part of the problem: in the window where divergence from the calibration distribution is largest, the majority of the apparent jump turns out to be seasonal.~~

~~A further difficulty is that a point forecast of curtailment is of limited use for decision-making. Curtailment is zero-inflated and heavy-tailed: on a large fraction of days a given plant curtails nothing, and on the days it does the amounts vary widely. A single predicted number conveys neither the probability that curtailment will occur nor the range it might take. What a decision-maker needs is a calibrated statement of uncertainty. Providing such intervals, and keeping them informative when the system changes regime, is the problem we address.~~ . A model calibrated on one part of that trajectory is miscalibrated on a later part.

~~This paper makes two~~ Adaptive conformal methods, such as the online update of Gibbs and Candès [2], are the standard response when exchangeability fails, and they are often evaluated by comparing an adaptive scheme with a static conformal split on one base model and reporting the width saved at nominal coverage. This paper shows that such a comparison can be misleading, and quantifies why. It makes three contributions.

The first is a ~~data contribution. We construct and~~ dataset. We release an open, plant-level dataset of curtailment in the SEN at daily and hourly resolution, spanning 53 continuous months and 300 plants across solar, wind and two hydropower technologies, built from the public reports of the system operator~~and reconciled against the official aggregate figures. In the course of building it we identified and corrected~~ , corrected for eight classes of errata ~~in the original public files, using deterministic and reproducible rules, and recovered two calendar days missing from the operator's own daily breakdown. We are not aware of an open curtailment dataset at this granularity for a Latin American power system.~~ and reconciled exactly with the official annual totals.

The second is ~~a diagnostic contribution, and it is the one that changed most in the course of this work. Adaptive conformal calibration is the standard recommendation when exchangeability fails, and on this dataset it appears to deliver: paired with a two-part zero-inflated lognormal~~ an evaluation of adaptive conformal calibration under a documented regime change. With a lognormal hurdle as base model, ~~a scheme that scores observations by their probability integral transform, transports the~~ an adaptive scheme that transports calibration scores onto the recent regime and ~~wraps the result in an online update~~ recalibrates online narrows the intervals of the most divergent window by about a third ~~while holding coveragewithin one point of nominal. We show that this apparent success is not a property of the calibration layer. Repeating every experiment on a multi-quantile gradient-boosting predictive fitted on the same rows and features,~~ at near-nominal coverage. We repeat every experiment over nine base predictives from two model families and find that (i) the gain does not transfer across base models; (ii) to first order it is a mechanical conversion of the static split's coverage error into interval width, which a constant-threshold oracle with the same ~~training cut-off, hyperparameters and seed, and calibrated with the same score, the same windows, the same seeds and the same embargo~~coverage reproduces without any adaptation, and which a controlled simulation reproduces with known ground truth; and (iii) holding the evaluation window fixed, the static ~~conformal splitalready attains nominal coverage in the divergent window and the adaptive scheme returns intervals that are slightly *wider*. Across eight base models and five evaluation windows, the width the adaptive layer recovers is linearly related to how much the static split over-covers. What the layer corrects, in this record, is over-dispersion~~ split's coverage error, and with it the apparent gain, depends on the base model. The apparent gain therefore measures how far the calibration of the base predictive ~~, not distribution shift~~drifts between windows, not how much distribution shift the layer absorbs. Three checks follow, which are cheap and apply to any base model: report the static split's coverage in the evaluation window, the width of the oracle re-levelled split, and the same comparison on a second base model.

~~We take that finding seriously rather than burying it. Section 5 reports it with its ablations, which localise the effect further: online adaptation is the component that earns its keep, score transport significantly *worsens* the interval score once adaptation is present, and the bootstrap tail shrinkage that is the most intricate part of the construction improves it by eight tenths of one per cent while consuming most of the runtime. A~~ The third is an operational assessment. We weight coverage by installed capacity and by curtailed energy, evaluate high-curtailment days defined both by the outcome and by information available at the forecast origin, and price the limits with a simple cost model in which the one-sided ~~conformalized quantile regression , a standard and far simpler construction, dominates the whole pipeline when paired with the better base model~~interval score is the expected cost of a plant owner. On these criteria the transport-based scheme is not a method we recommend; a flexible base predictive with a static conformal wrapper is. ~~We also treat the panel dependence of the data as a first-class problem rather than a caveat, and we separate three statements that are easy to conflate: the finite-sample marginal validity of the static split under exchangeability, the long-run coverage control of the adaptive update without exchangeability, and the empirical coverage actually observed in a dependent panel.~~

The ~~remainder of the paper is organised as follows~~evidence comes from one power system, two model families and a seven-day horizon, and we state the corresponding limits where they apply and in Section 7. Section 2 reviews related work~~. ~~, Section 3 describes the dataset and dates the ~~distributional change.~~ change, Section 4 presents the base ~~forecasters, the conformal layer in full implementation detail,~~ models, the calibration layers and the evaluation~~protocol. ~~ , Section 5 reports the ~~empirical results.~~ results, Section 6 discusses ~~what follows for practice~~them, Section 7 ~~states the limitations ,~~ lists the limitations and Section 8 concludes.

## 2. Related work

This work ~~sits at the junction of three literatures that rarely meet: the~~ draws on the applied literature on curtailment and ~~on~~ the network constraints that cause it, ~~the methodological literature on distribution-free uncertainty quantification, and the growing body of work on~~ on hybrid physical and data-driven modelling of power systems~~. We discuss each contribution individually rather than in groups, since what matters here is what each one establishes and~~ , and on the methodology of distribution-free uncertainty quantification and forecast evaluation. Each reference is discussed for what it establishes and for where our problem departs from it.

### 2.1. Curtailment: causes, measurement and forecasting

Curtailment is best understood as the visible output of a constrained network rather than as a property of the plants that suffer it. Henriot [3] formalises the economically efficient level of curtailment and shows that some curtailment is optimal~~rather than a failure~~, which frames ~~the forecasting problem as one of~~ forecasting as anticipating a rational system response. Joos and Staffell [4] quantify the short-term integration costs of that response in Britain and Germany and ~~show that~~ find curtailment and balancing costs ~~are~~ of comparable order, which is what makes anticipation commercially relevant. Tang et al. [5] document the Chinese experience, where solar curtailment ~~reached levels that~~ triggered a national policy response, and attribute it ~~primarily~~ mainly to transmission inadequacy~~rather than to oversupply as such~~. Schermeyer et al. [6] take the ~~complementary view from the network side, showing on German distribution and transmission data~~network view and show, on German data, how congestion management determines which plants are curtailed and when. Nycander et al. [7] ~~extend the accounting to a whole synchronous area, decomposing~~ decompose Nordic curtailment into transmission capacity, inertia and flexibility limits, ~~which is~~ the closest published analogue to the ~~concentrated,~~ corridor-limited situation studied here. Yildiz et al. [8] ~~provide the counterpart at the other end of the system, using~~ use metered data from low-voltage networks to show how distributed photovoltaics and behind-the-meter storage are curtailed in practice ~~,~~ and how poorly ~~that curtailment is captured by aggregate statistics~~ aggregate statistics capture it.

Forecasting curtailment~~specifically~~, as opposed to explaining it, has received ~~far~~ less attention. Bunodiere and Lee [9] predict curtailment in Kyushu with a rule-based method built on the operator's ~~own~~ dispatch logic, which works where that logic is published and ~~transferable and is difficult to generalise~~ is hard to transfer where it is not. ~~The wider forecasting literature has instead treated curtailment as a nuisance term inside net-load forecasting, an approach that sidesteps the plant-level question a portfolio owner actually faces.~~ Our problem differs~~from both~~: we forecast plant-level curtailment directly, ~~at a seven-day horizon~~seven days ahead, without access to the operator's dispatch model, and we ~~care about~~ are concerned with the uncertainty of ~~that~~ the forecast rather than with its point accuracy.

### 2.2. Storage, congestion and network capacity~~as the physical drivers~~

The physical mechanisms ~~that generate curtailment~~ matter here because they are the candidate explanations for the distributional change we document ~~,~~ and because they ~~set the ceiling on what any~~ bound what a statistical forecast can know. Denholm and Mai [10] establish how much storage, and ~~at~~ of what duration, is needed to absorb a given level of curtailment~~, a result that bounds how quickly storage entry can plausibly change a curtailment record~~. Wang et al. [11] embed ~~that~~ the question in unit commitment and show the trade-off between operating cost and wind curtailment when batteries participate in dispatch. Rayit et al. [12] evaluate the ~~economics from the~~ storage owner's ~~side, treating curtailed wind as~~ economics when curtailed wind is the input to ~~a grid-services business~~grid services. Mohamad et al. [13] ~~approach the same problem as one of siting rather than operation, optimising where storage should be placed on a network~~ optimise where storage is placed so as to minimise solar curtailment ~~subject to~~ under power-flow constraints, which ~~is the mechanism by which storage capacity~~ explains how storage in one region can fail to relieve curtailment in another.

On the network side, Vargas et al. [14] study wind curtailment and storage ~~jointly~~ as instruments of ~~transmission~~ congestion management, including ~~plant~~ ramp-rate limits, on a ~~system with the same~~ corridor-constrained ~~geometry as the one considered here~~system similar in geometry to the SEN. Fang et al. [15] characterise ~~how~~ locational marginal prices ~~behave when~~ under jointly uncertain renewable generation and demand~~are jointly uncertain, which is~~ , the price signal ~~underlying~~ behind the dispatch decisions that produce curtailment.

~~A distinct~~ A related line of work ~~attacks~~ relaxes the constraint itself ~~rather than its consequences,~~ by rating network elements dynamically~~instead of conservatively~~. Foss et al. [16] give the original dynamic ampacity algorithm~~, computing a conductor's real-time current-carrying capability from weather and thermal balance rather than from a fixed seasonal rating~~. Teh and Cotton [17] show that ~~applying~~ dynamic thermal rating in a wind-integrated network changes ~~system~~ reliability, not only capacity, because ~~the rating and the~~ rating and wind resource are correlated. Lai and Teh [18] ~~survey~~ review the field and ~~set out where the practical limitslie~~its practical limits. Glaum and Hofmann [19] quantify ~~what~~ the capacity that dynamic line rating would ~~deliver on the existing German transmission~~ unlock on the German grid, and ~~find capacity gains large enough to displace some network expansion.~~ Su et al. [20] ~~close the loop with our problem by forecasting~~ forecast short-term transmission capacity ~~itself~~ under dynamic line rating~~, which is the quantity that ultimately determines how much renewable output can be evacuated. For underground~~

and . For cable-limited corridors the binding constraint is thermalrather than electrical: Maximov et al. [21] show how non-uniform soil temperature changes cable ampacity, and OcłońOcłoń [22] quantifies how relates soil thermal conductivity trades off against material costin transmission designto material cost. We do not model any of these mechanisms. We cite them because they are the physical content of the shift our calibration layer has to survive, and because these mechanisms; a forecaster with access to them would be solving a different and better-posed problem than ours.

### 2.3. Hybrid physical and data-driven modelling

The natural A common response to a data-driven model that drifts is to anchor it in physics. Lei et al. [23] embed power-flow structure intoa learned optimal power flow and show that the physical constraints act as a regulariser that improves improve extrapolation. Kruse et al. [24] apply the same idea to grid frequency, where the governing dynamics are known and the disturbances are not. Kirchner-Bossi et al. [25] combine a physical wind-farm flow model with a data-driven correction for intra-day and day-ahead forecasting, and show and find that the hybrid beats either component. This route is closed to us for a specific and instructive reason: because the physics that determines curtailment is the operator's security-constrained dispatch, and neither the network model nor the constraint set is public. Our base model is models are therefore purely statisticalby necessity, and that is precisely why , and the calibration layer has to carry the burden of the is asked to cope with the shift.

### 2.4. Probabilistic forecasting and conformal prediction

Probabilistic forecasting in power systems has largely converged relies largely on quantile methods. Maciejowska et al. [26] introduce factor quantile regression averaging for electricity pricesand establish the benchmark that later work is measured against; , and Uniejewski and Weron [27] regularise it and show that the gain is robust. Van der Meer et al. [28] study prediction intervals for solar power and net load and document how coverage degrades with aggregation level and penetration, which is the same conditional-coverage question we face across plants of different size. Mangalova and Shesterneva [29] show that a nearest-neighbour construction is competitive in the GEFCom2014 probabilistic wind track, a useful reminder that simple non-parametric predictive distributions are hard to beat, and one our own benchmark results echo. None of these methods carries a finite-sample coverage guarantee.

Conformal prediction supplies one. Angelopoulos and Bates [30] give the modern account: prediction sets with finite-sample, distribution-free marginal coverage marginal coverage that holds in finite samples and without distributional assumptions under exchangeability, which is exactly the assumption a regime change violates. Two lines of work relax it. Tibshirani et al. [31] extend the guarantee to covariate shift by reweighting calibration points by with a density ratio, which presupposes that the an invariant conditional law $\mathbb{P}(Y \mid X)$is invariant and therefore fails when the change is in the conditional itself. Barber et al. [32] dispense with exchangeabilityaltogetherdrop exchangeability, using fixed non-uniform weights and bounding the coverage gap in total variation, which covers the regime-change case directly but at the cost of a bound that is not computable ; the bound covers regime change but cannot be computed from data. Gibbs and Candès [2] take the online route, updating the miscoverage level update the miscoverage level online from observed errors to and attain long-run coverage under arbitrary shifts, and ; their scheme is the adaptive component we useand ablate. Zaffran et al. [33] study the same scheme on time series, including electricity prices, show that its behaviour depends strongly on the step size and propose aggregating several step sizes; their finding motivates the per-model selection of the step in Section 4.7. Guan [34] localises the calibration set around the test point, a different way of buying an alternative route to conditional validity that we do not pursuehere but that is a natural competitor.

Two constructions are central to our implementation. Chernozhukov et al. [35] score observations by their probability integral transform, which is what makes a score makes scores comparable across plants of very different size and what handles the atom at zero; our score is theirs. Romano et al. [36] conformalize quantile regression directly, measuring nonconformity in the units of the responserather than in probability, and their methodis the benchmark that turns out to dominate our own pipeline in Section 5. Applications of these ideas ; we apply their method, in one-sided form, to every base model. Applications to power systems are recent and growing. Hu et al. [37] conformalize a temporal convolutional quantile network for wind power intervals, and Jensen et al. [38] show that ensembling conformalized quantile regressors improves both coverage and sharpness for time series. Neither addresses regime change , and neither, to our knowledge, asks the question this paper turns out to be about: or asks how much of the benefit attributed to a calibration layer is really an artefact of depends on the base model it is correctingcorrects.

Three further tools are used rather than extended. Peyré and Cuturi [39] ~~give~~ provide the computational treatment of optimal transport from which we take the monotone one-dimensional map~~.~~ ; Chernozhukov et al. [40] ~~establish that rearranging a set of estimated~~ show that rearranging crossing quantile curves into ~~a monotone one is never worse in an appropriate sense, which is what~~ monotone ones does not worsen them, which licenses the rearrangement step ~~in our multi-quantile base model.~~ of our quantile base models; and Killick et al. [41] give the PELT algorithm used to date the distributional change~~, and~~ .

### *2.5. Evaluating interval forecasts for decisions and under dependence*

Gneiting and Raftery [42] give the theory of proper scoring rules from which our ~~evaluation metric~~primary metric, the one-sided interval score, is derived. Gneiting and Ranjan [43] introduce threshold- and quantile-weighted versions of the continuous ranked probability score that emphasise a region of interest, such as the upper tail, while remaining proper; we use the threshold-weighted form to evaluate the base predictives in the tail. Lerch et al. [44] show that evaluating forecasts only on the cases in which an extreme outcome occurred, the forecaster's dilemma, rewards forecasters who overpredict extremes and can rank forecasts wrongly; this is why we report coverage on high-curtailment days defined both by the outcome and by information available when the forecast is issued. Our comparisons rest on a small number of base models and windows that share one panel. Cameron, Gelbach and Miller [45] show that a wild cluster bootstrap gives reliable inference with few clusters where cluster-robust standard errors do not; MacKinnon and Webb [46] extend the analysis to clusters of very different sizes; and Webb [47] proposes six-point weights for the case of very few clusters. We use these tools, together with a day-block bootstrap, to attach uncertainty to the cross-model relationships in Section 5.4.

## 3. Data

### *3.1. Source and scope*

The dataset consolidates the curtailment records published monthly by Chile's National Electricity Coordinator (Coordinador Eléctrico Nacional, CEN) into two validated relational tables. The daily table contains 293,678 records and the hourly table 6,906,761 records, both covering the continuous period from 1 January 2022 to 31 May 2026, that is 1,612 days built from all 53 monthly reports, with no missing days. Records span 300 generation plants across four technologies: solar photovoltaic (75 plants), wind (56), run-of-river hydropower (140) and reservoir hydropower (29). Twelve hydropower plants are labelled inconsistently across the operator's monthly reports, alternating between the run-of-river and reservoir categories; the plant registry resolves each of them to a single canonical technology, so that a count of distinct plant-technology pairs in the raw daily table returns 312 rather than 300. Hydropower curtailment records are present from June 2024 onward, with hourly hydropower detail from July 2024, following the coverage of the underlying public reports.

Each plant is linked to a metadata registry matched to the CEN installation catalogue, with 18 fields including the official identifier, installed capacity, owner, region, province, municipality, ~~geographic~~ coordinates, commissioning date, connection point and regulatory classification; the matching method is recorded for every plant. Days and hours without curtailment are stored explicitly as zero-valued records, which represent 49.8% of daily and 83.6% of hourly rows. ~~This~~ The explicit encoding of zeros ~~is deliberate: it~~ preserves the zero-inflated structure of the quantity, which ~~is essential for the modelling in~~ the models of Section 4 depend on. Figure 1 shows the geographic distribution of the plants~~, with the concentration of solar capacity in the northern regions and of hydropower in the south~~.

### *3.2. Construction and validation*

The 53 monthly reports were retrieved from the CEN public portal with a versioned download script, parsed and loaded into a relational database, and the original files were preserved unmodified with SHA-256 checksums. During construction we identified eight classes of errata in the original publications. Five were corrected or recovered deterministically in the curated tables and three are limitations of the source that we document without correction. Among the corrections, two omitted calendar days were recovered from the corresponding monthly hourly sheets and cross-checked to the ~~millesimal~~ third decimal against the source files' own total columns; one of ~~these~~ them was detectable only by a structural continuity check, since the omitted day carried zero energy and was therefore invisible to any energy-based validation.

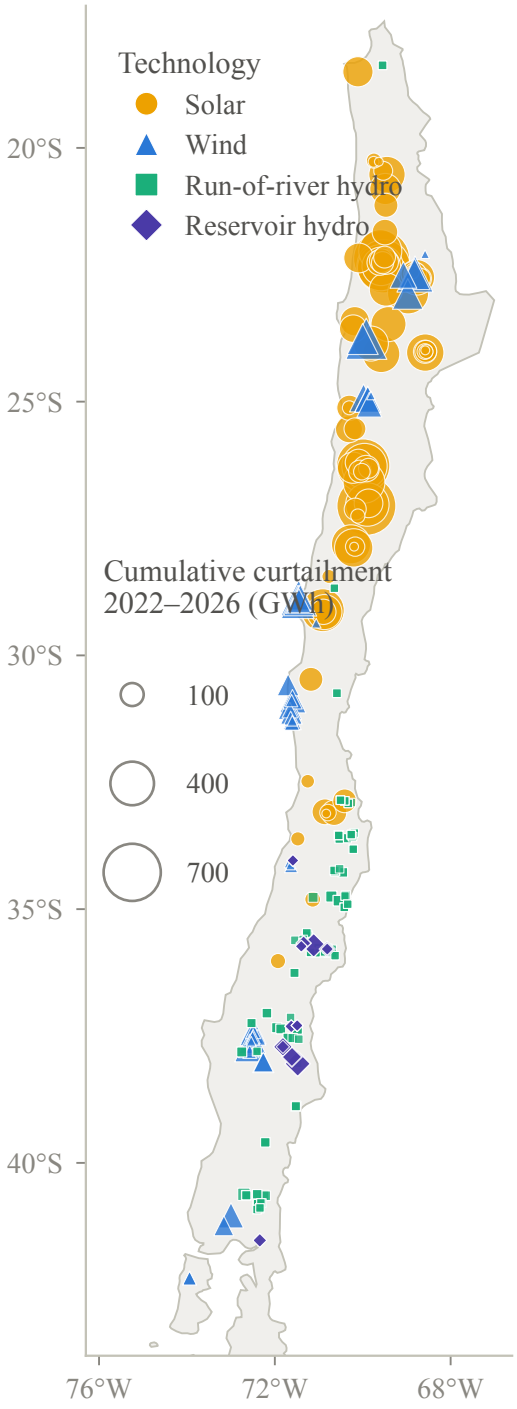



Figure 1: Geographic distribution of the 300 power plants in the dataset, coloured and shaped by technology, with marker area proportional to each plant's cumulative curtailed energy over January 2022 to May 2026.

Two independent checks establish the fidelity of the result. Annual totals from the daily table match the CEN official summary exactly, to ~~a difference of~~ 0.000 GWh, in every year from 2022 to 2026, and aggregate 2022 wind and solar curtailment reproduces the 1.47 TWh independently reported by Fraunhofer Chile for that year. At monthly resolution the agreement is close but not exact: in a ~~census~~ month-by-month reconciliation of hourly sums against daily totals over all 53 months, 50 months agree within 1% and 35 agree exactly, while the ~~three remaining months~~ remaining three correspond to documented inconsistencies within the operator's own publications. The residual sub-percent differences arise from the operator ~~re~~assigning the same monthly energy differently across sister run-of-river units ~~between its monthly~~ in its hourly sheets and in its annual closing, with a net effect of zero at system level~~, which is the same mechanism as errata classes six and seven~~. The dataset totals 19,030 GWh of recorded curtailment~~over the period~~.

The dataset is publicly available on Zenodo under a CC BY 4.0 licence [48], together with the data dictionary, the errata log and the integrity checksums. The processing pipeline that regenerates every curated table is available in the companion code repository.

### *3.3. Exploratory characterisation*

Three features of the data govern the modelling choices that follow.

First, curtailment is strongly zero-inflated and its occurrence is persistent~~in time~~, with a daily lag-one autocorrelation of roughly 0.85 at system level and a marked weekly cycle. Second, the positive magnitudes are heavy-tailed and approximately lognormal, with a ratio between the 99th and 50th percentiles of ~~the order of~~ about twenty-five across technologies ~~, and of~~ and about fifteen for solar alone ~~.~~ (Figure 2~~illustrates both~~).

Third, the change in the distribution of curtailment ~~over the study period~~ acts on scale ~~, not on occurrence and not on timing. Occurrence is stable: for~~ rather than on occurrence or timing. For solar plants in the northern regions, the probability of a positive day ~~,~~ averaged over each evaluation window ~~,~~ stays between 0.750 and 0.815 from January 2024 to May 2026, and a change-point search on the monthly series of $\mathbb{P}(Y > 0)$ returns no break after September ~~2022. At~~ 2022; at monthly resolution the ~~same~~ series ranges more widely, from 0.635 to 0.877~~, so the stability claim is one about window aggregates and not about individual months~~. The intraday ~~shape is nearly as stable: the peak hour of the normalised intraday profile is~~ profile is also stable: its peak is at hour 16 in every half-year from 2022 to the first half of 2025 and at hour 17 ~~in the last two half-years~~afterwards, two adjacent hours whose profile masses differ by less than 0.4 percentage points~~; the centroid of the profile moves by at most 22 minutes between consecutive half-years after 2023-H1~~, and the total variation distance between ~~the profiles of~~ consecutive half-years stays between 0.032 and 0.079 ~~over the same span.~~ after 2023. The positive magnitudes, in contrast, roughly triple: the median ~~of~~ positive daily curtailment ~~for~~ of solar plants in Antofagasta and Atacama rises from 92 MWh in the calibration window to 267 MWh in October to December 2024. ~~These three facts, zero inflation, heavy tails and a scale-type shift, motivate respectively the two-part model, the choice of evaluation metric and the adaptive calibration described next.~~

### *3.4. Dating the distributional change*

The timing of the change in scale is best read after removing the annual cycle. Figure 3 shows the monthly mean of $\log Y$ over positive plant-days for solar plants in Antofagasta and Atacama, together with the same series net of a calendar-month median. Applied to the deseasonalised series with the PELT algorithm~~of Killick et al.~~ [41] and with binary segmentation, both under an $L_2$ cost and a BIC penalty $\hat{\sigma}^2 \log n$ with a minimum segment length of three months, the two algorithms return ~~two breaks and agree on both~~the same two breaks: May 2023, with a jump of +1.05 in mean log-magnitude, and January 2024, with a jump of +0.76. Neither returns a break in October 2024 ~~, and neither returns any break~~ or in 2025.

~~That conclusion is robust to the analyst's choices. Repeating the detection over~~ Across 180 ~~configurations, crossing~~ detection configurations (two strata, three deseasonalisation methods~~(calendar-month median, calendar-month mean and a robust STL decomposition), ,~~ five penalty multipliers and three minimum segment lengths~~, )~~ no configuration returns a break in 2025 ~~and none returns a break~~ or in October 2024. The change is therefore a ramp of roughly eighteen months~~rather than a step, and it is~~ , essentially complete by the end of 2024: ~~comparing like with like across years,~~ the Wasserstein-1 distance between the positive log-magnitudes of October to December 2023 and ~~those~~ of October to December 2024 is 1.20, whereas between October to December 2024 and October to December 2025 it

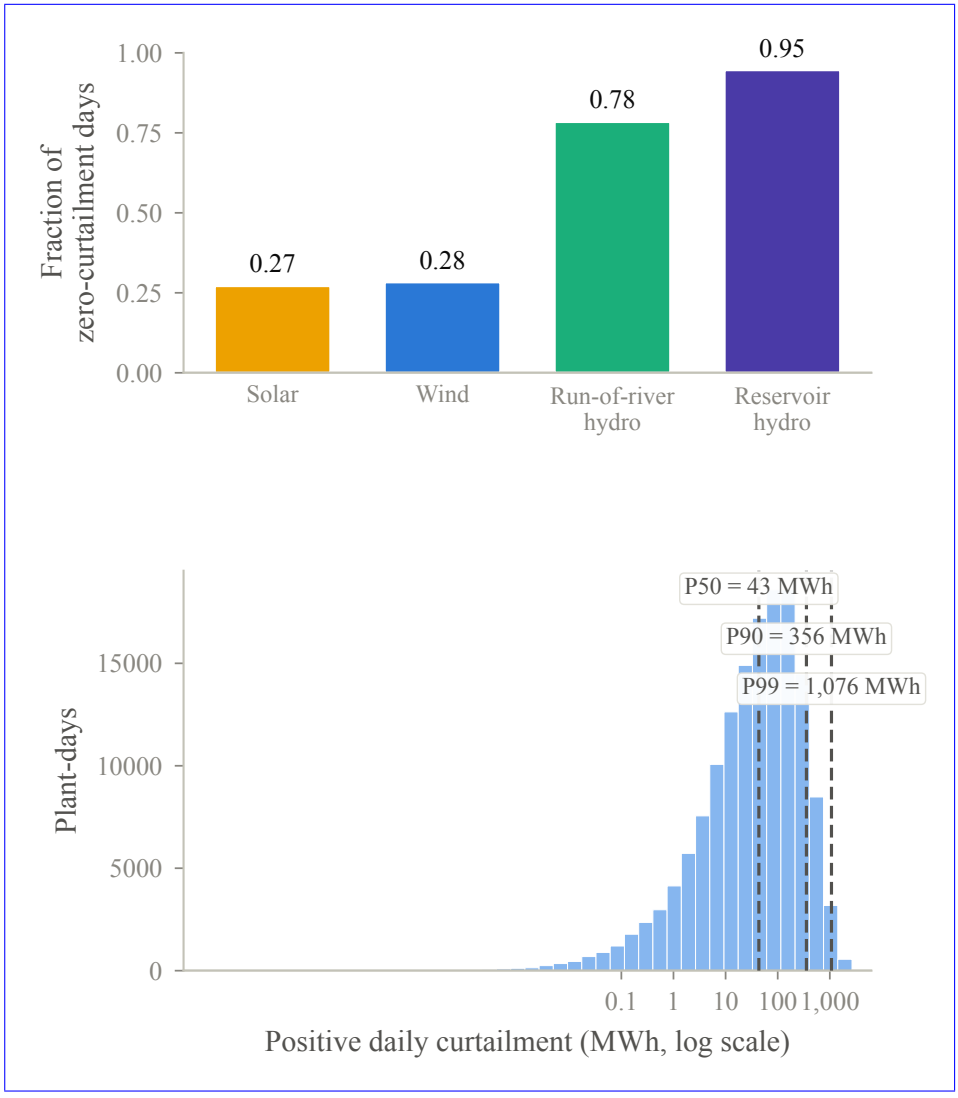



Figure 2: Distributional structure of plant-level curtailment. Top: share of zero-valued plant-days by technology. Bottom: distribution of positive magnitudes on a logarithmic scale, with the 50th, 90th and 99th percentiles marked.

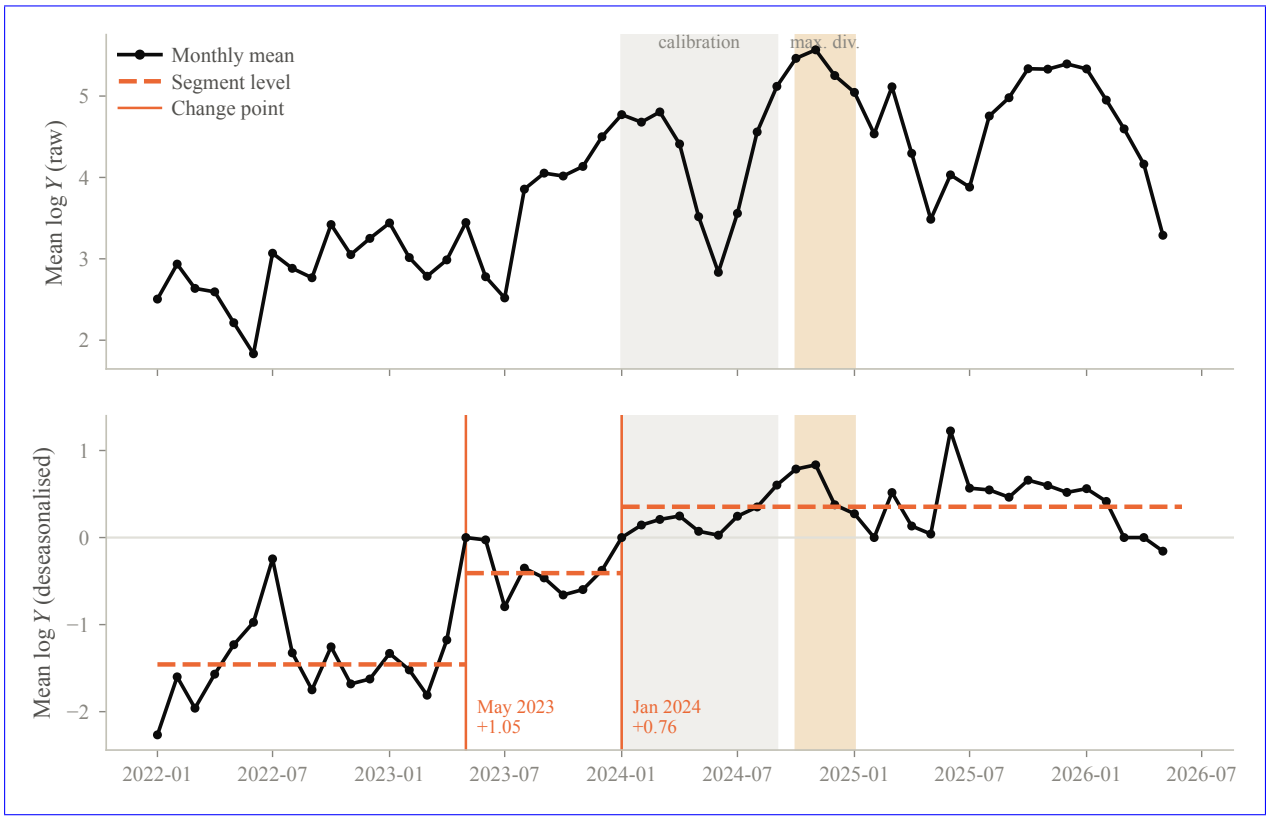



Figure 3: Monthly mean of $\log Y$ over positive plant-days, solar plants in Antofagasta and Atacama. Top: raw series, dominated by the austral annual cycle. Bottom: the same series net of a calendar-month median, with the two change points returned by PELT and binary segmentation (May 2023 and January 2024) and the calibration and transition windows of Section 4 shaded. The deseasonalised level peaks in October and November 2024 and then stops rising; the isolated spike of June 2025 is discussed in Section 6.

is 0.085, below the 0.115 threshold of a permutation null that resamples whole days~~and therefore not distinguishable from no change at all~~.

Because the annual cycle is strong, ~~any contrast between windows that fall~~ contrasts between windows in different seasons ~~registers a large distance even when no regime changeis present, and this bears directly on how the evaluation windows of Section 4 should be read~~register large distances even without a regime change. Table 1 decomposes the January–September to October–December shift in the monthly mean of $\log Y$ into a seasonal component, ~~given by the calendar-month effects and therefore~~ identical in every year, and a secular remainder. In 2024 the shift is the largest of the record, but 61% of it for northern solar, and 76% at system level, is the ~~seasonal component that recurs every year; the same contrast yields a Wasserstein-1 distance of 0.96 in 2023 and 0.88 in 2025, years for which no regime change is claimed. The~~ recurring seasonal component; the secular remainder of 2024, +0.46 for northern solar, is ~~real and is~~ the largest of the four years ~~, but it is~~ but of the same order as the +0.29 of 2023. ~~We report the raw contrast~~ The raw contrast is used throughout the results, and ~~flag here that it~~ should not be read as if it were entirely a regime effect.

## 4. Methods

### *4.1. Problem setup and base forecasters*

For each plant $i$ and target day $T$ we forecast the curtailed energy $Y_{iT} \geq 0$ from information available at the forecast origin $t = T - 7$~~, a direct seven-day-ahead formulation~~. The feature vector $x_{iT}$ has twelve components: the most recent

Table 1: Decomposition of the January–September to October–December shift in the monthly mean of log $Y$ over positive plant-days, in log units, by year. The seasonal component is the difference between the calendar-month effects of the two windows and is by construction the same in every year; the secular component is the remainder. The 2024 row is the contrast that defines the transition window used in the evaluation; the final row of each block repeats the decomposition on the calibration window itself (January to August against October to December 2024), which is the pair the evaluation actually uses.

| Year | Total shift | Seasonal | Secular | Seasonal share |
|---|---|---|---|---|
| *Solar, Antofagasta and Atacama* | | | | |
| 2022 | +0.64 | +0.72 | −0.09 | 113% |
| 2023 | +1.01 | +0.72 | +0.29 | 72% |
| **2024** | **+1.18** | **+0.72** | **+0.46** | 61% |
| 2025 | +0.90 | +0.72 | +0.17 | 81% |
| 2024, calibration window | +1.29 | +0.78 | +0.50 | 61% |
| *System (solar and wind)* | | | | |
| 2022 | +0.77 | +0.66 | +0.11 | 86% |
| 2023 | +0.89 | +0.66 | +0.23 | 74% |
| **2024** | **+0.86** | **+0.66** | **+0.20** | 76% |
| 2025 | +0.71 | +0.66 | +0.05 | 93% |
| 2024, calibration window | +0.94 | +0.69 | +0.24 | 74% |

Note: the calendar-month effects are estimated as the median of each calendar month over the whole record. A robust STL decomposition returns somewhat different values on a 53-month series carrying a strong trend; it does not change the order of magnitude of the split, and the secular components remain the smaller part of the 2024 contrast.

observation $y_{i,t}$, three lags $y_{i,t-1}$, $y_{i,t-7}$ and $y_{i,t-14}$, seven- and thirty-day trailing means ending at $t$, an occurrence indicator $\mathbb{1}\{y_{i,t} > 0\}$, the day of week and day of year of the target, the log of installed capacity, and the region and technology as categorical variables. Every dynamic feature is a shift of at least seven days on a calendar-continuous index, which is verified by assertion at construction time and when the features are built, which rules out leakage by design. . The modelling panel contains the 131 solar and wind plants. All base models are fitted on targets up to 31 December 2023 and never see the evaluation period; hyperparameters were fixed a priori and were not tuned against it.

The paper uses three base predictives , which differ only in the shape they impose on $\mathbb{P}(Y \mid x)$We use nine base predictives from two model families.

*Hurdle with constant dispersion*Lognormal hurdle. The base model of the submitted version, and the reference throughout. It factorises the predictive predictive factorises into occurrence and magnitude,

$$Y \mid x \;\sim\; (1 - p(x))\,\delta_0 \;+\; p(x)\,\mathrm{LogNormal}(\mu(x), \sigma^2), \tag{1}$$

with $p(x)$ from a gradient-boosting classifier and $\mu(x)$ from a gradient-boosting regression on $\log y$ over positive days. The dispersion $\sigma$ is estimated out of fold , by five-fold cross-fitting on the training period, to avoid which avoids the downward bias of the in-sample residual variance of a boosted model; on these data . Four variants are used. The *reference hurdle* has two models of 400 trees each, with hyperparameters fixed a priori (learning rate 0.05, depth 6, minimum child weight 5, row and column subsampling 0.9), and $\hat{\sigma} = 1.70$. The *early-stopped hurdle* holds out the last six months of the training period (July to December 2023) as a temporal validation set, fits each stage to a ceiling of 4,which we return to in Section 5 because it sets the sharpness floor of the whole pipeline.

*Hurdle with conditional dispersion.* The same construction with 000 trees with a patience of 50 rounds, and refits both stages on the full training period at the selected budget: 59 trees for occurrence and 46 for magnitude, 105 in total, with $\hat{\sigma} = 1.83$. The *conditional-dispersion hurdle* replaces $\sigma$ replaced by a fitted $\sigma(x)$, estimated out of fold from the squared stage-one residuals with a global scale correction and a floor at the fifth percentile of the training distribution. It is included to separate two possible explanations of the results: that the difficulty lies in the constant dispersion,or that it lies in the lognormal shape itselfThe *enlarged hurdle* has two models of 2,000 trees.

*Multi-quantile gradient boosting.* A predictive defined not by a parametric family but by a by a dense grid of 54 estimated quantiles, at levels 0.02 to 0.98 in steps of 0.02 plus 0.005, 0.01, 0.99, 0.995 and 0.999, fitted with

the pinball loss under ~~exactly~~ the same features, rows, training cut-off, per-model hyperparameters and seed as the hurdle. ~~One thing this does not hold fixed is capacity. The hurdle is two boosted models, a classifier and a regressor; the multi-quantile predictive is 54 boosted models of 400 trees each, one per quantile level. Matching the training protocol is not the same as matching the number of fitted parameters; Section 5.6 separates the two experimentally and Section 7 states what the separation does and does not licence. Estimated quantile curves may cross, and we restore monotonicity by rearrangement in the sense of Chernozhukov et al.~~Crossing quantile curves are rearranged into monotone ones [40] ~~,~~ by sorting each row and clipping at zero. ~~Two properties matter for what follows.~~ The predictive CDF is the interpolated inverse of the quantile function, so it is not committed to any tail shape~~. And~~ , and the mass at zero is ~~not imposed: it is~~ read off the grid as $F_x(0) = \max\{\tau_k : \hat{q}_{\tau_k}(x) = 0\}$~~, so zero inflation emerges from the fit rather than from the model form. Beyond the highest estimated level~~. Above the highest level, $\tau = 0.999$, the predictive expresses nothing, ~~so~~ and a requested quantile above ~~$\tau = 0.999$~~ it returns $+\infty$~~and is counted as an infinite interval under the same rule applied to every other method~~. Five variants differ only in the number of trees per quantile level: 15, 50, 100, 200 and 400. The variant with 400 trees per level is the *reference quantile model*.

~~This third model exists because of a specific question, raised in review and taken up in Section 5.3: whether the behaviour of the calibration layer is a property of the layer or of the predictive it is wrapped around~~Three base models appear in every main comparison: the reference hurdle, the early-stopped hurdle and the reference quantile model. The other six complete the cross-model analysis of Section 5.4.

### 4.2. *Conformal calibration with a PIT score*

~~A point forecast, or even the raw predictive, does not by itself guarantee that a stated interval contains the realised value with the nominal probability. We obtain that guarantee by conformal calibration. Rather than measuring nonconformity in energy units, which are not comparable across plants of very different size, we~~ We score each observation by its probability integral transform under the predictive law,

$$s = F_x(y), \tag{2}$$

~~following~~ the distributional-conformal ~~construction~~ score of Chernozhukov et al. [35]. Because $F_x$ has an atom at zero, the score is randomised there in the standard way for mixed distributions [49]: for $y = 0$ we set $s = F_x(0^-) + U(F_x(0) - F_x(0^-))$ with $U \sim \text{Uniform}(0, 1)$, which for the hurdle reduces to $s = (1 - p(x))\,U$~~; for $y > 0$ no randomisation is needed~~. Under the true predictive law $s$ is ~~then~~ Uniform(0, 1), ~~and it lives~~ on a common ~~[0, 1]~~ scale across plants of any size~~, which is why it handles the atom and the plant heterogeneity cleanly. The same two operations~~ . The calibration layer asks only two operations of the base model, the CDF and its generalised inverse, ~~are all the calibration layer ever asks of the base model , and this is what makes the comparison of Section 5.3 exact rather than approximate: the three predictives are substituted behind an identical interface~~ so every base model enters through the same interface and every comparison across base models changes nothing else.

~~For the hurdle this score has a further property that motivated the transport step. For a positive observation the score is a monotone transform of the standardised log-magnitude $z = (\log y - \mu(x))/\sigma$, namely $s = (1 - p) + p\,\Phi(z)$. A change of scale of the magnitude distribution by a factor $c$ is a location shift $\log c$ of $\log Y$, hence a translation of $z$ by $\delta = \log c/\sigma$. It is therefore a translation on the latent Gaussian scale, and only a monotone displacement of the law of $s$ on $[0, 1]$; it is not a translation of $s$ itself. This low-dimensional structure, established under the empirically supported assumption that the shift rescales the positive magnitudes while leaving occurrence and intraday timing essentially unchanged (Section 3.3), is what suggested that the displacement could be estimated from a short recent window while the shape of the tail is inherited from the large calibration set. That structure is a property of the base model rather than of the data: the single translation $\delta = \log c/\sigma$ exists because the hurdle models every positive magnitude as lognormal with one dispersion $\sigma$, so the motivation for the transport holds for the hurdle only. Under the conditional-dispersion hurdle the translation varies with $x$, and under the multi-quantile predictive the displacement of each score depends on the shape of its own estimated distribution, so for the other two base models of Section 5.3 the transport is the same algorithm without that motivation.~~

Given a calibration set of size $n$~~scored by (2)~~, the conformal quantile is the order statistic $\hat{q} = s_{(k)}$ with $k = \lceil (n + 1)(1 - \alpha) \rceil$, which yields ~~the finite-sample marginal guarantee~~ $\mathbb{P}(Y \le U(x)) \ge 1 - \alpha$ under exchangeability; when $k > n$ the quantile is $+\infty$~~and the interval is the whole line~~. For the hurdle the upper prediction limit $U(x) = F_x^{-1}(\hat{q})$ is

recovered in closed form: it is zero when $\hat{q} \le 1 - p(x)$, it is $+\infty$ when $\hat{q} \ge 1$, and otherwise

$$U(x) = \exp\Big(\mu(x) + \sigma\,\Phi^{-1}\big(\tfrac{\hat{q}-(1-p(x))}{p(x)}\big)\Big).$$

$$U(x) = \exp\Big(\mu(x) + \sigma\,\Phi^{-1}\big(\tfrac{\hat{q}-(1-p(x))}{p(x)}\big)\Big), \tag{3}$$

A short proof of the inversion is given in Appendix A, and it was additionally checked numerically against oracle coverageas shown in Appendix A. For the multi-quantile predictive the same quantile model the inversion is a linear interpolation on the quantile grid.

For the hurdle, a change of scale $c$ of the positive magnitudes is a translation by $\log c/\sigma$ on the latent Gaussian scale of the score, that is, a monotone displacement of the law of $s$ on $[0, 1]$. This structure motivated the transport step described below. It belongs to the hurdle, whose positive magnitudes share one dispersion; for the other base models the transport is the same algorithm without that motivation.

### 4.3. *Three guarantees, and which one applies where*

It is easy , and consequential, to conflate three statements that this paperkeeps separate.

Three statements are easy to conflate and are kept separate in this paper. *Finite-sample marginal validity*.The : the static split satisfies $\mathbb{P}(Y \le U(x)) \ge 1 - \alpha$ exactly, at the stated $n$ , provided if the calibration scores and the test score are exchangeable. This is a statement about ; this is an average over draws of the whole calibration set and the test point, not about any particular a statement about any plant, day or window.

*Long-run adaptive control*.The : the online update of Gibbs and Candès [2] guarantees $\frac{1}{T}\sum_{t \le T} \mathrm{err}_t \to \alpha$ for *any* sequence, with no distributional assumption whatever. It any sequence, without distributional assumptions, but says nothing about coverage at any finite time, and in exchange it discards the finite-sample guarantee of the split.

a finite time. *Empirical coverage in this panel*.What we actually measure. Neither of the two statements above describesit: what we measure, which neither statement describes, because exchangeability fails here twice over, and not only across the regime boundary and also within it. All plants share the daily state of the system, so the calibration scores arrive in strongly correlated daily blocks. We quantify this rather than assert it. On : on the calibration set , the intraclass correlation of the PIT score by date is 0.238, and with 108.8 plants per day this gives a design effect of the design effect is 26.7: , so the effective calibration sample is about 1,000 rows , not the rather than 26,552 that the finite-sample guarantee would nominally be stated at. 552. Every coverage figure we report therefore carries a standard error clustered by date, and every difference between methods carries a bootstrap interval that resamples whole days. Section 5.9 tests calibration schemes designed to restore exchangeability at the block, group and plant level, and reports how far each one actually moves conditional coverage.

### 4.4. *Responses to the shift*Calibration layers

We consider four responses, of which the last is the one the submitted version proposed. All layers produce a one-sided upper limit $[0, U]$ at nominal level $1 - \alpha = 0.90$ and use the same calibration window, January to August 2024.

*Weighted conformal.* Reweighting each calibration point by an estimate of the density ratio between calibration and test distributions is the standard remedy for *covariate* shift . It is the wrong tool here: the shift changes the conditional law $\mathbb{P}(Y \mid x)$, not the marginal of $x$, so no reweighting of old scorescan reflect the new conditional, and the density-ratio estimate degenerates because features that are monotone in time separate calibration from test almost perfectly. On a synthetic panel where covariate shift can be isolated, the per-point implementation returns infinite intervals for a large fraction of test points (`flagship/conformal_v2.py` in the companion repository); we report this as a negative result about the tool that motivates the alternatives. We take instead the fixed-recency-weight scheme of Barber et al. as the recency benchmark, in its simplest form: the sliding window below, which gives weight one to the *Static split.* The conformal quantile of the calibration scores, fixed for the whole test period.

*Sliding window.* The conformal quantile of the scores of the most recent last 60 daysbefore the embargo and zero to all earlier ones.

*Sliding-window recalibration.* We discard old calibration data and recalibrate on a trailing 60-day window , refreshed every 7 days. Because the base model forecasts at a 7-day horizon , the label forecast horizon is 7 days, the

outcome of an origin is ~~realised~~ known only 7 days later~~; to avoid using information unavailable at the forecast origin, every online calibration~~ , so every online window ends 7 days before its target ~~. We call this the horizon embargo and it applies to every online method. The static split does not use online calibration and is therefore unaffected, which serves as an implementation control.~~ date; this horizon embargo applies to all online layers.

*Adaptive conformal inference (ACI).* ~~We update the effective miscoverage level~~ The miscoverage level is updated online [2], $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \mathrm{err}_t)$~~with $\mathrm{err}_t = \mathbb{1}\{Y_t > U_t\}$ averaged over the~~ , with $\mathrm{err}_t$ the fraction of plants of day $t$ ~~, where the error feedback is delayed by the same~~ whose outcome exceeds its limit, fed back after the 7-day embargo. Left unbounded, the update can drive $\alpha_t$ to zero, at which point the conformal quantile does not exist and the limit is $+\infty$; we report both the unbounded update and an update bounded below at $\alpha_{\min} = 0.005$.

~~*Transport of scores.* Motivated by the monotone-displacement structure of Section 4.2, we transport the calibration~~ *Transport + ACI.* The calibration scores are mapped onto the distribution of the scores ~~from the old regime onto the new one before conformalising. Let $\hat{F}_{\mathrm{old}}$ and $\hat{F}_{\mathrm{new}}$ be the empirical CDFs of the calibration scores and of the scores in the recent window ;~~ in a recent window by the monotone one-dimensional ~~optimal-transport map is the increasing rearrangement~~ transport map $T = \hat{F}_{\mathrm{new}}^{-1} \circ \hat{F}_{\mathrm{old}}$ [39], ~~estimated nonparametrically by interpolation between empirical quantiles~~the conformal quantile is taken from the transported pool at the current $\alpha_t$, and $\alpha_t$ is updated as in ACI. In the upper tail, where the ~~short~~ recent window is noisy, the map is shrunk towards the identity in proportion to a bootstrap estimate of its variance~~, so that a noisy new quantile is replaced by the stable old one rather than extrapolated~~. Because the ~~transport map is data-dependent it breaks the exchangeability of~~ map depends on the data, the transported scores are not exchangeable with the test score; ~~we therefore restore validity operationally by wrapping the transported scores in the online adaptive scheme, so that~~ coverage is controlled by the ~~adaptive~~ online update and the ~~transport~~ map contributes only sharpness ~~. An idealised validity argument, exact under an oracle map and controlled through the total-variation framework of Barber~~ (Appendix B).

*One-sided conformalized quantile regression (CQR).* Following Romano et al. ~~under an estimated map, is given in Appendix B.~~ [36], the nonconformity score is $E = y - q_{1-\alpha}(x)$, measured in MWh, where $q_{1-\alpha}(x) = F_x^{-1}(1-\alpha)$ is the base model's own quantile, and the limit is $U(x) = \max\{q_{1-\alpha}(x) + \hat{Q}_{1-\alpha}(E), 0\}$ with $\hat{Q}$ the conformal quantile of $E$ on the calibration window. We apply it to every base model, so that the comparison between calibration layers is made on a common base model.

### 4.5. ~~*The adaptive scheme in full*~~

Weighted conformal prediction [31] is not used: the shift changes the conditional law $\mathbb{P}(Y \mid x)$, which no reweighting of old scores can reflect, and features that trend in time separate calibration from test almost perfectly, so the density-ratio estimate degenerates.

### 4.5. *The transport-based scheme in full*

~~The construction above is stated at the level of ideas. Because reproducibility of the exact procedure is the point,~~ Algorithm 1 gives ~~it completely~~the procedure, Figure 4 ~~shows the~~ its data flow, and Appendix C ~~tabulates every constant , including the seeds~~every constant and seed. Three implementation choices ~~deserve to be stated in the body because they are the ones~~ are stated here because a reader would otherwise have to guess them.

~~*The quantile*~~ *Quantile grid and* ~~*the*~~ *interpolation.* The map is estimated on an equispaced grid of ~~$g$ probability levelson $[0, 1]$, with $g = \min(100, \max(20, m/2))$~~ $g = \min(100, \max(20, m/2))$ probability levels, where $m$ is the number of scores in the recent window~~; the second term prevents a grid finer than the data supports. The map is~~ , and applied by two successive linear interpolations, ~~first~~ from a score to its level on the old empirical quantile function ~~, then~~ and from that level to the regularised new quantile~~, which is exactly the increasing rearrangement evaluated pointwise~~.

~~*The shrinkage function and its strength*~~*Shrinkage.* The recent window is bootstrapped $B = 150$ times; the map is the mean of the ~~$B$~~ replicate quantile functions, ~~which is a bagged estimate, and the replicate~~ and their standard deviation $\mathrm{sd}_{\mathrm{boot}}(u)$ at each level $u$ ~~is the width of the uncertainty band~~measures its uncertainty. The regularised quantile is

$$\hat{q}_{\mathrm{reg}}(u) = (1 - \lambda(u))\,\hat{q}_{\mathrm{old}}(u) + \lambda(u)\,\hat{q}_{\mathrm{new}}^{\mathrm{bag}}(u), \qquad \lambda(u) = \frac{1}{1 + (\mathrm{sd}_{\mathrm{boot}}(u)/\tau)^2}, \tag{4}$$

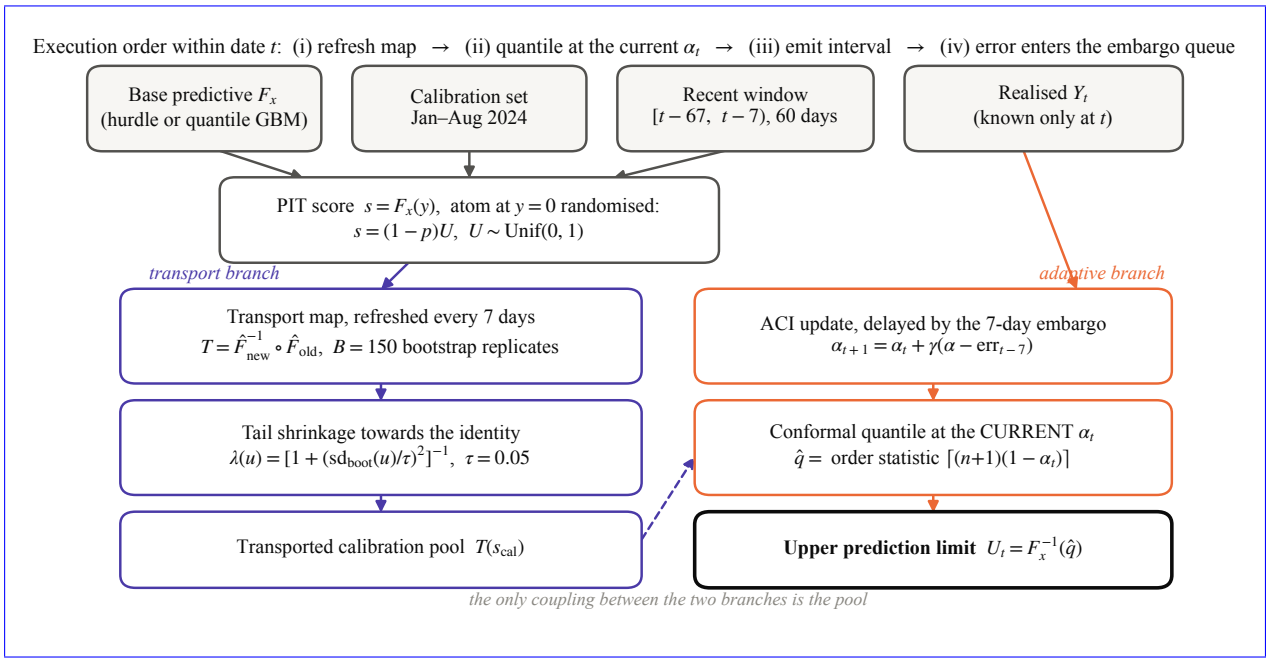



Figure 4: Data flow of the transport-based scheme, from the base predictive to the upper prediction limit. The transport branch (left) estimates and regularises the map; the adaptive branch (right) maintains the effective miscoverage level. The two are coupled only through the transported calibration pool; the order of operations within a date is stated at the top.

with $\tau = 0.05$ in PIT units, ~~a fixed regularisation scale that was never tuned against the test set~~fixed a priori. Where the bootstrap is stable ~~$\lambda \approx 1$ and~~ the map transports fully; where it is noisy ~~$\lambda \approx 0$ and the map~~ it reverts to the ~~identity, that is to the stable~~ old quantile. The result is monotonised by a running maximum and clipped to $[0, 1]$~~so that it remains a quantile function~~.

~~*The order*~~ *Order of execution.* On date $t$: (i) if a refresh is due, the map is recomputed from the ~~recent~~ window ending at $t - 7$ and the ~~entire~~ whole calibration pool is transported; (ii) the conformal quantile is taken ~~from the transported pool at the *current*~~ at the current $\alpha_t$~~, which does not yet incorporate the error of day $t$~~; (iii) the ~~interval is emitted~~limit is issued; (iv) the error of day $t$ enters the embargo queue and updates $\alpha$ ~~only~~ seven steps later. The map never uses the adaptive level~~and the adaptive~~ , and the update never uses the map~~; the sole coupling between the two branches is the pool they share~~.

### 4.6. What is ~~*new here,*~~ *taken from the literature and what is not*

~~Every ingredient of Algorithm 1 is taken from existing work, and it is worth being exact about which.~~

~~The probability-integral-transform score is the distributional-conformal score~~ The PIT score is that of Chernozhukov et al. [35]; the randomisation of the atom ~~for a mixed distribution is the classical~~ is the device of Smith [49]~~. The online update of the miscoverage level is the adaptive conformal inference~~ ; the online update is that of Gibbs and Candès [2]~~, unmodified. The~~ ; the monotone one-dimensional ~~transport~~ map is the increasing rearrangement ~~, standard in~~ of optimal transport [39]~~and, in the quantile-curve setting, the rearrangement of~~ , which Chernozhukov et al. [40] ~~. Bagging a bootstrap estimate and shrinking it~~ apply to quantile curves; bagging and shrinking towards a stable reference in proportion to ~~its~~ the bootstrap variance is ordinary shrinkage practice.

~~What is new~~ What is specific to the scheme is the combination~~and one specific device. The combination is applying~~ , a transport of the ~~*score*~~ distribution ~~, rather than of the response or the covariates,~~ score distribution used as a sharpness-only step inside an online ~~adaptive scheme~~ update that carries the coverage control, ~~so that the data-dependence of the map cannot damage validity. The device is the~~ and the variance-proportional tail shrinkage ~~(4), which uses the bootstrap dispersion of the estimated map to interpolate between transporting and not transporting, level by level.~~

**Algorithm 1** Transport + ACI with tail shrinkage. Constants and seeds in Appendix C.

**Require:** base predictive $F_x$; calibration scores $s_{\text{cal}}$ of size $n$; nominal level $\alpha$; step $\gamma$; window $W$; refresh period $R$; embargo $E$; bootstrap size $B$; shrinkage scale $\tau$; bounds $\alpha_{\min}, \alpha_{\max}$

$\alpha_1 \leftarrow \alpha$; $\mathcal{P} \leftarrow \text{sort}(s_{\text{cal}})$; queue $Q \leftarrow \emptyset$; $t_{\text{next}} \leftarrow t_0$
**for** each test date $t$ **do**
  **if** $t \geq t_{\text{next}}$ **then**
    $s_{\text{new}} \leftarrow$ scores with date in $[t - W - E,\ t - E)$
    **if** $|s_{\text{new}}| \geq 100$ **then**
      draw $B$ bootstrap resamples of $s_{\text{new}}$; set $\hat{q}^{\text{bag}}_{\text{new}}$ to their mean quantile function and $\text{sd}_{\text{boot}}$ to its replicate standard deviation
      form $\hat{q}_{\text{reg}}$ by (4); monotonise by running maximum; clip to $[0, 1]$
      $T \leftarrow$ interpolation from $\hat{q}_{\text{old}}$ to $\hat{q}_{\text{reg}}$; $\mathcal{P} \leftarrow \text{sort}(T(s_{\text{cal}}))$
    **end if**
    $t_{\text{next}} \leftarrow t + R$
  **end if**
  $k \leftarrow \lceil (n+1)(1-\alpha_t) \rceil$; $\hat{q}_t \leftarrow \mathcal{P}_{(k)}$ if $k \leq n$, else $+\infty$
  $U_t \leftarrow F_x^{-1}(\hat{q}_t)$ for every plant of date $t$ ▷ limit issued
  push $\text{err}_t = \frac{1}{|I_t|} \sum_{i \in I_t} \mathbb{1}\{Y_{it} > U_{it}\}$ onto $Q$
  **if** $|Q| > E$ **then**
    pop $\text{err}_{t-E}$ from $Q$; $\alpha_{t+1} \leftarrow \text{clip}(\alpha_t + \gamma(\alpha - \text{err}_{t-E}),\ \alpha_{\min},\ \alpha_{\max})$
  **end if**
**end for**

~~The two fare differently under ablation, and~~ of (4). Section 5.7 ~~reports both outcomes. The combination does not survive: once the online update is in place, adding the transport *worsens* the interval score by 68.6 MWh over the whole test, with an interval that excludes zero. The device does survive, but not its price: the variance-proportional shrinkage improves the interval score by 6.4 MWh, with a bootstrap interval that also excludes zero, for eight tenths of one per cent of the score and ninety-one per cent of the runtime. We regard stating both plainly as more useful than the claim of novelty they replace.~~ evaluates each component separately.

### *4.7. Hyperparameter selection*

The ~~submitted version reported the adaptive methods at $\gamma \in \{0.02, 0.05\}$ with a 60-day window, values taken from a synthetic prototyping stage and from common practice rather than from a formal selection on real data. That does not meet the standard the results should be held to, so the selection was redone.~~

step $\gamma$ and the length of the recent window are selected separately for each base model by rolling-origin validation inside the calibration window. Base-model predictions ~~exist only from~~ start on 1 January 2024, ~~since the base models are trained through 2023, so the only material available before the test period is the 244-day calibration window. We split it:~~ so January to April 2024 ~~becomes~~ serves as an inner calibration pool and May to August 2024 as an inner validation set, evaluated ~~in rolling origin~~ with the same seven-day refresh and ~~seven-day~~ embargo as the ~~real~~ test protocol. The ~~selection~~ criterion is the mean one-sided interval score of Section 4.8, which is ~~proper, is in MWh, and takes the value~~ $+\infty$ if ~~the~~ a configuration produces any infinite ~~interval, so that a configuration escaping into uninformative intervals is disqualified~~ limit, so such configurations are excluded by the criterion itself~~rather than by an ad hoc rule~~. The grids are $\gamma \in \{0.005, 0.01, 0.02, 0.05, 0.10, 0.20\}$ and ~~window lengths~~ windows of $\{30, 45, 60, 90, 120\}$ days.

~~The selection returns $\gamma = 0.005$ for pure adaptive conformal inference and $\gamma = 0.005$ with a 120-day window for the transported variant. Both are an order of magnitude gentler than the values reported in the submitted version, and both produce no infinite intervals at all in validation, whereas at $\gamma = 0.05$ between 34 and 42 per cent of validation intervals are infinite~~The selected values are reported in Appendix C. The test ~~set was not used in the selection and~~ period is used once, for the final evaluation~~. Sensitivity over the full $6 \times 5$ grid is reported~~ ; the full grid evaluated on the test ~~set in Section 5.8, Table G.19, as a display, never as a criterion, and we say so there~~period appears in

Appendix E for information only. Base-model hyperparameters were fixed a priori, except for the number of trees of the early-stopped hurdle, which is selected on the training period as described above.

### 4.8. Evaluation

~~The forecasts are upper prediction limits, that is one-sided intervals $[0, U(x)]$ at level $1 - \alpha = 0.90$, and the evaluation has to be appropriate to that. We report empirical coverage against the nominal level with standard errors clustered by date, and,for sharpness, the mean width over finite intervals together with the fraction of infinite ones. Reporting only those two, as the submitted version did, is misleading in a specific way: a method that escapes into infinite intervals removes its widest cases from the width average and therefore looks sharper for having failed. Some configurations here produce close to 10 per cent infinite intervals in~~ *Windows.* The calibration set is January to August 2024 (244 days, 26,552 plant-days). The test period, September 2024 to May 2026 (638 days, 77,868 plant-days), is split into five windows: September 2024; October to December 2024, the *transition window*; and the semesters 2025-S1, 2025-S2 and ~~2025-S2, so this is not a hypothetical concern.~~ 2026-S1. The transition window is defined by a measurable criterion, maximum divergence from the calibration distribution: the Wasserstein-1 distance between its positive log-magnitudes and those of the calibration set is 1.25, against 0.28 to 0.95 for the other windows. That distance is almost entirely a displacement of location, and the seasonal decomposition of Table 1 attributes 61 per cent of it to the annual cycle for northern solar and 74 per cent at system level. The window is the hardest case the calibration layer faces in this record rather than the signature of a single event; Appendix G shows that the conclusions do not depend on where its boundaries are drawn.

~~Our primary metric is therefore the interval score of the~~ *Primary metric.* The forecasts are one-sided limits $[0, U]$ at level $1 - \alpha$, ~~which follows from the scoring rules of Gneiting and Raftery for the quantile of that level,~~

~~$$\mathrm{IS}_\alpha(U, y) = U + \tfrac{1}{\alpha} \max(y - U, 0).$$~~

~~The multiplier is~~ scored by the one-sided interval score

$$\mathrm{IS}_\alpha(U, y) = U + \frac{1}{\alpha} \max(y - U, 0), \tag{5}$$

which is $1/\alpha$ ~~and not the $2/\alpha$ of the central interval score: in a one-sided limit all of the mass $\alpha$ lies in the upper tail, whereas the central interval places $\alpha/2$ in each tail, and with $2/\alpha$ the score would be minimised by the quantile of level $1 - \alpha/2$ rather than by the one every method calibrates~~times the quantile (pinball) loss of level $1 - \alpha$ plus a term that does not depend on $U$, and is therefore proper for that quantile [42]. It is ~~proper, it is~~ in MWh, ~~it~~ penalises width and non-coverage on the same scale, and ~~it~~ is $+\infty$ whenever $U$ is~~, so it cannot be improved by escaping. We report it in two forms and label them: IS restricted to finite intervals, which is comparable across methods but incomplete,~~ . Mean width over finite limits flatters a method that escapes into infinite limits, so we report three quantities: the interval score over finite limits, the unrestricted mean (infinite if any limit is infinite), and the fraction of infinite limits.

*Operational reading.* Consider a plant owner who must commit, seven days ahead, an allowance $U$ for curtailment, paying $c_h$ per MWh of allowance (for instance, energy that is not sold forward) and $c_p$ per MWh curtailed beyond the allowance (energy that must be bought back or settled). The expected cost $c_h\,U + c_p\,\mathbb{E}\max(Y - U, 0)$ is minimised by the quantile of level $1 - c_h/c_p$, and in units of $c_h$ it equals the one-sided interval score at $\alpha = c_h/c_p$. The interval score at $\alpha = 0.10$ is therefore the cost of an owner with $k = c_p/c_h = 10$. We also evaluate limits calibrated at $\alpha = 0.05$ and $\alpha = 0.01$, which correspond to $k = 20$ and ~~the unrestricted mean , which is the verdict and is infinite for any method that produces even one infinite limit. We also report the~~ $k = 100$; these two levels are also our tail-focused scores.

*Weighted coverage.* Plant-day coverage weights a 5 MW plant on a quiet day like a 300 MW plant on the most congested day. We therefore also report coverage weighted by installed capacity, coverage weighted by curtailed energy, $\sum_{i,t} y_{it}\mathbb{1}\{y_{it} \le U_{it}\} / \sum_{i,t} y_{it}$, the share of curtailed energy left uncovered, $\sum_{i,t} \max(y_{it} - U_{it}, 0) / \sum_{i,t} y_{it}$, and the mean exceedance given that the limit is exceeded.

*High-curtailment days.* A day is high if system curtailment, summed over the plants of the panel, is high. We use three definitions: (H1) the day's curtailment is above the 90th percentile of daily system curtailment in the calibration window, which marks 254 of the 638 test days because the series rose; (H2) the day is in the top decile of the test

period; (H3) the trailing seven-day mean of system curtailment known at the forecast origin, that is, ending seven days before the target, is above the same threshold. H1 and H2 condition on the outcome; H3 conditions only on information available when the forecast is issued. Evaluating forecasts only where the outcome turned out extreme favours forecasts that overpredict [44], so the outcome-based definitions describe exposure while H3 is the legitimate test of conditional calibration.

*Base predictive.* The continuous ranked probability score ~~of the base predictive~~ (CRPS), its threshold-weighted version [43] with weight $\mathbb{1}\{z \geq r_i\}$, where $r_i$ is the 90th percentile of plant $i$'s daily curtailment in the calibration window, and the Brier score of ~~the event~~ $\{Y > 0\}$ ~~; both are properties of~~ describe the base model ~~rather than of the calibration layer, which is precisely why they~~ and cannot arbitrate between calibration ~~schemes and why the interval score is needed~~layers on the same base model.

*Uncertainty.* Coverage carries a standard error clustered by date. Differences between methods ~~are reported with a bootstrap interval that resamples whole days,~~ carry a 95% percentile interval from 2,000 ~~replicates, since the day is the unit of dependence in this panel. Coverage is reported both marginally and conditionally, by technology, capacity tercile, region, and on high-curtailment days, defined as those above the ninetieth percentile of system-level daily curtailment in the calibration window.~~ bootstrap replicates that resample whole days.

~~The calibration set is January to August 2024. The first test month is September 2024. The following window, October to December 2024, is the transition window on which the calibration layeris stressed, and it is selected as the window of maximum divergence from the calibration distribution, not as the interval of entry of any particular technology: measured by the Wasserstein-1 distance between the positive log-magnitudes of each test window~~

### 4.9. *Decomposing an adaptive gain*

For a base model $m$ and a window $w$, let $C^{\mathrm{st}}_{mw}$ and ~~those of the calibration set, it scores 1.25 against 0.28 to 0.95 for the other four windows. That distance is almost entirely a displacement of location: the change is close to a pure translation of the distribution of log-magnitudes,~~ $W^{\mathrm{st}}_{mw}$ be the coverage and mean width of the static split, and $C^{\mathrm{ad}}_{mw}$ and ~~for a pure translation the Wasserstein-1 distance equals the difference in means,~~ $W^{\mathrm{ad}}_{mw}$ those of an adaptive layer. The quantities analysed across cells are the static over-coverage $\mathrm{OC}_{mw} = C^{\mathrm{st}}_{mw} - 90$, in percentage points, and the width change $\Delta_{mw} = 100\,(W^{\mathrm{ad}}_{mw}/W^{\mathrm{st}}_{mw} - 1)$, in per cent.

An adaptive layer that targets nominal coverage must narrow its limits when the static split over-covers and widen them when it under-covers, whatever the source of the ~~two agreeing to within 0.001 for this contrast. Carried out on the calibration window itself,~~ coverage error. To separate this mechanical effect from adaptation proper, we compute for each cell an oracle: the constant PIT threshold $q^{*}_{mw}$, chosen after the fact, at which the static limit $F_r^{-1}(q^{*}_{mw})$ attains in window $w$ exactly the coverage $C^{\mathrm{ad}}_{mw}$ of the adaptive layer. The oracle does not adapt in time; it only re-levels the static split. Its width change $\Delta^{\mathrm{mec}}_{mw} = 100\,(W^{*}_{mw}/W^{\mathrm{st}}_{mw} - 1)$ is the ~~seasonal decomposition of Table 1 splits that displacement 61 per centseasonal for northern solar and 74 per cent at system level. On that basis the window is best understood as the hardest case the calibration layer faces in this record, not as the signature of a single event, and Section 5.8 shows that~~ *mechanical* component, and the ~~conclusions do not depend on where exactly its boundaries are drawn. The remaining periods are taken by semester. The base forecaster is benchmarked against persistence, a seasonal-naive predictor, a point gradient-boosting model and a quantile gradient-boosting model.~~ residual $\Delta_{mw} - \Delta^{\mathrm{mec}}_{mw}$ measures what adaptation in time adds, or costs, relative to the best constant re-levelling at the same coverage. A negative residual would indicate that the layer tracks variation within the window; a positive one, that it pays for fluctuations of its level. The oracle uses the outcomes of the window and is a benchmark, not a method.

The cells share days, base models and windows, so they are not independent. We attach uncertainty in three complementary ways. First, a day-block bootstrap draws whole days with replacement once per replicate and recomputes every cell, including the oracle, on that draw, which preserves the dependence between cells that share days. Second, we fit the cross-cell regression with fixed effects for windows, for base models and for both, and with crossed random effects. Third, we test the slope with a restricted wild cluster bootstrap clustered by base model and, separately, by window, following Cameron et al. [45] and MacKinnon and Webb [46], with the six-point weights of Webb [47], which remain usable with five clusters. None of these makes the two model families a sample from a population of model families; statements about other families rest on the mechanism, not on these intervals. Finally, the early-stopped hurdle is held out: the relationship is estimated on the other eight base models and then used to predict its five cells.

### *4.10. Controlled simulation*

To separate the sources of a static split's coverage error with known ground truth, we simulate a zero-inflated panel of 100 plants observed for 240 calibration days and 638 test days split into five windows of the same lengths as the real ones. Occurrence follows a logistic model with plant effects and a common daily shock, and log-magnitudes follow a normal model with plant effects, a level path, a common daily shock and idiosyncratic noise with a time-varying standard deviation; both common shocks are first-order autoregressive, which induces the dependence across plants. Four scenarios specify the test period: no shift; a tripling of magnitudes, introduced as a ramp over the transition window; the same tripling together with a reduction of the idiosyncratic dispersion by 40 per cent; and the reduction of dispersion alone. Eight base predictives, all lognormal hurdles, span the relevant kinds of misspecification: a correctly specified oracle, constant over-dispersion by 30 per cent, partial tracking of the dispersion, a constant dispersion fixed at its calibration value with and without a constant scale error, and a level that lags the truth by 30 days with and without constant dispersion. Each replicate produces forty model-window cells, like the real analysis, which are passed through the same calibration code. Appendix F gives the full specification.

## 5. Results

### *5.1. Point forecasting~~adds little~~, within the scope tested*

Table 2 reports the mean absolute error of the ~~base models~~ point forecasts on the out-of-sample period. The seasonal-naive predictor, which averages the same weekday over ~~recent~~ the previous four weeks, is the most accurate~~of all and is not beaten by the~~ , and the medians of the gradient-boosting models ~~; the hurdle mixture median~~do not beat it; the median of the hurdle mixture, which is ~~optimised for the full predictive lawrather than for absolute error~~fitted for the whole predictive law, is the least accurate~~on this metric. This~~ , and the early-stopped hurdle, whose predictive is better by CRPS (Section 5.6), is worse still by this criterion. The ordering follows from the strong weekly persistence ~~documented in~~ of curtailment (Section 3.3~~: when a quantity is that persistent, sophisticated point models add little over a well-chosen calendar rule.~~

~~The scope of that conclusion should be stated precisely, because it is easy to over-read. It~~ ). The conclusion holds for these ~~five~~ model classes, this ~~twelve~~ feature set, this seven-day horizon and ~~mean absolute error as the criterion. It does not establish that point forecasting of curtailment is generally of little value. A~~ this criterion; a model with access to the ~~operator's~~ network model, ~~to~~ weather forecasts, ~~to~~ unit-commitment schedules or ~~to the~~ storage state of charge would be solving a better-posed problem, and a criterion ~~other than mean absolute error, in particular one that weighted the large-magnitude days that carry the economic mass, might well order these~~ that weights large days might order the same models differently. ~~What the table supports is narrower and still operationally useful: within the information set~~ Within the information available to a plant owner from public data, effort is better spent on quantifying uncertainty than on refining the predicted number.

~~One robustness check belongs here, because Section 5.6 reports that the hurdle used throughout this paper is not the best available on this problem. Refitting it with early stopping, which selects 105 trees and improves its CRPS by 5.7 per cent, worsens its mean absolute error from 104.1 to 108.6 MWh and would place it last in this table, 27.6 per cent behind the seasonal-naive predictor. The ordering here is therefore not an artefact of the particular hurdle we fitted. It is also a measured instance of the criterion-dependence noted above: the two fits are ordered one way by a proper scoring rule of the whole predictive and the other way by point accuracy, because the early-stopped fit is the wider and better-calibrated predictive and its conditional median sits further from the realised value. We draw nothing further from it.~~

### *5.2. With the reference hurdle~~base model~~, ~~the shift inflates intervals and~~ adaptation ~~recovers them~~appears to recover a third of the width*

~~On real data the~~ With the reference hurdle as base model, the shift does not break ~~coverage~~the coverage of the static split; it inflates it. ~~Under a static conformal split with the hurdle predictive, empirical coverage stays~~ Over the test period the static split covers between 89% and 95% in every ~~period~~ window (Table G.16 ~~), so the severe over-coverage seen on synthetic data is absent, because the base forecaster, through its lagged and moving-average features, absorbs much of the shift into its own predictive. The~~ in Appendix G), and the cost of the shift is paid in width~~rather than in coverage, and it is~~ , concentrated in the transition window~~:~~ , where the static split reaches 95.3% coverage with a

Table 2: Mean absolute error (MWh) of the base forecasters on the out-of-sample period, on the common row set. Lower is better. The comparison is between these model classes on this feature set at this horizon under this criterion; see the scope statement in the text.

| Model | 2024 | 2025 | 2026 | Total |
|---|---|---|---|---|
| Seasonal-naive (weekly) | 84.3 | 86.3 | 84.2 | 85.1 |
| Multi-quantile GBM (median) | 87.7 | 89.0 | 84.1 | 87.6 |
| Quantile GBM, 3 levels (median) | 88.5 | 89.9 | 84.3 | 88.3 |
| Persistence | 88.4 | 96.3 | 90.7 | 92.2 |
| Point XGBoost | 92.7 | 94.6 | 89.7 | 93.0 |
| Hurdle (mixture median) | 104.4 | 104.8 | 101.6 | 104.1 |
| Hurdle, early stopping (mixture median) | 110.1 | 108.8 | 104.6 | 108.6 |

mean width of 1203 MWhduring October to December 2024. Part of that widening is seasonal rather than secular, in the proportion given in Table 1, and the same qualitative pattern would be expected in the fourth quarter of any year in this record; what distinguishes 2024 is that the seasonal peak lands on top of the largest secular displacement of the four years. Figure 5 traces the behaviour over time: the static split drifts above the nominal level as the window advances, while the adaptive schemes track it more closely.

The adaptive schemes pay off precisely there. Transportof the calibration scores wrapped in online adaptation (drift. Transport+ACI with $\gamma = 0.05$ ) brings the transition intervals brings the transition-window limits down to 806 MWh at 90.9% coverage, a reduction of about a third 33% at a coverage within one point of nominal, and the sliding window gives 834 MWh at 92.0%. Outside that window the trade-off is ambiguous: with the static split already near nominal, the adaptive methods mostly exchange coverage for width and can under-cover, as in 2025-S1, and their gains are small and within the date-clustered standard error. That is what the data predict rather than a shortcoming of the adaptive layer. The first semester of 2025 . Read alone, this is the result that motivated the scheme. Over the whole test, however, the same configuration returns infinite limits on 5.9% of plant-days, and its interval score over finite limits, 813 MWh, is not distinguishable from the pre-transition regime: the Wasserstein-1 distance between the positive log-magnitudes of January to September 2024 and those of January to June 2025 is 0.1722, with a 95% bootstrap interval of [0.0703, 0.3665] resampling whole days, against a threshold of 0.1931 under a permutation null that also resamples whole days. There is no shift left to correct in that period.

Read on its own, this subsection is the result the submitted version reported826 MWh of the static split. The next one is why we no longer read it on its ownsubsections ask what the one-third reduction measures.

### 5.3. The *gain does not survive a change of same layer over different base models*

The calibration layer touches the base model through exactly two operations, the predictive CDF and its inverse, so the same layer can be run over adifferent predictive without changing a single constant. We do that, holding fixed the scoreTable 3 runs every calibration layer over the three main base models, and Figure 6(a) adds the conditional-dispersion hurdle. The calibration layer is identical across base models: same score, windows, embargo, the window boundaries, the seven-day embargo, the seeds , the values of $\gamma$, the 60-day recent window and the tail shrinkage, and varying only the base model among the three of Section 4.1. As a control on the comparison, the hurdle arm of this experiment reproduces Table G.16 in all thirty cells and four metrics, which is verified by assertion inside the script that produces it. Any difference between arms is therefore attributable to the base model and not to the protocol. *Base model* here means the pair (functional form, capacity) jointly: the arms differ both in the shape imposed on the predictive and in the number of boosted models fitted, and this design does not separate the two. That matters for the comparison of levels, and Section 5.6 separates them; it does not affect the diagnostic relationship, which is established *within* each arm by contrasting the static split against the adaptive scheme under one and the same predictive. seeds and constants.

Table 4 gives the result in the transition window and Figure 7 summarises it. The reduction of about a third disappears. With the conditional-dispersion hurdleThe one-third reduction does not transfer. In the transition window the static split already sits at 91.7%width change of Transport+ACI is −33.1% with the reference hurdle, −22.5% with the early-stopped hurdle, −4.8% with the conditional-dispersion hurdle and +3.6% with the quantile model, whose

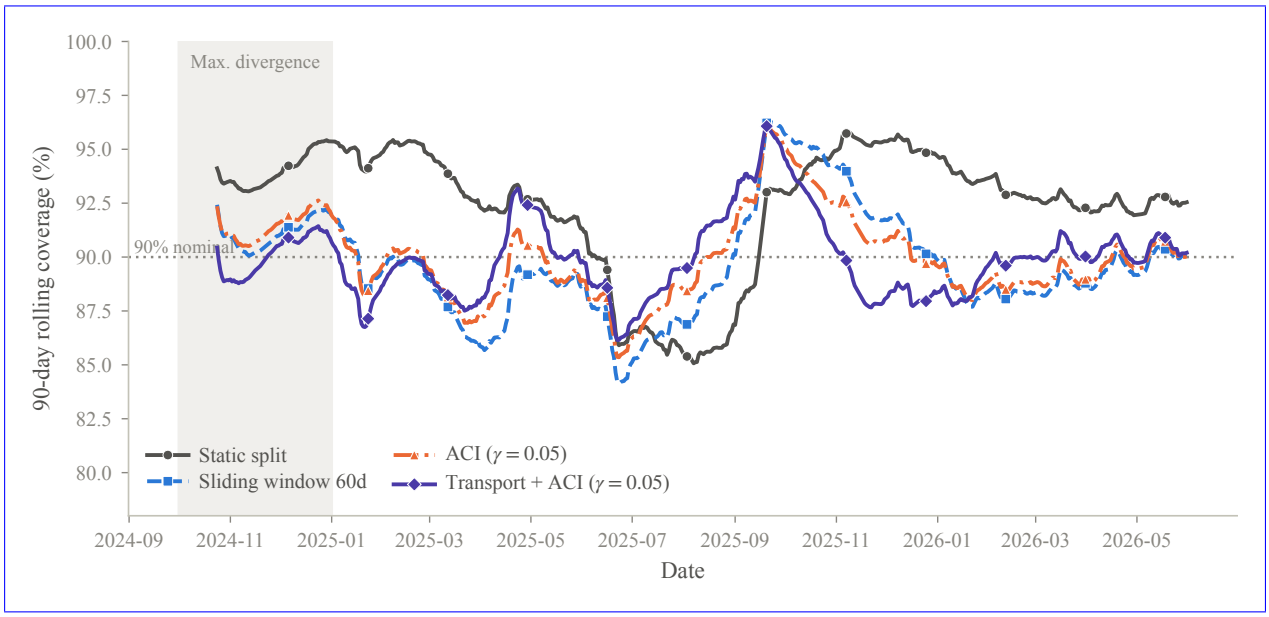



Figure 5: Rolling empirical coverage (90-day window) of the calibration layers with the reference hurdle as base model, with the 7-day horizon embargo. The dashed line marks the 90% nominal level and the shaded band the transition window, October to December 2024.

static split is already at 90.6% coverage. The static split's over-coverage in that window follows the same order: +5.3, +4.7, +1.7 and +0.6 percentage points.

The base model matters more than the layer. Over the whole test, the ~~adaptive scheme recovers only 4.8% of width while dropping to 89.5% coverage. With the multi-quantile predictive the static splitis at 90.6% coveragewith a mean width of 410 MWh, and~~ early-stopped hurdle with a plain static split has an interval score of 534 MWh, 35% below the static split on the reference hurdle and below every calibration layer on it, the ~~adaptive scheme returns intervals 3.6% *wider*, at 425 MWh~~best of which, one-sided CQR, reaches 714 MWh. Its static limits are 40% narrower than those of the reference hurdle at the same coverage (92.0% against 92.5%). On the early-stopped hurdle, adaptation adds little: the best online layer improves the interval score by 3%, Transport+ACI with $\gamma = 0.05$ worsens it by 3%, and the selected Transport+ACI configuration is within 1% of the static split. The quantile model is better again: its static split has an interval score of 433 MWh at 90.2% coverage. ~~The bootstrap interval of that width difference over whole days is +15.1 MWh with a 95% interval of [+0.0, +30.3], which only just excludes zero, and the corresponding difference in interval score, +14.5 with an interval of [+4.9, +24.0], also excludes it . On either reading the point estimate is on the opposite side from the one reported, and the reduction is absent~~

The ranking of the layers also depends on the base model. One-sided CQR is the best layer on the reference hurdle (714 MWh) and close to the best on the quantile model (438 MWh at 89.8% coverage), but on the early-stopped hurdle it over-covers (94.2%) and is worse than the static split (590 MWh). The reason is the scale of the nonconformity score. CQR corrects the base model's 0.9 quantile by a single number of MWh, the 90th percentile of $y - q_{0.9}(x)$ pooled over plants whose sizes differ by two orders of magnitude; on these data that percentile is dominated by small plants and zero days and lies close to zero, −1.1 MWh for the early-stopped hurdle. The early-stopped hurdle's own 0.9 quantile already covers 92.8% of the calibration window, and CQR leaves that over-dispersion in place, whereas the PIT score, which is free of scale, removes it by lowering the conformal level to 0.880.

~~The pattern is not confined to the transition window, and it is systematic. Table 5 lays out all fifteen cells given by three base models and five windows. The width change achieved by the adaptive layer~~

### 5.4. *How much of the gain is mechanical*

Across the 40 cells of eight base models by five windows, the width change of Transport+ACI is linearly related to ~~how far above nominal the static split sits in the same cell, with a Pearson correlation of −0.755~~ the static

Table 3: Calibration layers over the three main base models, nominal level 90%. Coverage in per cent with the date-clustered standard error, mean width over finite limits (MWh), one-sided interval score over finite limits (MWh) and fraction of infinite limits (%). In the whole-test interval score, an asterisk marks a method with infinite limits, whose unrestricted score is infinite. CRPS and twCRPS are those of the base predictive (MWh); the twCRPS weights the region above each plant's calibration-window 90th percentile. ΔIS: difference in whole-test interval score from the static split on the same base model, with a 95% day-block bootstrap interval, on plant-days where both limits are finite. No limit is infinite in the transition window. The selected configurations come from rolling-origin validation inside the calibration window (Appendix C).

| | Transition window (Oct–Dec 2024) | | | Whole test (Sep 2024–May 2026) | | | | |
|---|---|---|---|---|---|---|---|---|
| Method | Coverage | Width | IS | Coverage | Width | IS | % inf. | ΔIS vs. static [95% CI] |
| *Reference hurdle (2×400 trees)*: CRPS 133.5 / 89.5, twCRPS 78.9 / 40.7 | | | | | | | | |
| Static split (PIT) | 95.3 (0.45) | 1203 | 1395 | 92.5 (0.37) | 678 | 826 | 0.0 | – |
| Sliding window, 60 d | 92.0 (0.73) | 834 | 1112 | 89.9 (0.44) | 531 | 734 | 0.0 | −93.4 [−109.8, −76.4] |
| ACI ($\gamma = 0.05$) | 92.1 (0.67) | 822 | 1096 | 90.1 (0.43) | 544 | 753* | 4.2 | −95.9 [−117.7, −69.6] |
| Transport+ACI ($\gamma = 0.05$) | 90.9 (0.81) | 806 | 1107 | 90.1 (0.43) | 596 | 813* | 5.9 | −32.3 [−69.8, +9.3] |
| ACI (selected $\gamma$) | 94.8 (0.49) | 1086 | 1294 | 90.9 (0.41) | 590 | 763 | 0.0 | −63.0 [−69.0, −57.0] |
| Transport+ACI (selected) | 93.3 (0.54) | 924 | 1162 | 90.0 (0.45) | 564 | 761 | 0.0 | −65.4 [−82.8, −47.9] |
| One-sided CQR | 94.0 (0.55) | 937 | 1175 | 91.2 (0.38) | 532 | 714 | 0.0 | −112.4 [−120.8, −103.7] |
| *Early-stopped hurdle (105 trees)*: CRPS 124.0 / 84.4, twCRPS 70.3 / 37.2 | | | | | | | | |
| Static split (PIT) | 94.6 (0.51) | 689 | 823 | 92.0 (0.35) | 409 | 534 | 0.0 | – |
| Sliding window, 60 d | 92.6 (0.69) | 575 | 758 | 89.8 (0.39) | 350 | 517 | 0.0 | −16.6 [−22.4, −10.7] |
| ACI ($\gamma = 0.05$) | 92.3 (0.61) | 553 | 736 | 90.1 (0.37) | 354 | 524* | 3.3 | −19.8 [−27.9, −11.0] |
| Transport+ACI ($\gamma = 0.05$) | 91.1 (0.82) | 534 | 748 | 90.0 (0.38) | 377 | 552* | 3.3 | +9.4 [−7.8, +27.7] |
| ACI (selected $\gamma$) | 94.3 (0.53) | 655 | 797 | 90.8 (0.37) | 377 | 520 | 0.0 | −13.8 [−16.0, −11.6] |
| Transport+ACI (selected) | 93.8 (0.52) | 613 | 761 | 89.9 (0.40) | 367 | 527 | 0.0 | −6.6 [−12.9, −0.2] |
| One-sided CQR | 96.2 (0.39) | 840 | 932 | 94.2 (0.30) | 501 | 590 | 0.0 | +57.2 [+52.9, +61.6] |
| *Reference quantile GBM (54×400)*: CRPS 93.0 / 68.0, twCRPS 51.6 / 28.4 | | | | | | | | |
| Static split (PIT) | 90.6 (0.85) | 410 | 588 | 90.2 (0.38) | 288 | 433 | 0.0 | – |
| Sliding window, 60 d | 90.1 (0.94) | 409 | 598 | 89.9 (0.41) | 288 | 438 | 0.0 | +5.7 [+4.0, +7.5] |
| ACI ($\gamma = 0.05$) | 90.6 (0.87) | 412 | 588 | 90.2 (0.41) | 294 | 447* | 2.8 | +8.5 [+5.5, +11.4] |
| Transport+ACI ($\gamma = 0.05$) | 90.5 (0.99) | 425 | 602 | 90.2 (0.44) | 305 | 459* | 4.4 | +19.1 [+14.1, +24.2] |
| ACI (selected $\gamma$) | 90.6 (0.86) | 411 | 588 | 90.3 (0.39) | 289 | 434 | 0.0 | +0.8 [+0.4, +1.1] |
| Transport+ACI (selected) | 89.9 (0.95) | 411 | 598 | 90.1 (0.41) | 294 | 444 | 0.0 | +11.0 [+8.3, +13.6] |
| One-sided CQR | 89.3 (0.90) | 389 | 602 | 89.8 (0.38) | 276 | 438 | 0.0 | +5.7 [+4.2, +7.2] |

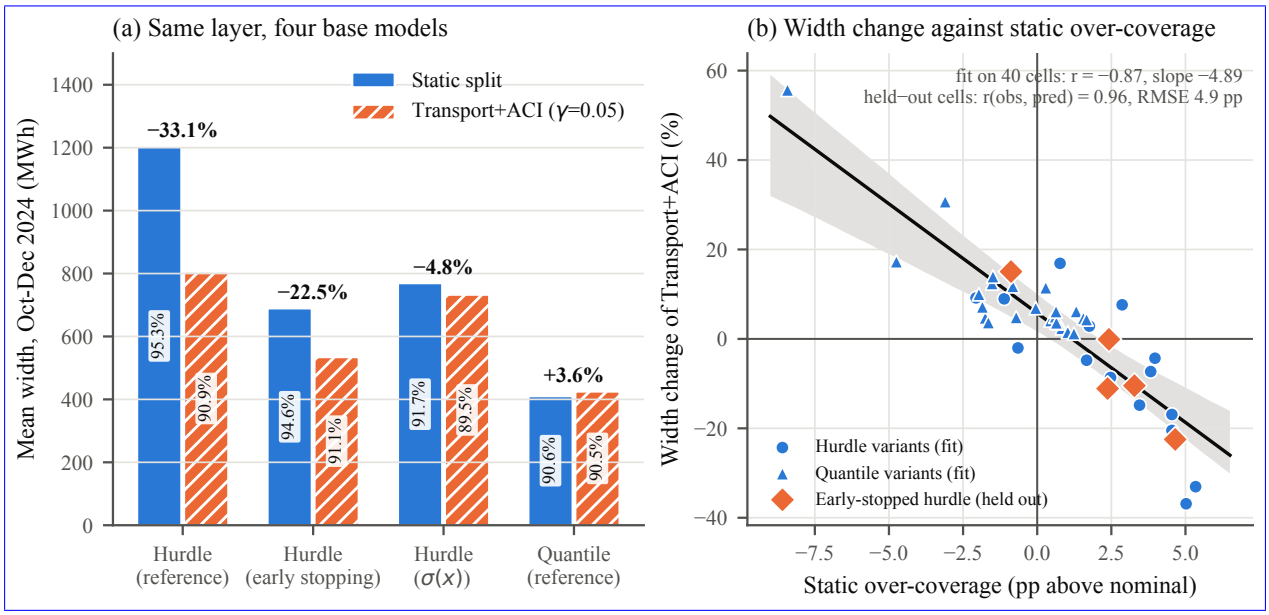



Figure 6: (a) Mean width in the transition window of the static split and of Transport+ACI ($\gamma = 0.05$) over four base models, with coverage above each bar and the width change in bold. (b) Width change of Transport+ACI against the over-coverage of the static split in the same cell, for the eight base models of the cross-model analysis (blue) and the held-out early-stopped hurdle (orange); line and band: least-squares fit on the 40 cells with its 95% day-block bootstrap band.

over-coverage, with $r = -0.87$ and a slope of ~~−4.36 per cent~~ −4.89% of width per percentage point (Figure 6b). The relationship predicts the five cells of the held-out early-stopped hurdle with a correlation of 0.96 between observed and predicted changes and a root mean squared error of 4.9 points. A relationship of ~~over-coverage. The intercept is +5.45 per cent: when the static split is already at the nominal level, the adaptive layer makes intervals about five per cent wider. That is the diagnostic. What the layer recovers is not distributional displacement; it is the over-dispersion the base model built into every interval, and the size of the apparent gain is a measurement of how badly the base model was calibrated in the first place.~~

~~The relationship can be estimated on more than these fifteen cells. Each arm of the capacity ladder of Section 5.6 was run through the same calibration layer in the same five windows, so eight base models are available rather than three, giving forty cells. We exclude the coarse-grid arm discussed in Section 7, whose static split returns infinite limits throughout and therefore yields no width. The addition matters for a specific reason: the fifteen cells span over-coverage only over [−2.06, +5.34] percentage points, and the low-capacity arms reach −8.42, filling the region below nominal where the three-model design has no point at all~~and on which the intercept depends~~this kind~~ is expected from any layer that targets nominal coverage, so the question is how much of it requires adaptation at all.

~~Table 6 reports the result~~ The constant-threshold oracle answers it (Table 4 and Figure ~~8 shows it. The relationship strengthens: $r$ goes from −0.755 to −0.873 and the slope from −4.36 to −4.89 per percentage point. The bootstrap intervals, computed by resampling whole days once per replicate and recomputing all forty cells~~on that resample so that the crossed dependence between arms and windows is preserved, exclude zero for all three statistics, and the intercept, which on fifteen cells was [+0.07, +11.25] and barely excluded zero, is now [+1.60, +9.84]. Leaving out any single cell moves $r$ within [−0.889, −0.820], and leaving out any entire base model within [−0.899, −0.792], so no arm carries the result.

~~One qualification belongs here. Fitting the two sides of the nominal level separately improves the residual sum of squares by more than chance ($F(2, 36) = 6.778$7). Over the 45 cells, $p = 0.0032$), so we do not state the relationship symmetrically. The improvement is one of level and not of rate: allowing the intercept alone to differ accounts for it ($F(1, 37) = 12.982$~~the oracle's width change has a slope of −4.80 against −4.98 for the observed change of Transport+ACI with $\gamma = 0.05$, and it explains 84% of the variance of the observed change. The residual, what

Table 4: How much of the adaptive width change is mechanical. For each adaptive configuration, the cross-cell regression of the width change on the static over-coverage (observed), the same regression for the constant-threshold oracle that attains the adaptive layer's coverage (mechanical), the share of the variance of the observed change explained by the mechanical one, and the mean residual. 40 cells: eight base models by five windows; 45 cells: the same plus the early-stopped hurdle. Intervals are 95% day-block bootstrap intervals, recomputing every cell on each of 2,000 replicates.

| Configuration | Cells | Observed slope (%/pp) | Mechanical slope (%/pp) | $R^2$ | Mean residual (%) | $r$(res., OC) |
|---|---|---|---|---|---|---|
| Transport+ACI, $\gamma = 0.05$ | 40 | -4.89 [-5.60, -3.20] | -4.95 [-5.67, -3.34] | 0.87 | +7.0 [+5.3, +8.9] | +0.03 |
| | 45 | -4.98 [-5.66, -3.36] | -4.80 [-5.53, -3.21] | 0.84 | +6.9 [+5.3, +8.6] | -0.07 |
| Transport+ACI, $\gamma = 0.05$, $\alpha_{\min} = 0.005$ | 40 | -6.28 [-7.26, -4.18] | -5.15 [-5.86, -3.50] | 0.28 | +11.7 [+8.6, +15.1] | -0.15 |
| | 45 | -6.35 [-7.33, -4.31] | -4.98 [-5.69, -3.33] | 0.30 | +11.2 [+8.1, +14.5] | -0.19 |
| ACI, $\gamma = 0.05$, $\alpha_{\min} = 0.005$ | 40 | -4.98 [-5.52, -3.52] | -4.13 [-4.73, -2.82] | 0.78 | +4.3 [+2.6, +6.2] | -0.33 |
| | 45 | -4.89 [-5.45, -3.45] | -3.98 [-4.59, -2.71] | 0.79 | +4.0 [+2.4, +5.9] | -0.36 |
| Transport+ACI, selected | 40 | -2.52 [-3.36, -1.32] | -2.17 [-3.05, -0.99] | 0.99 | +1.1 [+0.4, +1.8] | -0.47 |
| | 45 | -2.27 [-3.20, -1.07] | -1.88 [-2.85, -0.72] | 0.98 | +0.8 [+0.2, +1.5] | -0.50 |
| ACI, selected | 40 | -1.38 [-1.71, -0.85] | -1.26 [-1.58, -0.74] | 0.99 | -0.1 [-0.3, +0.1] | -0.49 |
| | 45 | -1.30 [-1.64, -0.76] | -1.17 [-1.51, -0.66] | 0.99 | -0.1 [-0.3, -0.0] | -0.49 |

adaptation in time adds beyond re-levelling, averages +6.9% with a 95% interval of [+5.3, +8.6] and is unrelated to the over-coverage ($r = -0.07$). The other configurations tell the same story. With the update bounded at $\alpha_{\min} = 0.005$ the mechanical slope is unchanged (−4.98), but the residual is larger and more variable (+11.2%), ~~$p = 0.0009$)and allowing the slope to differ as well adds nothing ($F(1, 36) = 0.685$, $p = 0.413$). The rate at which the adaptive layer converts a departure from nominal coverage into width is the same on both sides, −6.50 per percentage point; what differs is~~ because when $\alpha_t$ reaches its bound the limits become very wide, a cost the oracle does not pay; this is why the oracle explains less of the variance in that configuration ($R^2 = 0.30$). Pure ACI with the same step has a residual of +4.0%. With the small steps selected by rolling origin the layers move little and almost entirely mechanically ($R^2$ of 0.98 and 0.99), and the ~~offset. We therefore keep the claim scoped to~~ residual is within a fraction of a per cent of zero (+0.8% for Transport+ACI and −0.1% for ACI). The slope of the cross-cell relationship is therefore mechanical. On average over the cells, adaptation in time contributes at most a fraction of a per cent of width beyond the best constant level chosen after the fact, and with the step of Algorithm 1 it costs several per cent.

What is not mechanical is where the over-coverage ~~, which is the region the three-model design covers densely, and record the under-covered arms as evidence that the mechanism does not stop at the nominal level rather than as a symmetric law. Restricted to the twenty-five cellsin which the static split over-covers, the region to which the claim is scoped, the slope is steeper, −6.97 with a bootstrap interval of [−7.90, −3.60] against −4.89 over all forty cells, and the relationship keeps its sign and its strength, $r = -0.870$, [−0.928, −0.528]~~ comes from. Within a window all base models face the same shift, and there the width change still follows the over-coverage (Table ~~6); the figures quoted in the abstract are those of all forty cells.~~

~~We probed that break rather than accepting it, because the obvious worry is that one arm carries it . Of the 15 cells below nominal, 5 come from the lowest-capacity quantile arm, and 3 of the 4 cells below −2 percentage points come from the same arm. Removing it does not remove the break: on the remaining 35 cells the level shift grows from −11.80~~ 5): in four of the five windows the correlation across the nine base models lies between −0.99 and −0.80; in the first semester of 2025, when the static splits of all base models are close to nominal and their over-coverage spans only 1.4 points, it is −0.18. In the transition window the static over-coverage ranges from −8.4 to ~~−13.59 per cent and the statistic rises rather than falls. Dropping any single arm leaves the shift between −10.63 and −13.59. Absorbing a separate intercept for each base model~~+5.3 percentage points across base models, and in a two-way analysis of variance of the over-coverage over all cells the base model accounts for 54% and the window for 18%. Figure 8 shows the mechanism for the three main base models: the PIT histogram of the reference hurdle loses mass in the upper tail between the calibration and the transition windows, so that the ~~break is identified only from variation within arms, leaves it at −11.48 per cent. The break is not the artefact of one arm~~calibrated quantile $\hat{q}$ over-covers, whereas that of the quantile model barely moves.

~~It is, however, an artefact of the specification in a different sense,~~

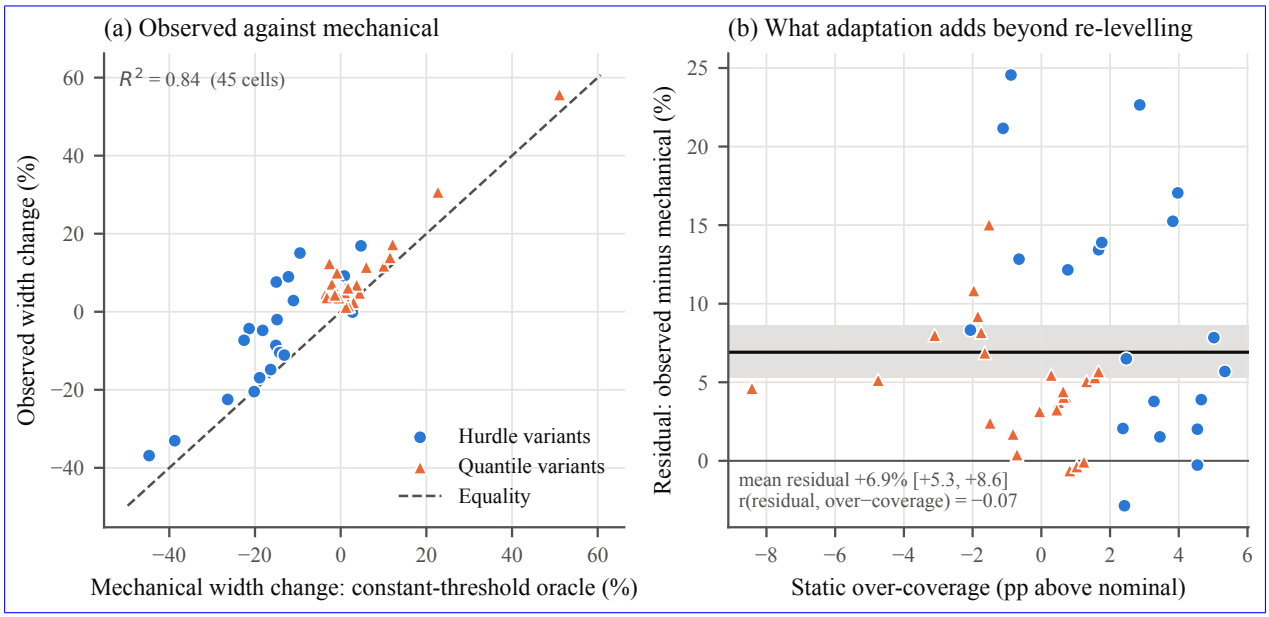



Figure 7: Decomposition of the width change of Transport+ACI ($\gamma = 0.05$) over the 45 base-model-by-window cells. (a) Observed width change against the width change of the constant-threshold oracle with the same coverage. (b) Residual against the static over-coverage; line and band: mean residual and its 95% day-block bootstrap interval.

The cells share days, base models and windows, and Table 5 reports the slope under estimators that treat that dependence differently. Fixed effects for windows, for base models or for both, and crossed random effects, give slopes between −6.14 and ~~this is what decides the wording. Fitting the 35 cellswithout the extreme arm and predicting that arm out of sample, the specification with a break predicts its most extreme cell almost exactly, +54.2 against +55.7 observed where the single line predicts +43.0, but it is worse on the other four, and the out-of-sample root mean squared error ends up a tie, 8.85 against 8.82. A quadratic term tells the same story from the other side: it is indistinguishable from zero over the forty cells ($p = 0.71$) and strongly significant over the thirty-five ($F(1, 32) = 27.9$). A design whose preferred functional form flips when one arm is removed does not determine that form, and the honest conclusion is that these data locate the relationship but not its shape below nominal. That is the reason~~ −4.50. Over the 45 cells, the day-block bootstrap interval, which recomputes every cell on each resample of days, is [−5.66, −3.36]. The wild cluster bootstrap rejects a zero slope clearly when clustering by base model ($p = 0.0006$) and only narrowly when clustering by window ($p = 0.048$), with five clusters. These intervals describe uncertainty over the days of this record; with two model families they cannot describe uncertainty over model families, and we do not ~~generalise the claim, and it is a stronger reason than the break statistic alone~~extend the attribution beyond the families tested.

~~These $F$ tests treat the forty cells as independent observations, which they are not: they come in blocks of five per base model and eight per window and share the underlying panel, so the $p$ values are optimistic. We report them only as grounds for *not* widening the claim, a direction in which optimistic $p$ values are conservative. The intervals that govern the statements this paper does make are the day-block bootstraps of Table 6.~~

~~Two further observations sharpen the reading~~

### 5.5. *Controlled simulation*

Table 6 and Figure 9 report the simulation of Section 4.10. Three results matter. First, ~~the multi-quantile predictive is not merely different, it is better on every proper metric and in every window: its CRPS is 18 to~~ a correctly specified base model keeps the static split close to nominal coverage under every scenario, including a tripling of magnitudes: its mean over-coverage after the shift is +0.6 points in the level scenario and +0.1 points when level and dispersion change together, so an adaptive layer has nothing to recover. Second, a constant over-dispersion of 30 per cent~~lower than the hurdle's, 92.9 against 133.5 MWh in the transition window, an advantage that Section 5.6 decomposes into a~~

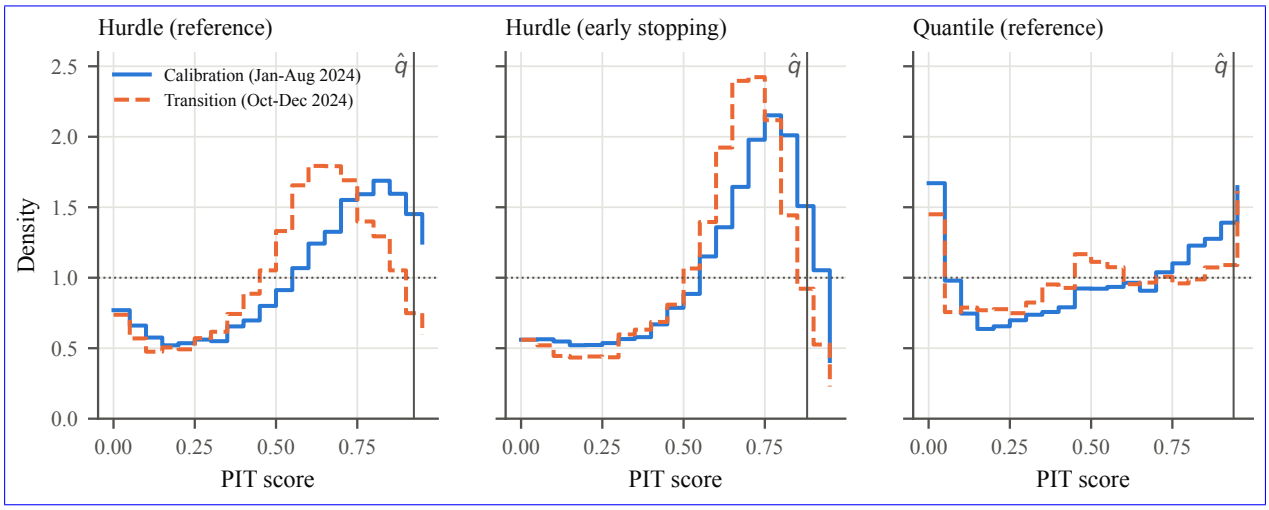



Figure 8: Histograms of the PIT score in the calibration window (January to August 2024) and in the transition window (October to December 2024) for the three main base models. The dotted line is the uniform density; the vertical line marks the static conformal quantile $\hat{q}$ at the 90% level.

~~contribution of the~~ , present in calibration and test alike, is absorbed by the conformal quantile: the over-coverage of that model is +0.1 points without a shift. Third, coverage drifts when the base model fails to follow the change: the constant-dispersion model ~~form and one of the fitted budget~~over-covers by +4.7 points when the dispersion falls, and the ~~best of the eighteen model-by-method combinations by interval score is the multi-quantile predictive in all five windows, in four of them paired with~~ lagged-level model under-covers during the ramp. In the three scenarios with a shift the cross-cell relationship appears in every replicate, and the constant-threshold oracle reproduces it: the median share of the variance of the observed width change explained by the mechanical one is 0.90 in the level-and-dispersion scenario, and the median residual is +10.7%, positive as in the real panel. The simulation also shows how much a single dataset can vary: with common daily shocks, the ~~plain static split and no adaptive layer at all. Its intervals in the transition window are 410 MWh at 90.6% coverage, against 806 MWh at 90.9% for the full pipeline of the submitted version. Second, the hurdle retains one advantage that should be recorded: the Brier score of the occurrence event is better under the hurdle, 0.0721 against 0.1027 in the transition window. That comparison is not like for like, since the occurrence probability of the multi-quantile predictive is read off a quantile grid at 0.02 resolution rather than produced by a dedicated classifier, and we report it as descriptive~~over-coverage of the correctly specified model in a single post-shift window ranges from −5.7 to +4.0 points across replicates (2.5th to 97.5th percentiles).

### 5.6. *Model form~~against~~ , fitted capacity and early stopping*

The comparison of ~~Section 5.3~~ base models holds features, rows, training cut-off, per-model hyperparameters and seed fixed~~across arms, but it does not hold capacity fixed: the hurdle is two boosted models of 400 treesand the multi-quantile predictive is fifty-four of them. A reader is entitled to ask whether the second wins on its freedom from a parametric tail shape or simply on having more budget to spend. The question is answerable and we answer it.~~ , but not the number of fitted trees: the reference hurdle has 800, the reference quantile model 21,600. Table 7 ~~varies the tree budget at a fixed 54-level grid and, in the other direction, gives the hurdle five times its own.~~

~~Three things follow, and the second is the one that matters.~~

~~At a strictly matched budget the advantage nearly disappears. Fifty-four quantile heads sharing 810 trees, against the hurdle's~~ separates the two. At a matched budget of about 800 ~~, improve the CRPS of~~ trees the quantile model improves the CRPS over the whole test by only 2.4~~per cent and that of the transition window by 1.8 per cent, not by the eighteen to thirty per cent of Section 5.3. The headline figure is not available at the hurdle's own budget, and stating it without that qualification would attribute to the model form what is partly a matter of capacity.~~

Table 5: Inference on the cross-cell relationship under dependence, Transport+ACI with $\gamma = 0.05$. Slope of the width change (%) on the static over-coverage (pp). Rows differ in what they hold fixed and in how they treat the dependence between cells that share days, base models and windows.

| Estimator | 40 cells | 45 cells |
|---|---|---|
| Pooled OLS | -4.89 (0.44) | -4.98 (0.41) |
| Window fixed effects | -5.62 (0.33) | -5.64 (0.34) |
| Base-model fixed effects | -4.23 (0.68) | -4.50 (0.62) |
| Two-way fixed effects | -6.00 (0.55) | -6.14 (0.59) |
| Crossed random effects (REML) | -5.58 (0.35) | -5.56 (0.35) |
| Day-block bootstrap, 95% interval | [-5.60, -3.20] | [-5.66, -3.36] |
| Wild cluster bootstrap by base model, $p$ | 0.0027 (8 cl.) | 0.0006 (9 cl.) |
| Wild cluster bootstrap by window, $p$ | 0.0560 (5 cl.) | 0.0484 (5 cl.) |
| Within-window correlation, range over 5 windows | [-0.99, -0.58] | [-0.99, -0.18] |
| Family × window means (10 cells), $r$ | -0.84 | -0.89 |
| *Variance of the static over-coverage explained (two-way ANOVA)* | | |
| base model | 55% | 54% |
| window | 17% | 18% |

Standard errors in parentheses are the usual OLS or REML ones and treat cells as independent; they are shown for comparison with the bootstrap and wild cluster results. Wild cluster bootstrap: restricted, six-point Webb weights, 9,999 replicates.

Table 6: Controlled simulation, 50 replicates per scenario. Left: static over-coverage (pp) averaged over the four post-shift windows and the replicates, for four of the eight base models (Appendix F). Right: cross-cell regression of the Transport+ACI width change on the static over-coverage within each replicate (40 cells), median and 2.5–97.5 percentiles over replicates; mechanical slope from the constant-threshold oracle; share of the variance of the observed change explained by the mechanical one; mean residual.

| | Static over-coverage (pp) | | | | Cross-cell regression, Transport+ACI | | | |
|---|---|---|---|---|---|---|---|---|
| Scenario | M1 | M2 | M4 | M7 | Slope | Mech. slope | $R^2$ obs.∼mech. | Residual (%) |
| S0, no shift | +0.1 | +0.1 | +0.1 | +0.1 | -7.18 [-17.09, -1.15] | -5.03 [-12.32, 0.20] | 0.72 [0.03, 0.96] | 10.7 [6.1, 35.9] |
| S1, level ×3 | +0.6 | +0.6 | +0.6 | -0.7 | -7.88 [-16.39, -2.73] | -5.21 [-9.68, -0.18] | 0.78 [0.10, 0.98] | 12.3 [7.2, 32.7] |
| S2, level ×3 and dispersion ×0.6 | +0.1 | +0.1 | +4.7 | -1.6 | -6.21 [-8.73, -4.14] | -5.22 [-7.15, -3.85] | 0.90 [0.66, 0.99] | 10.7 [4.9, 18.6] |
| S3, dispersion ×0.6 | +0.0 | +0.0 | +4.7 | +0.0 | -5.59 [-8.46, -3.48] | -5.13 [-7.53, -3.18] | 0.89 [0.66, 0.99] | 10.8 [5.3, 17.3] |

M1: correctly specified oracle. M2: oracle with dispersion inflated by 30% in every period. M4: dispersion fixed at its calibration value. M7: level lagged by 30 days. In S0 there is no shift and the regression has no systematic variation to explain.

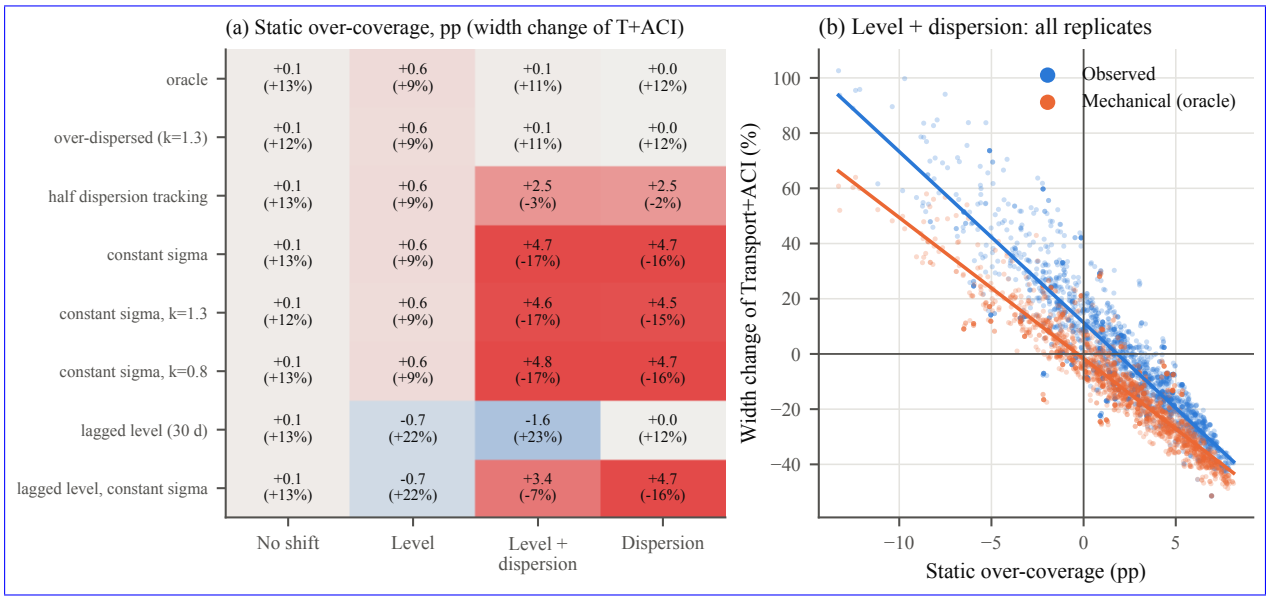



Figure 9: Controlled simulation. (a) Static over-coverage averaged over the post-shift windows and replicates, by base model and scenario, with the mean width change of Transport+ACI in parentheses. (b) Level-and-dispersion scenario, all cells of all replicates: observed width change of Transport+ACI (blue) and width change of the constant-threshold oracle (orange) against the static over-coverage, with least-squares lines.

~~The hurdle cannot buy parity with capacity . Given five times its budget, four thousand trees against eight hundred, it gets *worse*: its CRPS rises from 89.5 to 95.5 MWh over the whole test and from 133.5 to 141.1 in the transition window, and its mean interval width rises from 678 to 970 MWh. The deficit of the hurdle on this problem is not a deficit of budget.~~

~~The advantage saturates early. By~~ %; the advantage grows with capacity and saturates near 5,400 trees~~the multi-quantile CRPS is within 0.5 MWh of its value at 21, 600, so the fitted budget of the model used throughout this paper is roughly four times larger than it needs to be. At 2, 700 trees, a budget of the same order as the enlarged hurdle 's~~ where it reaches 23.7%. The hurdle does not convert capacity into accuracy: with 4,000 ~~, the CRPS is 100.6 against 141.1: at comparable budget the gap is about thirty per cent.~~

~~One design choice is worth stating because it bears on how the enlarged hurdle should be read. No arm of the ladder uses early stopping. Every model is fitted to its full tree budget, with no validation split held back to halt it, which is the same protocol under which the base models of the submitted version were fitted. That choice is what allows the enlarged hurdle to overfit rather than to stop, so we measured what stopping it would buy rather than assuming.~~

~~Holding out the last six months of the training period as a temporal validation set, fitting each stage of the hurdle to a ceiling of 4,000~~ trees ~~with a patience of 50 rounds, and refitting on the full training period at the selected budget, early stopping~~its CRPS worsens by 6.6%, and early stopping, which selects 105 trees~~in total, 59 for the occurrence stage and 46 for the magnitude stage. That is an eighth of the 800 trees the base model of this paper uses. The stopped hurdle improves on it : CRPS falls from 89.52 to 84.38 over the whole test, -5.7 per cent, and from 133.52 to 124.03 in the transition window. Its estimated dispersion rises from 1.7016 to 1.8274. Dispersion is estimated out of fold inside the training period, where additional trees continue to reduce the residual, so $\hat{\sigma}$ falls monotonically along the budget axis while the CRPS on the evaluation period rises along it. The divergence is the mechanism identified above: what the extra trees buy is fit to the pre-2024 regime, and it does not transfer.~~

~~Two things follow. The stopped hurdle does better than restore the 800-tree performance, so the hurdle used throughout this paper is not the best hurdle available, and readers should discount our base model accordingly. But the ordering is unchanged and the margin remains large: the stopped hurdle scores 124.03 in the transition window against the multi-quantile predictive's 92.92, still 33 per cent behind . What the experiment establishes is that the~~

Table 7: Model form and fitted capacity. Every model shares features, rows, training cut-off, per-model hyperparameters and seed; only the number of trees and the model form change. CRPS of the base predictive (MWh); change over the whole test relative to the reference hurdle; static-split coverage and mean width over the whole test.

| | | CRPS | | | Static split | |
|---|---|---|---|---|---|---|
| Base model | Trees | Transition | Whole test | vs. ref. | Coverage | Width |
| Hurdle, 2×400 | 800 | 133.5 | 89.5 | +0.0% | 92.5 | 678 |
| Hurdle, early stopping (59+46) | 105 | 124.0 | 84.4 | -5.7% | 92.0 | 409 |
| Hurdle, 2×2000 | 4,000 | 141.1 | 95.5 | +6.6% | 92.1 | 970 |
| Quantile GBM, 54×15 | 810 | 131.2 | 87.4 | -2.4% | 87.0 | 247 |
| Quantile GBM, 54×50 | 2,700 | 100.6 | 71.4 | -20.3% | 89.4 | 263 |
| Quantile GBM, 54×100 | 5,400 | 94.0 | 68.3 | -23.7% | 90.1 | 277 |
| Quantile GBM, 54×200 | 10,800 | 92.8 | 67.8 | -24.2% | 90.2 | 285 |
| Quantile GBM, 54×400 | 21,600 | 93.0 | 68.0 | -24.1% | 90.2 | 288 |

~~hurdle form saturates very early on this problem, at roughly a hundred trees, and that neither adding budget nor spending it more carefully closes the gap to the quantile form.~~

~~The reading we take from this is narrow and we state it narrowly.~~ , improves it by 5.7%. The early-stopped hurdle remains 24% behind the quantile model. On this panel, and between these two ~~model~~ families, the ~~multi-quantile form converts additional~~ quantile form converts capacity into predictive accuracy and the two-part lognormal form ~~does not. We~~ saturates at about a hundred trees; we make no claim about ~~quantile regression as a model class or about other problems.~~

~~The mechanism of the second point is worth recording because it corroborates a result the paper reports elsewhere. When the hurdle is enlarged its estimated dispersion *falls*, from $\hat{\sigma} = 1.7016$ to 1.6641, while its intervals widen by 292 MWh. Dispersion cannot be the cause: at a fixed conformal quantile the smaller $\hat{\sigma}$ would narrow the intervals by 34 MWh. What widens them is the conformal quantile itself, which moves from 0.9242 to 0.9532, and it moves because the upper tail of the score distribution is worse placed. Crossing the two fitted components identifies which one is responsible: substituting only the enlarged $\mu(x)$ moves the conformal quantile to 0.9512, that is 93 per cent of the total displacement, while substituting only the smaller $\hat{\sigma}$ moves it to 0.9270. The degradation is one of location, not of dispersion. This is the experimental counterpart of the finding reported in Section 6 for the conditional-dispersion variant, where modelling $\sigma(x)$ could not correct a shift that acts on the mean: there the conclusion came from a model that improved dispersion and did not help, here from one that improved dispersion and hurt~~the two model classes in general.

### 5.7. *Which components of the transport-based scheme earn their cost*

~~The diagnostic above says the gain is attributable to the base model. The ablations say which parts of the calibration layer are doing anything at all. We ran five nested configurations, adding the pure static split and transport without adaptation to the three groups requested in review, so that each component can be isolated: static split, adaptive conformal inference alone, transport alone at a fixed level, transport plus adaptation without tail shrinkage, and the full scheme. Everything else is held fixed.~~ Table 8 ~~reports the marginal contribution of each component to the interval score with a bootstrap interval over whole days, and Figure 9 shows it window by window.~~

~~Three findings follow, and only the first is favourable.~~

~~Online adaptation earns its cost. Adding it to~~ attributes the interval score of Transport+ACI to its components, for the three main base models. Online adaptation helps where the static split ~~improves the~~ over-covers: it improves the whole-test interval score by ~~297.9 MWh in the transition window, with a bootstrap interval of [−348, −251], and by~~ 95.9 MWh ~~over the whole test~~on the reference hurdle and by 19.8 MWh on the early-stopped hurdle, at a runtime ~~cost of two tenths~~ of a fraction of a second.

~~Score transport does not, once adaptation is present. On top of the adaptive update it *worsens* the interval score by 68.6 MWh over the whole test, with an interval of [+41, +100] that excludes zero, and it worsens it in four of the five windows individually, significantly in three. Transport without adaptation does improve~~ , but it costs 8.5 MWh on the quantile model, whose static split is already at nominal coverage. Score transport without adaptation improves on the static split ~~, on the two hurdles~~ by an amount ~~statistically indistinguishable from what adaptation aloneachieves,~~

Table 8: Marginal contribution of each component of the transport-based scheme to the mean one-sided interval score (MWh; negative is an improvement), with 95% day-block bootstrap intervals, for the three main base models. All adaptive steps use $\gamma = 0.05$ and no bound on $\alpha_t$, as in Algorithm 1. Differences are computed on the plant-days where both limits are finite. Runtime is the additional wall-clock time of the component over the 638 test dates.

| | Reference hurdle | | Early-stopped hurdle | | Reference quantile GBM | | |
|---|---|---|---|---|---|---|---|
| Component | Transition | Whole test | Transition | Whole test | Transition | Whole test | Runtime (s) |
| Online adaptation (ACI over static) | -297.9† | -95.9† | -85.1† | -19.8† | +0.5 | +8.5† | +0.22 |
| Transport alone (over static) | -282.9† | -94.2† | -64.2† | -16.9† | +10.3† | +5.7† | +7.45 |
| Transport added to ACI | +12.0 | +68.6† | +12.3† | +30.3† | +14.1† | +11.5† | +0.46 |
| Tail shrinkage added to transport | -0.6 | -6.4† | -0.6† | -2.7† | -0.1 | -0.3† | +6.36 |
| Full scheme (over static) | -286.5† | -32.3 | -73.4† | +9.4 | +14.5† | +19.1† | +7.04 |

† the 95% interval excludes zero. Intervals for every cell are in the result files of the companion repository; runtime measured on the reference hurdle.

~~at thirty-seven times the runtime. The two components are substitutesrather than complements, and the cheaper one is at least as good.~~

~~The tail shrinkage does not pay for itself. Its effect on the whole test is −6.4 MWhwith a bootstrap interval of [−10, −3], which excludes zerobut amounts to eight tenths of~~ similar to that of adaptation alone, so the two are substitutes. Once adaptation is present, adding the transport worsens the interval score on every base model, by +68.6, +30.3 and +11.5 MWh, each with an interval that excludes zero. The tail shrinkage changes the score by less than one per cent ~~of an interval score of about 810, and it accounts for 6.16 of the 6.79 seconds the full pipeline costs, that is ninety-one per cent of the computational budget of the method.~~

~~Taken together the full pipeline improves on~~ while accounting for 90% of the runtime of the scheme. The full scheme is not distinguishable from the static split on the two hurdles and is worse than it on the quantile model, by ~~286.5 MWh in the transition window and shows no demonstrable loss in any semester: in 2025-S1 and 2025-S2 the differences, +58.8 and +14.3 MWh , have intervals that contain zero, and in 2026-S1 it improves by 14.7 MWh with an interval that excludes zero. Over the whole test it nets −32.3 MWh with an interval of [−70, +9] that contains zero: the complete method is not distinguishable from the static split it replaces, and we do not read that as a positive result for it.~~

### 5.8. ~~*Metrics, infinite intervals and probabilistic benchmarks*~~

~~Table 9 evaluates every method on the whole test period under the full metric set, alongside four probabilistic benchmarks that the submitted version did not include: a per-plant empirical quantile over a trailing year, a per-plant rolling historical distribution over 60 days, a one-sided conformalized quantile regression on the multi-quantile predictive, and a per-plant zero-inflated lognormal fitted without covariates. All four respect the same nominal level, the same embargo and the same evaluation rows~~19.1 MWh [+14.1, +24.2].

~~The benchmarks change the picture. All four attain a better interval score than every method built on~~

### 5.8. *Operational evaluation*

Table 9 reports coverage weighted by capacity and by curtailed energy, the share of curtailed energy left above the limit, and the expected cost at three cost ratios; Table 10 and Figure 10 report the behaviour on high-curtailment days.

Plant-day coverage overstates protection. Weighted by curtailed energy, the coverage of the methods built on the three main base models falls to between 63 and 84%, because limits are exceeded mostly on large days; weighting by installed capacity lowers it by one to three points. The transport-based variants on the reference hurdle leave between 13.3 and 14.2% of the curtailed energy above the limit, the highest shares among these methods, against 10.0% for the static split on the same base model and 9.8% for the static split on the quantile model. With every base model, Transport+ACI leaves more energy uncovered than the static split on the same model.

On high-curtailment days defined by the outcome, every method loses coverage relative to its average, and only methods that over-cover on average stay at or above the nominal level. On the 254 days of definition H1 the static split on the reference hurdle covers 91.9% and Transport+ACI 86.7%; the ~~hurdle, and each differs from the~~ static split ~~by a day-block bootstrap interval that excludes zero. Three of them buy that score with coverage, and should be~~

read accordingly: the rolling 60-day distribution reaches an interval score of 433 at 86.5% coverage, the trailing-year quantile 486 at 87.2% and the zero-inflated lognormal 639 and CQR on the quantile model cover 86.7% and 85.9%, with limits less than half as wide as those of the reference hurdle's static split. Conditioning only on information available at the forecast origin (H3), the methods on the quantile model are close to nominal, at 91.0% for the static split and 90.4% for CQR, and Transport+ACI on the reference hurdle covers 89.6%. The shortfall on outcome-defined days is the forecaster's dilemma [44]: a forecaster that is calibrated on average under-covers on the days that turned out extreme, unless its limits are wide everywhere.

At every cost ratio, the methods built on the quantile model are the cheapest among those that are close to nominal on average, and among them the calibration layer changes the cost by at most 5%. Transport+ACI on the reference hurdle costs 813 at 82.4% , all several points below nominal, so an operator who needs the stated level cannot use them. The conformalized quantile regression is the exception and is $k = 10$ against 433 for the static split on the quantile model, and the most informative row of the table. It attains 89.8% coverage, within half a point of nominal selected Transport+ACI on the reference hurdle costs 1302 and 4621 at $k = 20$ and $k = 100$, against 539 and 929. In the tail the layers on the quantile model differ slightly: at $k = 100$ the selected Transport+ACI, whose step is small, is the cheapest of them (896, with a mean width of 276 MWh and an interval score of 438, against 596 MWh and 813 for 3.5% below the full pipeline of the submitted version. A standard construction from 2019, applied to a better base predictive, dominates the method this paper proposed.

The treatment of infinite intervals matters and is now explicit. Three configurations of the submitted version produce infinite limits, up to 5.9 per cent of the test period, static split) and the only one to reach 99% coverage. Two per-plant empirical quantiles that use no base model, over the last 60 days and over the last year, cost about as much as the methods on the quantile model, and under the width-over-finite-intervals convention they appear sharper than they are. The unrestricted interval score reports them as infinite, which is the correct verdict for a method that returns uninformative intervals.

The cause is structural rather than incidental: the adaptive update is allowed to drive $\alpha_t$ to zero or below, at which point the required order statistic does not exist and the conformal quantile is $+\infty$. Bounding the update from below removes the problem entirely. Table G.17 shows the effect of $\alpha_{\min} \in \{0.005, 0.01, 0.02\}$. At $\alpha_{\min} = 0.005$ the fraction of infinite intervals falls to zero in all five windows and all four adaptive configurations, at a cost of at most half a point of marginal coverageyearly one is the cheapest of all methods at $k = 100$ (873). Both under-cover at every level, at 86.5, 91.1 and 95.5% for the 60-day quantile and 87.2, 92.4 and the unrestricted interval score becomes finite and therefore comparable. We recommend the bound as a default. It is worth noting what it does not fix: with infinities removed and the score made comparable, the transported variants remain worse than pure adaptation, which is the finding of Section 5.7 rather than an artefact of the infinities97.2% for the yearly one at the 90, 95 and 99% levels, which an owner who needs the stated level cannot accept.

### 5.9. *Panel dependence and conditional coveragesensitivity*

Clustered standard errors quantify the uncertainty that daily dependence induces in a reported coverage figure. They do not restore exchangeability, and the finite-sample guarantee of the split rests on exchangeability rather than on the width of a confidence interval. The distinction matters here because the dependence is severe: as reported in Section 4.3, the The intraclass correlation of the calibration scoresby date is 0.238 , giving a design effect of 26.7 and an effective calibration sample of about 1,000 rows rather than 26,552.

We tested three families of remedy, and report what each one actually buys. Table G.20 gives marginal performance together with the worst absolute deviation from nominal on each conditioning axis, which is the operative measure of conditional coverage.

*Block conformal.* Subsampling (Section 4.3). Calibrating on one plant per datemakes the calibration , which makes the scores approximately independentat the cost of reducing them from 26, 552 to 244. Averaged over 200 replicates with a fixed seed, , moves the conformal quantile moves of the reference hurdle from 0.9242 to 0.9263 , a difference of two thousandths of a PIT unit, while the with a replicate standard deviation of that quantile is 0.0112. The point estimate is therefore close to unbiased; what : the pooled construction misstates is not the level but the precision , since the guaranteeis nominally stated at a sample size twenty-seven times larger than the effective one. Marginal performance is essentially unchanged and conditional coverage is marginally worse, so this is a correction to the interpretation rather than an improvement to the method.

Table 9: Operational evaluation over the whole test period. Coverage by plant-day (date-clustered standard error), weighted by installed capacity and weighted by curtailed energy; share of curtailed energy above the limit; mean width (MWh); fraction of infinite limits. Cost: expected cost per plant-day in units of the allowance cost $c_h$, for a ratio $k = c_p/c_h$ of the penalty to the allowance cost, with the limit calibrated at the decision-optimal level $1 - 1/k$; at $k = 10$ it is the one-sided interval score at 90%. Costs over finite limits; an asterisk marks methods with infinite limits.

| | Coverage and exposure (90% limits) | | | | | | Cost per plant-day | | |
|---|---|---|---|---|---|---|---|---|---|
| Method | Plant-day | Capacity | Energy | Uncovered | Width | % inf. | $k = 10$ | $k = 20$ | $k = 100$ |
| *Reference hurdle (2×400 trees)* | | | | | | | | | |
| Static split (PIT) | 92.5 | 90.9 | 80.2 | 10.0% | 678 | 0.0 | 826 | 1320 | 3629 |
| ACI (selected $\gamma$) | 90.9 | 89.1 | 76.8 | 11.7% | 590 | 0.0 | 763 | 1221 | 4225 |
| Transport+ACI (selected) | 90.0 | 88.0 | 74.0 | 13.3% | 564 | 0.0 | 761 | 1302 | 4621 |
| Transport+ACI ($\gamma = 0.05$) | 90.1 | 88.0 | 73.0 | 13.8% | 596 | 5.9 | 813* | – | – |
| One-sided CQR | 91.2 | 88.7 | 74.8 | 12.4% | 532 | 0.0 | 714 | 1181 | 3426 |
| *Early-stopped hurdle (105 trees)* | | | | | | | | | |
| Static split (PIT) | 92.0 | 90.7 | 78.8 | 8.4% | 409 | 0.0 | 534 | 743 | 1693 |
| ACI (selected $\gamma$) | 90.8 | 89.2 | 75.9 | 9.7% | 377 | 0.0 | 520 | 721 | 1748 |
| Transport+ACI (selected) | 89.9 | 88.1 | 73.1 | 10.9% | 367 | 0.0 | 527 | 741 | 1769 |
| Transport+ACI ($\gamma = 0.05$) | 90.0 | 88.1 | 72.0 | 11.5% | 377 | 3.3 | 552* | – | – |
| One-sided CQR | 94.2 | 93.3 | 84.3 | 6.1% | 501 | 0.0 | 590 | 1055 | 3634 |
| *Reference quantile GBM (54×400)* | | | | | | | | | |
| Static split (PIT) | 90.2 | 88.0 | 66.1 | 9.8% | 288 | 0.0 | 433 | 539 | 929 |
| ACI (selected $\gamma$) | 90.3 | 88.1 | 66.2 | 9.8% | 289 | 0.0 | 433 | 541 | 945* |
| Transport+ACI (selected) | 90.1 | 87.9 | 65.6 | 10.1% | 294 | 0.0 | 444 | 556 | 896* |
| Transport+ACI ($\gamma = 0.05$) | 90.2 | 88.0 | 66.7 | 10.0% | 305 | 4.4 | 459* | – | – |
| One-sided CQR | 89.8 | 86.7 | 63.0 | 11.0% | 276 | 0.0 | 438 | 546 | 927 |
| *Benchmarks without a base model* | | | | | | | | | |
| Empirical quantile per plant, 365 days | 87.2 | 87.0 | 61.9 | 11.5% | 317 | 0.0 | 486 | 583 | 873 |
| Rolling daysistribution per plant, 60 days | 86.5 | 86.3 | 64.9 | 10.2% | 282 | 0.0 | 433 | 531 | 912 |
| Zero-inflated lognormal per plant | 82.4 | 81.5 | 55.5 | 20.4% | 338 | 0.0 | 639 | 942 | 3011 |

Table 10: Coverage on high-curtailment days and conditional coverage, 90% limits. H1: system curtailment above the calibration-window 90th percentile; H2: top decile of test days; H3: trailing seven-day mean of system curtailment known at the forecast origin above the same threshold. H1 and H2 condition on the outcome, H3 only on information available when the forecast is issued. Coverage on each subset with 95% day-block bootstrap interval; energy-weighted coverage on the same subset. The last three columns give the worst absolute deviation from 90% across technologies, capacity terciles and regions (pp).

| | H1, ex post (254 days) | | H2, ex post (64 days) | | H3, ex ante (260 days) | | Worst deviation | | |
|---|---|---|---|---|---|---|---|---|---|
| Method | Coverage | Energy | Coverage | Energy | Coverage | Energy | Techn. | Size | Region |
| *Reference hurdle (2×400 trees)* | | | | | | | | | |
| Static split (PIT) | 91.9 [90.6, 93.1] | 79.6 | 90.7 [88.0, 92.9] | 77.8 | 94.2 [93.4, 94.8] | 82.1 | 2.5 | 3.6 | 4.8 |
| ACI (selected $\gamma$) | 90.2 [88.7, 91.4] | 76.1 | 88.8 [85.8, 91.2] | 74.2 | 92.6 [91.7, 93.5] | 78.7 | 1.0 | 2.0 | 3.7 |
| Transport+ACI (selected) | 88.0 [86.4, 89.5] | 72.1 | 85.6 [81.8, 88.9] | 69.4 | 90.0 [88.9, 91.1] | 73.4 | 0.1 | 1.8 | 4.9 |
| Transport+ACI ($\gamma = 0.05$) | 86.7 [85.2, 88.1] | 69.2 | 84.1 [80.2, 87.4] | 66.6 | 89.6 [88.4, 90.7] | 72.2 | 0.3 | 1.6 | 4.8 |
| One-sided CQR | 89.9 [88.5, 91.1] | 74.0 | 88.2 [85.3, 90.6] | 71.7 | 92.6 [91.8, 93.4] | 77.2 | 1.9 | 3.7 | 5.9 |
| *Early-stopped hurdle (105 trees)* | | | | | | | | | |
| Static split (PIT) | 91.4 [90.3, 92.4] | 78.9 | 89.9 [87.6, 91.9] | 76.8 | 94.4 [93.8, 94.9] | 83.3 | 2.2 | 3.2 | 4.6 |
| ACI (selected $\gamma$) | 90.1 [88.9, 91.2] | 76.1 | 88.4 [85.9, 90.6] | 73.9 | 93.3 [92.6, 93.9] | 80.8 | 0.9 | 2.0 | 3.7 |
| Transport+ACI (selected) | 88.0 [86.7, 89.3] | 72.1 | 86.2 [83.0, 89.1] | 70.4 | 90.9 [90.1, 91.8] | 75.7 | 0.1 | 1.4 | 3.4 |
| Transport+ACI ($\gamma = 0.05$) | 86.5 [85.3, 87.8] | 68.5 | 83.1 [79.6, 86.2] | 64.8 | 89.9 [88.9, 90.9] | 72.9 | 0.1 | 1.5 | 3.1 |
| One-sided CQR | 93.9 [92.9, 94.8] | 84.5 | 92.8 [90.8, 94.4] | 82.6 | 96.1 [95.7, 96.6] | 88.1 | 4.4 | 5.0 | 6.0 |
| *Reference quantile GBM (54×400)* | | | | | | | | | |
| Static split (PIT) | 86.7 [85.3, 88.0] | 62.5 | 82.2 [79.3, 84.9] | 55.3 | 91.0 [90.2, 91.8] | 69.8 | 1.8 | 2.4 | 5.9 |
| ACI (selected $\gamma$) | 86.6 [85.1, 87.9] | 62.4 | 82.1 [79.0, 84.9] | 55.5 | 90.9 [90.1, 91.7] | 69.6 | 1.9 | 2.4 | 6.0 |
| Transport+ACI (selected) | 85.5 [84.1, 86.9] | 60.5 | 80.4 [76.8, 83.5] | 53.1 | 89.9 [89.0, 90.8] | 67.6 | 1.7 | 2.5 | 6.1 |
| Transport+ACI ($\gamma = 0.05$) | 85.8 [84.2, 87.3] | 61.6 | 81.5 [77.7, 85.0] | 55.8 | 90.5 [89.5, 91.5] | 69.2 | 1.9 | 2.3 | 6.1 |
| One-sided CQR | 85.9 [84.5, 87.1] | 59.2 | 81.0 [78.2, 83.5] | 51.4 | 90.4 [89.6, 91.2] | 66.5 | 1.8 | 4.3 | 6.2 |

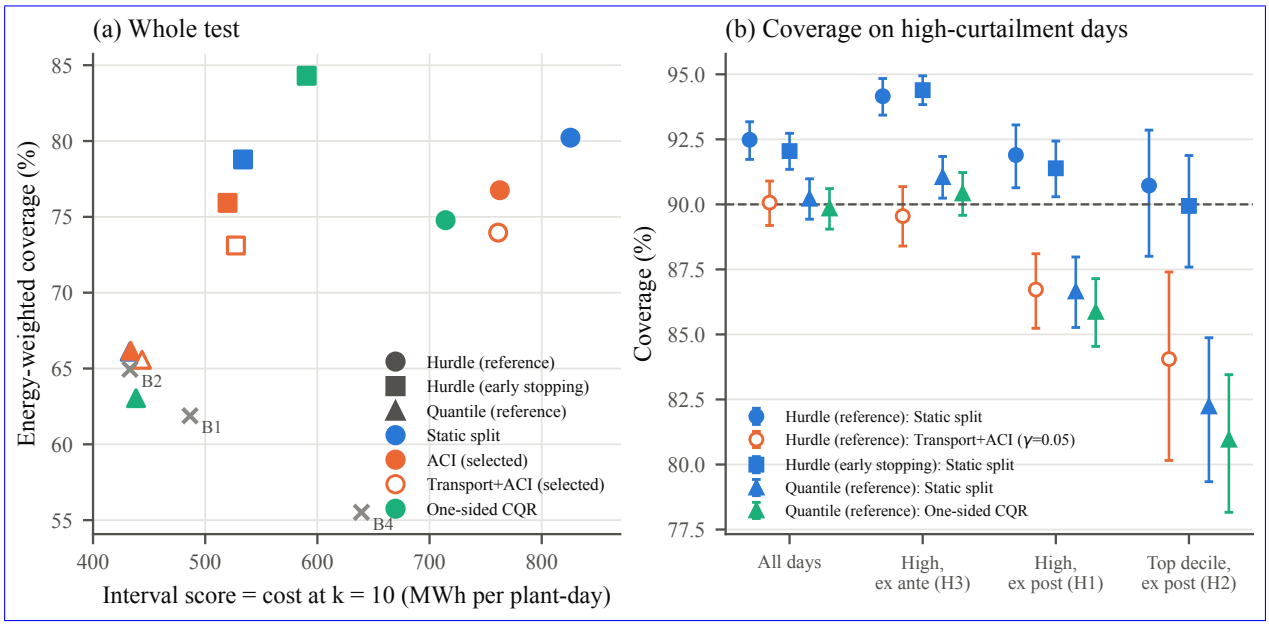



Figure 10: (a) Energy-weighted coverage against the 90% interval score, whole test; marker shape identifies the base model and colour the calibration layer, crosses are the benchmarks without a base model. (b) Coverage on all days and on high-curtailment days under the three definitions, with 95% day-block bootstrap intervals; dashed line: nominal 90%.

~~*Group conformal.* Mondrian calibration with a separate conformal quantile per group restores group-conditional coverage by construction only if the group quantiles are well estimated. By technology and~~ the precision of the guarantee, not its level. Mondrian calibration by technology or by capacity tercile ~~the effect is small. By region it is harmful:~~ changes little, and by region or by plant it worsens the worst regional deviation ~~rises~~ from 4.8 to 7.2 ~~percentage~~ points, because ~~several regions contribute~~ some groups have too few calibration days for a stable quantile ~~and the resulting estimates are noisy. Plant-level calibration, the finest partition, has the same problem in a stronger form and the same 7.2-point worst regional deviation , although it and Mondrian by region have the two best marginal interval scores of the group. This is a negative result worth recording: partitioning the calibration set is not a free way to buy conditional validity when the partition is fine relative to the effective sample size~~(Table G.20 in Appendix G). Among the main methods, the worst deviation across technologies, capacity terciles and regions ranges from 3.1 to 6.2 points (Table 10); the regional axis is the hardest for every method.

~~*The adaptive schemes.* The best conditional coverage in the table belongs to plain adaptive conformal inference, with a worst deviation of 4.7 percentage points, better than every Mondrian variant. That is a point in favour of online adaptation, and it is consistent with the ablation result: adaptation is the component that works.~~

~~One conditional result deserves emphasis because it runs against the interest of the method. On~~ Appendix E varies every parameter of the calibration layer and of the evaluation. The step $\gamma$ matters most: with $\alpha_{\min} = 0.005$, coverage stays within half a point of nominal for $\gamma \leq 0.05$ and falls to between 88 and 89% at $\gamma = 0.20$, and without a bound steps of 0.05 and above produce infinite limits on several per cent of plant-days. The length of the recent window comes next: a 30-day window raises the interval score of Transport+ACI on the early-stopped hurdle by 7%. The shrinkage scale, the bootstrap size, the grid of the map, the refresh period and the minimum number of recent scores move it by at most 2.5%. On the quantile model the bound does not remove every infinite limit of Transport+ACI, because the transported pool can exceed the highest estimated quantile level, 0.999. The conclusions are unchanged with calibration windows of four and six months, with quantile grids of 27 and 14 levels, with other thresholds for high-curtailment ~~days, defined by the calibration-window ninetieth percentile of system-level daily curtailment and covering 254 of the 638 test days, the static split covers at 91.9 per cent while Transport~~days, and across ten seeds of the conformal layer, whose standard deviation of coverage is at most 0.04 points. Across depths, learning rates and

training seeds of the early-stopped hurdle, its static split has an interval score between 517 and 558 MWh, always below Transport+~~ACI covers at 86.7 per cent. The economically important days are exactly where the adaptive scheme under-covers most, and a portfolio owner sizing an exposure on the largest days would be worse served by it than by the static split. Adaptive conformal inference sits between the two at 88.3 per cent .~~ ACI on the same model and far below the reference hurdle. The results for the transition window do not depend on where its boundaries are drawn (Table G.18).

### *5.10. ~~Sensitivity to the window boundary, the change-point dates and the calibration hyperparameters~~*

~~The transition window is defined by a measurable criterion, but the criterion still has to be applied to some boundary, and one might ask whether the results are an artefact of that choice. Table G.18 re-evaluates the three representative methods on five alternative definitions of the window besides the one used, ranging from two months to seven. The Wasserstein-1 divergence from the calibration distribution varies from 0.99 to 1.34 across them, and the ordering of the three methods by coverage is the same in every case: the static split over-covers by 4.0 to 5.3 points, adaptive conformal inference sits between, and the transported variant covers least. By the interval score over finite intervals the static split is the worst of the three in all six; the transported variant returns infinite limits in three of them, 5.5 to 8.1 per cent, where its unrestricted score is infinite. The apparent gain of the transported variant follows how far the static split over-covers in each window (Pearson $r = -0.844$ across the six), which is the same relationship established in Section 5.3, here traced in the boundary rather than across base models.~~

~~The chronology itself is likewise insensitive to the analyst's choices, as reported in Section 3.4: across 180 detection configurations, none returns a break in 2025 and none returns one in October 2024. The two dates that do appear, May 2023 and January 2024, are returned by both algorithms under the default penalty and are the two most frequent across the whole sensitivity grid.~~

~~*Calibration hyperparameters*. The selection of Section 4.7 used only the inner validation set and was closed before the test set was touched. Table G.19 reports the full $6 \times 5$ grid on the test set, as a display, never as a criterion. For the transported variant coverage is flat in $\gamma$ and in the window, between 89.9 and 90.1 per cent over all thirty configurations, and what varies is the fraction of infinite intervals: none at $\gamma \leq 0.01$, and between 21.9 and 23.0 per cent at $\gamma = 0.20$.~~

## 6. Discussion

### *6.1. What the ~~diagnostic means~~ evaluation implies for practice*

The ~~central empirical result of this paper is negative, and it is worth stating what it does and does not license.~~

~~It does not say that adaptive conformal calibration is useless. Online adaptation is the one component that improves the interval score reliably in our ablations, it costs almost nothing to run, and it delivers the best conditional coverage of any scheme we tested. It does say that the headline gain of a calibration layer , measured as width recovered~~ width that an adaptive conformal layer saves against a static split ~~,~~ is not a measure~~ment~~of how much distribution shift the layer absorbed. ~~On these data it is a measurement of how over-dispersed the base predictive was, and the relationship between the two is close enough to linear to be used as a diagnostic: a large apparent gain should prompt an examination of the base model before it is reported as a property of the calibration method.~~

~~The practical ordering that follows is the reverse of the one the submitted version implied. Fixing the base predictive dominates fixing the calibration layer. Replacing a constant-dispersion lognormal hurdle by a multi-quantile predictive fitted under the same protocol halves the interval width at the same coverage and improves the CRPS by a fifth to a third in every window, at a fitted budget of the order of five thousand trees against~~ To first order it is the conversion of the ~~hurdle's eight hundred, a budget the hurdle cannot use (Section 5.6), whereas the entire adaptive apparatus, run on the worse base model, recovers less and costs more. Once~~ static split's coverage error into width, which any layer that targets nominal coverage performs: the constant-threshold oracle, which adapts nothing in time, reproduces 84% of the variation of the observed change across cells, in the real panel and, with known ground truth, in the simulation. The static split's coverage error in turn measures how far the calibration of the base predictive drifted between the calibration and the evaluation window. With the window held fixed, so that every base model faces the same shift, that drift spans 13.8 percentage points across base models in the transition window, and the base

model explains 54% of its variance against 18% for the window. The simulation shows the same structure: a correctly specified predictive keeps the static split near nominal coverage under a tripling of magnitudes, a dispersion that is wrong by the same factor in every period is absorbed by the conformal quantile, and coverage drifts only when the base model ~~is adequate, the simplest available calibration layeris competitive: a one-sided conformalized quantile regression on the multi-quantile predictive attains 89.8 per cent coverage at 276 MWh.~~ fails to follow a change in the data.

~~There is~~ These results suggest three checks before a gain is attributed to an adaptive calibration layer: report the coverage of the static split in the evaluation window; report the width of the constant-threshold oracle at the adaptive layer's coverage, so that the residual, which is what adaptation adds, is visible; and repeat the comparison on a second base model. All three are cheap, because a ~~methodological point here that generalises beyond curtailment. A calibration layer evaluated against a single base modelcannot distinguish between correcting a shift and correcting the model. The experiment that separates them is cheap, since a~~ well-designed layer touches the base model only through ~~a CDF and its inverse, and we would argue it should be standard in this literature. Our own construction did not survive it .~~ its CDF and quantile function.

### 6.2. ~~*Where the operational value actually lies*~~

The ordering of effort follows. Replacing the reference hurdle by its early-stopped version, without touching the calibration layer, lowers the whole-test interval score of the static split by 35%, and replacing it by the quantile model lowers it by 48%. The best calibration layer on the reference hurdle lowers it by 13%. On a good base model the choice of layer matters little, and the ranking of layers itself depends on the base model: one-sided CQR is among the best layers on the quantile model and the worst on the early-stopped hurdle, because its correction is a single number of MWh pooled across plants of very different size (Section 5.3).

~~For plant-level curtailment at a seven-day horizon, and within the information set available from public data, effort is better spent on reliable uncertainty quantification than on chasing point accuracy, which a calendar baseline already delivers. That conclusion survives the revision. What changes is the recommendation about how to obtain the uncertainty: through a flexible , well-calibrated predictive distribution and~~ For an operator or asset owner working with public data at this horizon, the evidence supports the following practice.

1. Invest first in the base predictive: a ~~simple conformal wrapper, rather than through an elaborate adaptive layerover a rigid parametric one.~~
   ~~Two operational details from Section 5 are worth carrying forward. Bounding the online update from below is a free improvement: at~~ flexible distributional model, regularised or early-stopped, whose PIT histogram is checked on the calibration window and on recent data.
2. Wrap it in a static conformal layer, a PIT split or a one-sided CQR, recalibrated periodically.
3. If drift is expected, add ACI with a step selected out of sample and a lower bound on the miscoverage level, $\alpha_{\min} = 0.005$~~it removes infinite prediction limits entirely, at a coverage~~ , which removes infinite limits at a cost of at most half a point ~~, and it should be the default in any deployment where an uninformative interval is worse than a slightly wrong one. And conditional reliability on the largest days is the property a portfolio owner should check first, because it is where the schemes differ most and where the money is: on~~ of coverage.
4. Check conditional coverage on regimes known when the forecast is issued, such as the trailing system curtailment, and report energy-weighted coverage next to plant-day coverage.
5. Do not use the transport-based scheme of Algorithm 1: once ACI is present the transport worsens the interval score, its tail shrinkage consumes most of the runtime for a negligible gain, and on outcome-defined high-curtailment days ~~the adaptive scheme under-covers by more than three points relative to~~ it leaves more curtailed energy uncovered than the static split.

### 6.2. *The change in the record, and what we do not claim about it*

The distributional change documented in Section 3.4 is ~~real, is~~ a ramp of roughly eighteen months between mid-2023 and late ~~2024, and is essentially complete before 2025 begins. On its causeswe are deliberately agnostic, and the dataset released here cannot settle the question,~~ 2024. The dataset cannot identify its causes, because it records curtailment and not the assets or ~~the~~ network states that absorb it. Several mechanisms ~~operate over the same~~

~~months and cannot be separated with these data: the~~ overlap: growth of solar capacity behind a saturated north-~~to~~-south corridor, reinforcements of that corridor, changes in northern mining demand, the entry of battery storage, and the ~~mechanical~~ saturation of curtailment ~~itself~~ once the corridor is full ~~throughout the solar window. The literature discussed in Section 2 makes clear how much would be needed to identify any of them:~~ through the solar hours. Mohamad et al. [13] show that the effect of storage on curtailment depends on ~~where it sits~~ its position relative to the binding constraint, and Denholm and Mai [10] ~~show~~ that it depends on duration as much as on power~~. We~~ ; we have neither the ~~locational~~ network model nor the charge-discharge records that such an analysis requires.

A second ~~and distinct~~ phenomenon appears at the turn of 2024 to 2025: a break in the rate of growth. The year-over-year growth of the monthly mean log-magnitude falls to +0.09 per year for northern solar from January 2025 and to +0.01 at system level from December 2024, while the northern solar fleet keeps expanding by 8 per cent through ~~2025 and curtailment per installed MW in those regions falls by 15 per cent in the fourth quarter.~~ 2025. Storage entry is one candidate for that flattening and is ~~quantitatively~~ of the right order, but the supporting energy balance is an upper bound that ~~holds only under assumptions of~~ assumes one full daily cycle, unit efficiency and that every charged megawatt-hour ~~having otherwise been curtailed;~~ would otherwise have been curtailed, and the same exercise ~~repeated~~ for January to May 2026 does not reproduce it. We record storage as one ~~bounded~~ hypothesis among several and make no causal claim. ~~A registry of storage systems~~ The supplementary material lists the storage systems of the SEN with their commercial operation dates~~is provided in the supplementary material~~ , so that the hypothesis can be tested ~~independently~~ by anyone with ~~the~~ network and dispatch data~~we lack~~.

~~It is worth noting that the calibration layer responds~~ The calibration layers respond to only one of these two kinds of change. A displacement of the level of the predictive law ~~, such as the ramp of 2023 and 2024,~~ appears as a displacement of the score distribution and is what the transport map and the online update are built to absorb. A break in the rate of growth ~~is not, and none of the methods evaluated here detects it. That does not damage their coverage, because a trend that flattens~~ leaves no displacement to correct ~~, but it matters for interpretation:~~ and is invisible to coverage, so reliability under shifts of level carries no guarantee under changes of trend~~, and the two are not distinguishable from coverage alone~~.

### *6.3. The mid-2025 excursion*

The coverage dip visible in Figure 5 ~~can be given a date and a signature. It~~ is concentrated in June 2025, the largest deseasonalised monthly anomaly of the ~~entire~~ post-calibration record~~: the monthly mean log-magnitude for northern solar stands well above the deseasonalised level of the calibration window, and the~~ . The Wasserstein-1 distance between May to July 2025 and August to December 2025 is 1.334, larger than the 1.147 of the transition contrast~~itself~~. Its signature is the reverse of the 2024 window: magnitudes rise while the occurrence probability falls~~, whereas in October to December 2024 the two rose together~~. It appears in solar and in wind ~~separately~~ and across four regions, which rules out a single plant or a residual erratum, and the operator's ~~monthly~~ report for that month records no storage system entering operation. A one-month excursion ~~with an occurrence signature opposite to that of the main transition is the kind of event that~~ of this kind is absorbed only slowly by a continuously adapting scheme ~~absorbs only slowly and that~~ and is the case a change-point trigger would catch~~, which is one of the directions we identify below~~.

### *6.4. Open directions*

~~Three follow directly~~ Four directions follow from the results. A change-point detector could trigger recalibration at the onset of a shift~~rather than adapting continuously~~, which would address ~~the~~ excursions like that of June 2025~~excursion; the machinery used offline in~~ ; the offline machinery of Section 3.4 ~~to date the breaks is a natural starting pointfor an online version~~ is a starting point. A scale-invariant version of CQR, for instance with the nonconformity score normalised by a predicted scale or computed on the log scale, would remove the dependence of the correction on the mix of plant sizes and on changes in level. Localised conformal prediction~~in the sense of Guan~~ [34], which weights the calibration set by proximity to the test point, is a more principled route to conditional validity than the Mondrian partitions ~~that failed in Section 5.9, and it does not suffer from the small-group problem that defeated them. And extending the evaluation to the multivariate setting, across~~ of Appendix G. And the evaluation can be extended to portfolios of plants that share a transmission corridor and whose curtailment is ~~therefore jointly determined, is the natural next step for an operator who manages a portfolio rather than an asset~~jointly determined.

## 7. Limitations

~~Both reviewers of the submitted version noted that its limitations were understated. They were, and the following are the ones we consider material.~~

~~The scope of~~

1. *One system, one public data source.* All evidence comes from the SEN between 2022 and mid-2026. The mechanical part of the decomposition (Section 5.4) follows from how a coverage-targeting layer works and should hold elsewhere; its size, the ~~point-forecasting conclusion~~coefficients of the cross-cell relationship and the ranking of methods should not be transported to other systems without checking.
2. *Two model families.* ~~Section 5.1 compares five model classes on a twelve-feature set at a seven-day horizon under mean absolute error. It does not establish anything about point forecasting of curtailment in general, and in particular says nothing about models with access to network topology, dispatch schedules, weather forecasts or storage state of charge, nor about criteria that weight large-magnitude daysmore heavily.~~
   The nine base predictives are four variants of a lognormal hurdle and five capacity variants of a multi-quantile gradient-boosting model. The attribution of the coverage drift to the base model is established within this design; it is an empirical association for these families on this panel, and the intervals of Section 5.4 quantify sampling uncertainty over days, not uncertainty over model families.
3. ~~*Exchangeability fails, and the finite-sample guarantee*~~ *The oracle is* ~~*correspondingly weaker than it looks*~~*a benchmark.* The ~~static split's guarantee is marginal and requires exchangeability, which does not hold here either~~ constant-threshold oracle uses the outcomes of each window and cannot be deployed. It separates the part of a width change that requires no adaptation from the rest; it does not provide an alternative method.
4. *Dependence.* Exchangeability fails across the regime boundary ~~or within a regime. The~~ and within it; the effective calibration sample is about 1,000 rows rather than 26,552 (Section 4.3)~~, so the nominal finite-sample statement is stated at a sample size the data do not supply. None of the block, group or~~ . Block, group and plant-level ~~schemes of Section 5.9 repairs this ; the block scheme quantifies it, and the group schemes make conditional coverage worse.~~
   ~~*The diagnostic rests on two model families and one dataset.* The relationship between over-coverage and width recovery is established over forty cells from eight predictives on one panel, but the eight are not eight model classes: three are variants of the hurdle family and five are budget variants of a single quantile family. The design therefore spans two functional forms, not eight. We regard it as a mechanism that generalises, since it follows from what a conformal quantile does, but we have not demonstrated that it holds in other systems or with other model families, and the specific coefficients should not be transported.~~
   ~~*The separation of model form from fitted capacity rests on one panel and two model families.* Section 5.6 settles, for this problem, a question that the comparison of Section 5.3 leaves open: the multi-quantile advantage is not available at the hurdle's own tree budget, the hurdle does not recover it when given five times that budget, and at comparable budgets the gap is about thirty per cent. What that licenses is a statement about these two families on this panel. It does not license a statement about quantile regression against two-part models in general, and we do not make one. Two features of the ladder bound it further. It varies budget only through the number of trees, holding depth, learning rate and the rest fixed, so it separates budget from form along one axis and not all of them. The ladder itself uses no early stopping, so the enlarged hurdle is free to overfit rather than to halt. We fitted an early-stopped hurdle separately, reported in Section 5.6: it selects 105 trees, an eighth of our base model's budget, and beats that base model by 5.7 per cent of CRPS. Our base model is therefore not the best hurdle available on this problem, and every level we report for it, and every difference computed from it, should be read with that discount: Table G.16, the ablations of Table 8, the benchmark comparison of Table 9, the $\alpha$ bound of Table G.17, the panel-dependence schemes of Table G.20, the boundary sensitivity of Table G.18, and Figures 5, 6 and 9. None of those tables was re-run with the stopped hurdle, so we claim no direction for the discount there. The one comparison that was re-run is the point forecasting of Table 2, which is deliberately not in that list, and there the discount runs the other way: the stopped fit is better by CRPS but worse by mean absolute error, enough to place it last (Section 5.1). What does hold is the margin to the quantile form, which stays above twenty per cent over the whole test. A reader comparing model families on our numbers should compare against the stopped hurdle rather than against ours.~~

~~*The multi-quantile predictive has its own boundaries.* It expresses nothing above its highest estimated quantile level, $\tau = 0.999$, and requests beyond it return an infinite limit. In this evaluation no interval hit that boundary, but a more extreme test period could, and the failure mode is not hypothetical: an arm fitted on a nine-level grid, whose top level is $\tau = 0.9$, returns an infinite limit for every test row under the static split, because the conformal quantile of 0.938 falls above the highest level the predictive expresses. That arm is excluded from Table 7 and from the diagnostic of Section 5.3; it is recorded here as a mechanical demonstration of where a quantile-grid predictive breaks. Its implied occurrence probability is also coarse, read off the quantile grid at 0.02 resolution, which is why its Brier score is worse than the hurdle's and why that comparison is reported as descriptive rather than as a verdict.~~

~~Differences of one to three points of coverage~~ calibration (Appendix G) quantify this but do not repair it. Differences of one to three points of coverage are within the date-clustered standard error and are ~~within~~ not interpreted.

5. *High-curtailment days.* Coverage on days selected by the outcome is a description of exposure, not a test of calibration [44]; the ex-ante definition H3 is the ~~date-clustered standard error~~legitimate test, and all definitions depend on a threshold, whose effect is reported in Appendix E.
6. *A stylised cost.* The cost model is linear in the allowance and in the exceedance, with constant unit costs, and ignores prices, contracts and the timing of settlement. It ranks methods for three cost ratios; it does not value them in money.
7. *A stylised simulation.* The simulated panel is generated from lognormal hurdles with common autoregressive shocks and linear ramps. It demonstrates the mechanism with known ground truth; its magnitudes are not estimates of the real ones.
8. *Selection on a short window.* ~~and we have avoided interpreting them. Where a difference matters to a conclusion we report a bootstrap interval over whole days and state whether it excludes zero.~~

   ~~*The hyperparameter selection uses a short inner validation set.* Because base-model predictions begin~~ Base-model predictions start in January 2024, so the rolling-origin selection ~~of Section 4.7 runs on~~ uses four months of inner validation. That is enough to reject ~~the aggressive settings, which produce infinite intervals in a third of validation cases, but it is~~ steps that produce infinite limits, but a thin basis for fine distinctions among the gentler ones.
9. ~~*The causes of the distributional change are not identified and cannot be with this dataset*~~*Point forecasting*. ~~We record storage as one bounded hypothesis among several for the trend break at the turn of 2024 to 2025, and we make no causal claim. The supplementary registry is offered so that others can test it with data we do not have.~~ The comparison of Section 5.1 covers five model classes, twelve features, a seven-day horizon and mean absolute error. It says nothing about models with access to network topology, dispatch schedules, weather forecasts or storage state of charge, nor about criteria that weight large days.
10. ~~*The*~~ *What the dataset* ~~*records curtailment only*~~*contains*. ~~It contains~~ The dataset records curtailment only, with no network state, ~~no~~ dispatch decision, ~~no price and no~~ price or storage operation, so ~~every mechanism discussed in Section 6 is outside what it can adjudicate.~~ the causes of the distributional change are not identified (Section 6.2).

## 8. Conclusion

We have ~~presented~~ released an open, errata-corrected, plant-level dataset of renewable curtailment in Chile's National Electricity System, covering 53 continuous months and 300 plants ~~, whose annual totals reconcile~~ and reconciling exactly with the ~~system~~ operator's ~~published figures, and we have~~ annual totals, and used it to evaluate conformal calibration of seven-day-ahead upper prediction limits under a documented ~~distributional change~~ change in the distribution of curtailment.

The ~~empirical findings are these. Within the models, features, horizon and metric tested, point forecasting adds little over a calendar baseline, so the operationally relevant problem is uncertainty quantification. A regime shift of~~ main findings are four. First, the ~~kind observed here inflates standard conformal intervals into over-covered, wider bands rather than breaking their coverage. A substantial share of what reads as regime change is the annual cycle and has to be separated from it before the shift can be sized. And the gain that an adaptive conformal layer appears to~~

deliver against that shift is , on these data, a measurement apparent benefit of an adaptive calibration layer depends on the base model: the scheme that narrows the reference hurdle's intervals by a third in the most divergent window leaves those of a multi-quantile model essentially unchanged. Second, to first order that benefit is the mechanical conversion of the static split's coverage error into width, which an oracle without any adaptation reproduces and a controlled simulation confirms; it measures the drift of the base model's over-dispersion rather than predictive's calibration, not the size of the shift: it does not replicate when the base predictive is replaced by a . Third, improving the base model dominates refining the calibration layer: early stopping alone narrows the hurdle's static limits by 40% at the same coverage, more than any calibration layer achieves on the reference hurdle. Fourth, on high-curtailment days defined by the outcome every method loses coverage and only those that over-cover on average stay at the nominal level, while on days identified from information available at the forecast origin the methods built on the quantile model stay close to nominal.

The transport-based adaptive pipeline is not the preferred operational method. On these data it under-covers on high-curtailment days (86.7% against 91.9% for the static split on the same base model), leaves more curtailed energy uncovered than the static split, and is dominated, at every cost ratio considered, by a static conformal wrapper (PIT split or one-sided CQR) around a multi-quantile alternative fitted under the same protocol, it is linearly related to how much the static split over-covers, and its two most elaborate components, score transport and bootstrap tail shrinkage, do not survive ablation. A conformalized quantile regression on the better base modelpredictive, which attains near-nominal coverage at less than half the widthof the full pipelineits width.

Three guarantees should be kept apart when such results are reported, and we have kept them apart throughout. The static split provides finite-sample *marginal* marginal coverage under exchangeability, an assumption which this panel violates both across the regime boundary and within it, with an effective calibration sample twenty-seven times smaller than the nominal one. The adaptive update provides long-run average coverage control without any distributional assumptionwithout distributional assumptions, and nothing at finite time. Empirical coverage in a dependent panel is a third thing, : it is what we measure, and is what it varies across technologies, plant sizes, regions and high-curtailment days in the ways reportedhere.

as reported. The dataset and the code that regenerates every tableand figure are publicly available, and we hope both the resource and the negative result are useful to operators, asset owners and researchers working on the integration of variable renewable energy, figure and number are public.

## CRediT authorship contribution statement

**Pablo Reyes Cerda:** Conceptualization, Data curation, Software, Validation, Investigation, Writing – original draft. **Kerven Cea Morales:** Methodology, Formal analysis, Validation, Writing – review and editing.

## Declaration of competing interest

Pablo Reyes Cerda is the founder of Synapta SpA and of CurtailmentIQ, a commercial forecasting service that uses data sources related to those described in this article. Kerven Cea Morales declares no competing interest.

## Declaration of generative AI and AI-assisted technologies in scientific writingthe manuscript preparation process

During the preparation of this work the authors used an AI-based assistant to improve the readability and language support the drafting and language editing of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

## Data availability

The dataset is openly available on Zenodo under a CC BY 4.0 licence (DOI: 10.5281/zenodo.21198816, which resolves to the latest version of the deposit). The processing pipeline, the modelling code and the scripts that regenerate every tableand figure , figure and number in this article are available in the companion repository at `https://github.com/Pablosebastian-reyes/curtailmentiq-research`. The supplementary material accompanying

this article contains a registry of the energy storage systems of the SEN with their commercial operation dates and sources, compiled from the public reports of the system operator, the Ministry of Energy and the National Energy Commission; it is offered so that the bounded hypothesis of Section 6 can be tested independently, and it is not part of the released dataset.

## Appendix A. Inversion of the PIT limit

Fix $x$ and write $p = p(x)$, $\mu = \mu(x)$. The mixture CDF is $F_x(y) = 0$ for $y < 0$, $F_x(0) = 1 - p$, and $F_x(y) = (1 - p) + p\,\Phi((\log y - \mu)/\sigma)$ for $y > 0$, which is continuous and strictly increasing on $(0, \infty)$ from $1 - p$ to 1. Given a level $\hat{q} \in [0, 1]$ we seek the smallest $u \geq 0$ with $F_x(u) \geq \hat{q}$. If $\hat{q} \leq 1 - p$ the atom already reaches the level, so $u = 0$. If $\hat{q} \geq 1$ no finite $u$ attains it and $u = +\infty$. For $1 - p < \hat{q} < 1$, strict monotonicity gives a unique solution of $(1 - p) + p\,\Phi((\log u - \mu)/\sigma) = \hat{q}$, that is $\Phi((\log u - \mu)/\sigma) = (\hat{q} - (1 - p))/p \in (0, 1)$, hence $\log u = \mu + \sigma\,\Phi^{-1}((\hat{q} - (1 - p))/p)$ and

$$U(x) = \exp\Big(\mu + \sigma\,\Phi^{-1}\big(\tfrac{\hat{q}-(1-p)}{p}\big)\Big),$$

, which is (3). The two boundary cases are recovered as the limits of the clipped argument $(\hat{q} - (1 - p))/p \to 0^+$ and $\to 1^-$, since $\Phi^{-1}(0^+) = -\infty$ gives $U = 0$ and $\Phi^{-1}(1^-) = +\infty$ gives $U = +\infty$. □

For the multi-quantile predictive quantile model the same object is obtained without a closed form. Let $\tau_1 < \cdots < \tau_K$ be the estimated levels and $\hat{q}_{\tau_1}(x) \leq \cdots \leq \hat{q}_{\tau_K}(x)$ the rearranged quantiles, and augment the grid with the node $(0, 0)$. Then $F_x$ is the piecewise-linear interpolant of the augmented pairs, its atom is $F_x(0) = \max\{\tau_k : \hat{q}_{\tau_k}(x) = 0\}$, and $F_x^{-1}(\hat{q})$ is the piecewise-linear interpolant in the other direction, equal to 0 for $\hat{q} \leq F_x(0)$ and to $+\infty$ for $\hat{q} > \tau_K$.

For both families, and for $y > 0$, $y \leq F_x^{-1}(q)$ if and only if $F_x(y) \leq q$. The coverage of a constant threshold $q$ over a set of plant-days is therefore the fraction of zero outcomes plus the fraction of positive outcomes with $F_x(y) \leq q$, which is how the oracle of Section 4.9 is computed exactly.

## Appendix B. Coverage under score transport

*Static split.* If the calibration scores $s_1, \ldots, s_n$ and the test score $s_{n+1}$ are exchangeable, then with $\hat{q} = s_{(k)}$, $k = \lceil (n + 1)(1 - \alpha) \rceil$, one has $\mathbb{P}(s_{n+1} \leq \hat{q}) \geq k/(n + 1) \geq 1 - \alpha$, and since $F_x$ is monotone, $s_{n+1} \leq \hat{q}$ if and only if $Y \leq U(x)$; hence $\mathbb{P}(Y \leq U(x)) \geq 1 - \alpha$. This is a *marginal* statement , and it is The statement is marginal and exact only under exchangeability; the serial and cross-sectional dependence of the panel make it approximate. Section 4.3 quantifies how approximate: with an intraclass correlation by date of 0.238 and 108.8 plants per day, the effective $n$ calibration sample is about 1,000 rather than 26,552, so and the guarantee should be read at the smaller sample that size.

*Transported scores.* Let $P_{\text{old}}$ and $P_{\text{new}}$ be the laws of the score in the old and new regimes and $T = F_{\text{new}}^{-1} \circ F_{\text{old}}$ the monotone transport map. Under an *oracle* oracle map, $T(s) \sim P_{\text{new}}$ for $s \sim P_{\text{old}}$, so the transported calibration scores are exchangeable with the test score and the static guarantee transfersexactly. With an *estimated* estimated map $\hat{T}$ the transported scores are only approximately distributed as $P_{\text{new}}$, and the coverage gap is bounded by the total-variation distance $d_{\text{TV}}(\hat{T}_{\#}P_{\text{old}}, P_{\text{new}})$ through the weighted-exchangeability framework of Barber et al. [32]. We do not rely on this bound in practice: because $\hat{T}$ is data-dependent and the gap is not directly controllable, we obtain validity operationally The bound cannot be computed from data, so validity is obtained by feeding the transported scores into the adaptive online update of Gibbs and Candès [2], whose long-run coverage $\frac{1}{T}\sum_t \text{err}_t \to \alpha$ holds for any sequence, so that ; the transport contributes sharpness and the online control update contributes coverage. Section 5.7 reports shows that on these data the sharpness contribution is negative once the online control update is present.

## Appendix C. Hyperparameters, constants and seeds

Table C.11 lists every constant the constants of the pipeline . It is generated directly from the source code rather than transcribed, by a script that imports the modules and reads the constants and default arguments, so that it cannot drift from the implementationand the values selected by rolling origin for each base model.

Table C.11: Hyperparameters, constants, selected values and seeds.

| Item | Value | Detail |
|---|---|---|
| *Base models* | | |
| Forecast horizon | 7 days | direct; every dynamic feature lags the target by at least 7 days |
| Last training target | 2023-12-31 | no model sees the evaluation period |
| Trees per model / learning rate / depth | 400 / 0.05 / 6 | fixed a priori; reference models |
| Min. child weight / subsample / colsample | 5 / 0.9 / 0.9 | all models |
| Features | 12 | lags 7, 8, 14, 21 d; 7- and 30-d means; occurrence; day of week and year; log capacity; region; technology |
| Hurdle dispersion | out of fold, 5 folds | $\hat{\sigma} = 1.70$ (reference), 1.83 (early stopping) |
| Early stopping of the hurdle | ceiling 4,000, patience 50 | validation Jul–Dec 2023; refit on full training period: 59 + 46 trees |
| Quantile grid | 54 levels, 0.005–0.999 | 0.02–0.98 by 0.02, plus 0.005, 0.01, 0.99, 0.995, 0.999; rearranged |
| Capacity ladder | 15, 50, 100, 200, 400 | trees per quantile level |
| *Calibration layers* | | |
| Nominal level | $1 - \alpha = 0.90$ | one-sided upper limit $[0, U]$; tail levels 0.95 and 0.99 |
| Score | PIT, $s = F_x(y)$ | atom at zero randomised |
| Calibration window | 2024-01-01 to 2024-08-31 | 244 days, 26,552 plant-days |
| Horizon embargo | 7 days | online windows end at $t-7$; error feedback delayed 7 steps |
| Sliding window | 60 days, refreshed every 7 | |
| Transport window / refresh | 60 days / 7 days | at least 100 recent scores per refresh |
| Transport quantile grid | $g = \min(100, \max(20, m/2))$ | $m$ recent scores; two linear interpolations |
| Bootstrap of the map / shrinkage scale | $B = 150$ / $\tau = 0.05$ | $\lambda(u) = [1 + (\mathrm{sd}_{\mathrm{boot}}(u)/\tau)^2]^{-1}$ |
| ACI bound | none, or $\alpha_{\min} = 0.005$ | both reported |
| One-sided CQR | $U = \max\{q_{0.9}(x) + \hat{Q}_{0.9}(y - q_{0.9}(x)), 0\}$ | on each base model's own quantile |
| *Selection by rolling origin* | | |
| Inner calibration / validation | Jan–Apr 2024 / May–Aug 2024 | same refresh and embargo as the test |
| Grids | step $\gamma$: 0.005, 0.01, 0.02, 0.05, 0.10, 0.20 | windows: 30, 45, 60, 90, 120 days |
| Criterion | mean one-sided interval score | $+\infty$ if any limit is infinite |
| Selected, reference hurdle | ACI $\gamma = 0.005$ | Transport+ACI $\gamma = 0.005$, window 120 d |
| Selected, early-stopped hurdle | ACI $\gamma = 0.005$ | Transport+ACI $\gamma = 0.005$, window 120 d |
| Selected, hurdle with $\sigma(x)$ | ACI $\gamma = 0.005$ | Transport+ACI $\gamma = 0.005$, window 120 d |
| Selected, hurdle 2×2000 | ACI $\gamma = 0.005$ | Transport+ACI $\gamma = 0.005$, window 120 d |
| Selected, quantile GBM 54×15 | ACI $\gamma = 0.005$ | Transport+ACI $\gamma = 0.005$, window 45 d |
| Selected, quantile GBM 54×50 | ACI $\gamma = 0.005$ | Transport+ACI $\gamma = 0.005$, window 45 d |

Table C.11, continued.

| Item | Value | Detail |
|---|---|---|
| Selected, quantile GBM 54×100 | ACI $\gamma = 0.005$ | Transport+ACI $\gamma = 0.005$, window 45 d |
| Selected, quantile GBM 54×200 | ACI $\gamma = 0.005$ | Transport+ACI $\gamma = 0.005$, window 45 d |
| Selected, reference quantile GBM | ACI $\gamma = 0.005$ | Transport+ACI $\gamma = 0.005$, window 45 d |
| *Evaluation* | | |
| Primary metric | $U + \alpha^{-1} \max(y - U, 0)$ | proper for the $1 - \alpha$ quantile; $+\infty$ if $U$ is |
| Coverage standard error | clustered by date | |
| Bootstrap of differences | 2,000 replicates of whole days | 95% percentile interval |
| CRPS / twCRPS | 300 samples per plant-day | twCRPS threshold: plant's calibration-window 90th percentile |
| *Seeds* | | |
| Base-model training | 42 | XGBoost hist, 4 threads |
| PIT atom and map bootstrap | 20260720 | one generator, consumed in a fixed order |
| CRPS sampling | 11 | |
| Day-block bootstrap and wild cluster bootstrap | 20260901 | |
| Simulation | 20261103 | seed per scenario and replicate |

## Appendix D. Reproducibility

Every table~~and figure~~, figure and number in the text is regenerated by ~~a versioned script~~ versioned scripts from the frozen dataset; the numbers quoted in the text are read from the result files by a script rather than transcribed. The environment is ~~python~~ Python 3.12.3 with pandas 3.0.3, numpy 2.4.6, scipy 1.17.1, scikit-learn 1.9.0, xgboost 3.2.0, ruptures 1.1.10~~and~~ , statsmodels 0.15.0 and matplotlib 3.11.0. The seeds are 42 for base-model training, 20260720 for the randomisation of the PIT atom and the bootstrap of the transport map, 11 for the CRPS sampling~~and~~ , 20260901 for the day-block ~~bootstrap and~~ bootstraps, the permutation nulls and the wild cluster bootstrap, and 20261103 for the simulation. The order in which the conformal generator is consumed is fixed and documented, ~~which is what allows the hurdle arm of the model-agnosticism experiment to reproduce Table G.16~~ so that the reference hurdle reproduces its published results bit for bit. ~~The command sequence that regenerates the entire set of results, together with~~ All nine base models were refitted from the training data with the published code: the reference models reproduce the stored predictions to a relative difference below $10^{-5}$, which is the rounding of the stored files, and the early-stopped hurdle reproduces its tree counts and dispersion exactly. The command sequence and a log of every run ~~, is~~ are in the companion repository.

~~The model-agnosticism experiment of Section 5.3 was reproduced independently from a clean clone of the repository, in an environment resolved from the specification file rather than copied, in which four of the five principal libraries resolved to versions other than the reference ones. The retrained multi-quantile base model matched the versioned one by SHA-256 checksum,~~

## Appendix E. Sensitivity analysis

All analyses vary one parameter at a time around a reference configuration, Transport+ACI with $\gamma = 0.05$, a 60-day window, weekly refresh, $B = 150$, $\tau = 0.05$, a map grid of 100 levels, at least 100 recent scores and $\alpha_{\min} = 0.005$, and report the whole test period. The embargo is fixed at seven days by the forecast horizon and is not a free parameter. Table E.12 covers the calibration layer, Table E.13 the nominal level, Table E.14 the calibration window, the quantile grid and the threshold defining high-curtailment days, and Table E.15 the seeds and the base-model hyperparameters. The full $\gamma$×window grid for the reference hurdle is in Table G.19 of Appendix G.

Table E.12: Sensitivity of Transport+ACI to the parameters of the calibration layer, whole test, for the early-stopped hurdle and the reference quantile GBM. Coverage (%), mean width (MWh), fraction of infinite limits (%) and one-sided interval score over finite limits (MWh).

| | | Early-stopped hurdle | | | | Reference quantile GBM | | | |
|---|---|---|---|---|---|---|---|---|---|
| Parameter | Value | Cov. | Width | % inf. | IS | Cov. | Width | % inf. | IS |
| Reference | ref | 89.8 | 375 | 0.0 | 547 | 89.9 | 302 | 1.4 | 455 |
| Shrinkage scale $\tau$ | 0.01 | 89.8 | 375 | 0.0 | 546 | 89.9 | 302 | 0.9 | 454 |
| Shrinkage scale $\tau$ | 0.025 | 89.8 | 375 | 0.0 | 547 | 89.9 | 302 | 1.4 | 454 |
| Shrinkage scale $\tau$ | 0.1 | 89.8 | 375 | 0.0 | 547 | 89.9 | 301 | 1.5 | 454 |
| Shrinkage scale $\tau$ | 0.25 | 89.8 | 375 | 0.0 | 547 | 89.9 | 301 | 1.5 | 454 |
| Shrinkage scale $\tau$ | none | 89.8 | 375 | 0.0 | 547 | 89.9 | 301 | 1.5 | 454 |
| Bootstrap size $B$ | 25 | 89.8 | 376 | 0.0 | 547 | 89.9 | 301 | 1.7 | 454 |
| Bootstrap size $B$ | 50 | 89.8 | 375 | 0.0 | 547 | 89.9 | 300 | 1.8 | 454 |
| Bootstrap size $B$ | 300 | 89.8 | 375 | 0.0 | 547 | 89.9 | 302 | 1.4 | 454 |
| Map grid | 25 | 89.8 | 378 | 0.0 | 550 | 89.9 | 303 | 0.6 | 454 |
| Map grid | 50 | 89.8 | 377 | 0.0 | 548 | 89.9 | 303 | 0.6 | 455 |
| Map grid | 200 | 89.8 | 374 | 0.0 | 546 | 89.9 | 303 | 0.9 | 455 |
| Refresh period (days) | 1 | 89.8 | 380 | 0.0 | 551 | 89.9 | 302 | 1.2 | 455 |
| Refresh period (days) | 14 | 89.7 | 366 | 0.0 | 540 | 89.9 | 302 | 1.1 | 455 |
| Refresh period (days) | 28 | 89.7 | 359 | 0.0 | 533 | 89.9 | 302 | 0.2 | 454 |
| Min. recent scores | 50 | 89.8 | 375 | 0.0 | 547 | 89.9 | 302 | 1.4 | 455 |
| Min. recent scores | 500 | 89.8 | 375 | 0.0 | 547 | 89.9 | 302 | 1.4 | 455 |
| Min. recent scores | 2000 | 89.8 | 375 | 0.0 | 547 | 89.9 | 302 | 1.4 | 455 |
| Window (days) | 30 | 89.8 | 415 | 0.0 | 584 | 89.7 | 308 | 1.8 | 465 |
| Window (days) | 45 | 89.7 | 389 | 0.0 | 561 | 89.8 | 305 | 1.8 | 459 |
| Window (days) | 90 | 89.7 | 372 | 0.0 | 545 | 89.9 | 295 | 1.9 | 451 |
| Window (days) | 120 | 89.8 | 364 | 0.0 | 534 | 90.0 | 298 | 1.5 | 451 |
| Transport without ACI | gamma=0 | 89.8 | 350 | 0.0 | 517 | 90.0 | 288 | 0.0 | 438 |
| Step $\gamma$ | 0.005 | 89.9 | 357 | 0.0 | 524 | 90.2 | 293 | 0.0 | 441 |
| Step $\gamma$ | 0.01 | 90.0 | 362 | 0.0 | 531 | 90.2 | 295 | 0.0 | 443 |
| Step $\gamma$ | 0.02 | 90.0 | 381 | 0.0 | 551 | 90.2 | 299 | 0.0 | 447 |
| Step $\gamma$ | 0.1 | 89.5 | 409 | 0.0 | 578 | 89.5 | 293 | 3.7 | 450 |
| Step $\gamma$ | 0.2 | 88.5 | 498 | 0.0 | 681 | 88.0 | 286 | 9.5 | 470 |
| Bound $\alpha_{\min}$ | none | 90.0 | 379 | 3.3 | 554 | 90.2 | 304 | 4.4 | 459 |
| Bound $\alpha_{\min}$ | 0.001 | 89.8 | 400 | 0.0 | 572 | 89.9 | 301 | 1.9 | 454 |
| Bound $\alpha_{\min}$ | 0.01 | 89.7 | 365 | 0.0 | 537 | 89.9 | 302 | 0.5 | 454 |
| Bound $\alpha_{\min}$ | 0.02 | 89.7 | 355 | 0.0 | 528 | 89.8 | 299 | 0.0 | 451 |
| ACI, step $\gamma$ | 0.005 | 90.8 | 377 | 0.0 | 520 | 90.3 | 289 | 0.0 | 434 |
| ACI, step $\gamma$ | 0.01 | 90.4 | 367 | 0.0 | 517 | 90.2 | 289 | 0.0 | 434 |
| ACI, step $\gamma$ | 0.02 | 90.2 | 361 | 0.0 | 519 | 90.2 | 290 | 0.0 | 437 |
| ACI, step $\gamma$ | 0.1 | 89.7 | 386 | 0.0 | 552 | 89.5 | 298 | 0.0 | 452 |
| ACI, step $\gamma$ | 0.2 | 88.9 | 445 | 0.0 | 619 | 88.8 | 315 | 0.0 | 474 |
| ACI, step $\gamma$ | 0.05 | 89.8 | 355 | 0.0 | 521 | 90.0 | 294 | 0.0 | 444 |

Table E.13: Sensitivity to the nominal level, whole test. Coverage (%) and one-sided interval score at the same level (MWh). Adaptive layers use $\gamma = 0.05$ and $\alpha_{\min} = \alpha/20$.

| | | Early-stopped hurdle | | Ref. quantile GBM | |
|---|---|---|---|---|---|
| Level | Method | Cov. | IS | Cov. | IS |
| 0.80 | Static split | 83.8 | 380 | 82.1 | 343 |
| 0.80 | ACI | 80.2 | 382 | 80.3 | 348 |
| 0.80 | Transport+ACI | 80.0 | 394 | 80.2 | 356 |
| 0.80 | One-sided CQR | 82.4 | 372 | 81.1 | 348 |
| 0.90 | Static split | 92.0 | 534 | 90.2 | 433 |
| 0.90 | ACI | 89.8 | 521 | 90.0 | 444 |
| 0.90 | Transport+ACI | 89.8 | 547 | 89.9 | 455* |
| 0.90 | One-sided CQR | 94.2 | 590 | 89.8 | 438 |
| 0.95 | Static split | 96.0 | 743 | 94.3 | 539 |
| 0.95 | ACI | 94.8 | 701 | 94.7 | 555* |
| 0.95 | Transport+ACI | 94.9 | 748 | 94.8 | 558* |
| 0.95 | One-sided CQR | 98.0 | 1055 | 94.2 | 546 |
| 0.99 | Static split | 99.1 | 1693 | 98.4 | 929 |
| 0.99 | ACI | 99.0 | 1470 | 98.9 | 913* |
| 0.99 | Transport+ACI | 98.9 | 1566 | 98.9 | 910* |
| 0.99 | One-sided CQR | 99.8 | 3634 | 98.4 | 928 |

* the method returns infinite limits; the score is over finite limits.

## Appendix F. Specification of the controlled simulation

*Panel.* $N = 100$ plants observed for 240 calibration days followed by 638 test days, split into windows of 30, 92, 181, 184 and ~~the regenerated tables matched the published ones cell by cell, the only difference being wall-clock timing columns. The dataset files, which are distributed through the archive rather than the repository, were verified against the checksums versioned here before use~~151 days, the lengths of the real windows; the second window plays the role of the transition window.

*Data-generating process.* For plant $i$ and day $t$, occurrence is $Z_{it} \sim \text{Bernoulli}(\text{logit}^{-1}(a_i + \eta_t))$ and, when $Z_{it} = 1$, $\log Y_{it} = m_i + \ell_t + \xi_t + s_t \varepsilon_{it}$ with $\varepsilon_{it} \sim N(0,1)$. Plant effects are $a_i \sim N(1, 0.8^2)$ and $m_i \sim N(4, 0.8^2)$. The common shocks $\eta_t$ and $\xi_t$ are stationary first-order autoregressive processes with coefficient 0.7 and marginal standard deviations 0.8 and 0.5. They are the source of dependence across plants.

*Scenarios.* The level path $\ell_t$ and the idiosyncratic dispersion $s_t$ are constant at 0 and 1 before the transition window, change linearly over it and stay constant afterwards. S0: no change. S1: $\ell$ rises by $\log 3$, a tripling of magnitudes. S2: as S1, and $s$ falls from 1.0 to 0.6. S3: $s$ falls from 1.0 to 0.6 with no change in level.

*Base predictives.* All are lognormal hurdles with occurrence probability equal to the true probability marginalised over $\eta_t$, which a forecaster seven days ahead cannot predict. They differ in location and dispersion. Writing $\sigma_t = (s_t^2 + 0.5^2)^{1/2}$ for the true marginal dispersion of $\log Y$ given occurrence and $\sigma_{\text{cal}}$ for its calibration value, the predictive dispersion is $\kappa\, \sigma_{\text{cal}}^{1-\phi} \sigma_t^{\phi}$ and the location is $m_i + \ell_{t-L}$. M1, the oracle: $\phi = 1$, $\kappa = 1$, $L = 0$. M2: $\phi = 1$, $\kappa = 1.3$. M3: $\phi = 0.5$. M4, constant dispersion: $\phi = 0$. M5: $\phi = 0$, $\kappa = 1.3$. M6: $\phi = 0$, $\kappa = 0.8$. M7: $\phi = 1$, $L = 30$ days. M8: $\phi = 0$, $L = 30$ days.

*Calibration and evaluation.* The same code as for the real data: static PIT split on the calibration days; ACI and Transport+ACI with $\gamma = 0.05$, $\alpha_{\min} = 0.005$, a 60-day window, weekly refresh and the 7-day embargo; nominal level 0.90. For each model-window cell we compute the static over-coverage, the width change of each adaptive layer and the width change of the constant-threshold oracle at the adaptive layer's coverage. Fifty independent replicates per scenario. The seeds of the data, of the randomisation of the PIT atom and of the map bootstrap are fixed functions of the scenario, replicate and model indices, so every replicate can be regenerated on its own.

## Appendix G. Additional tables

Tables G.16–G.20 complete the main text. Table G.16 gives the reference hurdle by window; Table G.17 the effect of bounding $\alpha_t$; Table G.18 the sensitivity of the transition-window results to its boundaries; Table G.19 the full

Table E.14: Sensitivity to the calibration window (all windows end on 31 August 2024), to the density of the quantile grid of the reference quantile GBM, and to the threshold that defines high-curtailment days (H1). Whole test; coverage (%) and one-sided interval score over finite limits (MWh).

| Analysis | Value | Method | Coverage | IS |
|---|---|---|---|---|
| Calibration window | 8 months (Jan-Aug) (early-stopped hurdle) | Static split | 92.0 | 534 |
| Calibration window | 8 months (Jan-Aug) (early-stopped hurdle) | Transport+ACI | 89.8 | 547 |
| Calibration window | 8 months (Jan-Aug) (early-stopped hurdle) | One-sided CQR | 94.2 | 590 |
| Calibration window | 6 months (Mar-Aug) (early-stopped hurdle) | Static split | 92.7 | 546 |
| Calibration window | 6 months (Mar-Aug) (early-stopped hurdle) | Transport+ACI | 89.8 | 548 |
| Calibration window | 6 months (Mar-Aug) (early-stopped hurdle) | One-sided CQR | 94.3 | 591 |
| Calibration window | 4 months (May-Aug) (early-stopped hurdle) | Static split | 93.3 | 559 |
| Calibration window | 4 months (May-Aug) (early-stopped hurdle) | Transport+ACI | 89.8 | 546 |
| Calibration window | 4 months (May-Aug) (early-stopped hurdle) | One-sided CQR | 94.3 | 591 |
| Calibration window | 8 months (Jan-Aug) (quantile GBM) | Static split | 90.2 | 433 |
| Calibration window | 8 months (Jan-Aug) (quantile GBM) | Transport+ACI | 89.9 | 455 |
| Calibration window | 8 months (Jan-Aug) (quantile GBM) | One-sided CQR | 89.8 | 438 |
| Calibration window | 6 months (Mar-Aug) (quantile GBM) | Static split | 89.9 | 433 |
| Calibration window | 6 months (Mar-Aug) (quantile GBM) | Transport+ACI | 89.9 | 455 |
| Calibration window | 6 months (Mar-Aug) (quantile GBM) | One-sided CQR | 89.6 | 438 |
| Calibration window | 4 months (May-Aug) (quantile GBM) | Static split | 89.6 | 433 |
| Calibration window | 4 months (May-Aug) (quantile GBM) | Transport+ACI | 89.9 | 455 |
| Calibration window | 4 months (May-Aug) (quantile GBM) | One-sided CQR | 89.2 | 439 |
| Quantile grid | 54 levels (quantile GBM) | Static split | 90.2 | 433 |
| Quantile grid | 54 levels (quantile GBM) | Transport+ACI | 89.9 | 454 |
| Quantile grid | 54 levels (quantile GBM) | One-sided CQR | 89.8 | 438 |
| Quantile grid | 27 levels (quantile GBM) | Static split | 90.2 | 433 |
| Quantile grid | 27 levels (quantile GBM) | Transport+ACI | 89.9 | 455 |
| Quantile grid | 27 levels (quantile GBM) | One-sided CQR | 89.8 | 438 |
| Quantile grid | 14 levels (quantile GBM) | Static split | 90.2 | 433 |
| Quantile grid | 14 levels (quantile GBM) | Transport+ACI | 89.9 | 455 |
| Quantile grid | 14 levels (quantile GBM) | One-sided CQR | 89.8 | 438 |
| High-day threshold | P75 (425 days), reference hurdle | Static split (PIT) | 92.2 | – |
| High-day threshold | P75 (425 days), reference hurdle | Transport+ACI (selected) | 88.9 | – |
| High-day threshold | P75 (425 days), reference hurdle | Transport+ACI ($\gamma = 0.05$) | 88.1 | – |
| High-day threshold | P75 (425 days), reference hurdle | One-sided CQR | 90.5 | – |
| High-day threshold | P90 (254 days), reference hurdle | Static split (PIT) | 91.9 | – |
| High-day threshold | P90 (254 days), reference hurdle | Transport+ACI (selected) | 88.0 | – |
| High-day threshold | P90 (254 days), reference hurdle | Transport+ACI ($\gamma = 0.05$) | 86.7 | – |
| High-day threshold | P90 (254 days), reference hurdle | One-sided CQR | 89.9 | – |
| High-day threshold | P95 (197 days), reference hurdle | Static split (PIT) | 92.4 | – |
| High-day threshold | P95 (197 days), reference hurdle | Transport+ACI (selected) | 88.4 | – |
| High-day threshold | P95 (197 days), reference hurdle | Transport+ACI ($\gamma = 0.05$) | 87.0 | – |
| High-day threshold | P95 (197 days), reference hurdle | One-sided CQR | 90.4 | – |
| High-day threshold | P75 (425 days), early-stopped hurdle | Static split (PIT) | 91.6 | – |
| High-day threshold | P75 (425 days), early-stopped hurdle | Transport+ACI (selected) | 88.6 | – |
| High-day threshold | P75 (425 days), early-stopped hurdle | Transport+ACI ($\gamma = 0.05$) | 88.1 | – |
| High-day threshold | P75 (425 days), early-stopped hurdle | One-sided CQR | 93.9 | – |
| High-day threshold | P90 (254 days), early-stopped hurdle | Static split (PIT) | 91.4 | – |
| High-day threshold | P90 (254 days), early-stopped hurdle | Transport+ACI (selected) | 88.0 | – |
| High-day threshold | P90 (254 days), early-stopped hurdle | Transport+ACI ($\gamma = 0.05$) | 86.5 | – |
| High-day threshold | P90 (254 days), early-stopped hurdle | One-sided CQR | 93.9 | – |
| High-day threshold | P95 (197 days), early-stopped hurdle | Static split (PIT) | 92.2 | – |
| High-day threshold | P95 (197 days), early-stopped hurdle | Transport+ACI (selected) | 88.8 | – |
| High-day threshold | P95 (197 days), early-stopped hurdle | Transport+ACI ($\gamma = 0.05$) | 86.7 | – |
| High-day threshold | P95 (197 days), early-stopped hurdle | One-sided CQR | 94.6 | – |
| High-day threshold | P75 (425 days), reference quantile GBM | Static split (PIT) | 88.0 | – |
| High-day threshold | P75 (425 days), reference quantile GBM | Transport+ACI (selected) | 87.3 | – |
| High-day threshold | P75 (425 days), reference quantile GBM | Transport+ACI ($\gamma = 0.05$) | 87.7 | – |
| High-day threshold | P75 (425 days), reference quantile GBM | One-sided CQR | 87.4 | – |
| High-day threshold | P90 (254 days), reference quantile GBM | Static split (PIT) | 86.7 | – |
| High-day threshold | P90 (254 days), reference quantile GBM | Transport+ACI (selected) | 85.5 | – |
| High-day threshold | P90 (254 days), reference quantile GBM | Transport+ACI ($\gamma = 0.05$) | 85.8 | – |
| High-day threshold | P90 (254 days), reference quantile GBM | One-sided CQR | 85.9 | – |
| High-day threshold | P95 (197 days), reference quantile GBM | Static split (PIT) | 87.1 | – |
| High-day threshold | P95 (197 days), reference quantile GBM | Transport+ACI (selected) | 85.6 | – |
| High-day threshold | P95 (197 days), reference quantile GBM | Transport+ACI ($\gamma = 0.05$) | 85.8 | – |
| High-day threshold | P95 (197 days), reference quantile GBM | One-sided CQR | 86.1 | – |

Table E.15: Sensitivity to seeds and to base-model hyperparameters. Top: ten seeds of the conformal layer (PIT atom and map bootstrap), mean and standard deviation over seeds. Bottom: early-stopped hurdle refitted with other depths, learning rates and training seeds; trees selected by early stopping, CRPS over the whole test, and whole-test coverage and interval score of the static split and of Transport+ACI.

| Analysis | Method | Coverage | | IS | |
|---|---|---|---|---|---|
| | | mean | sd | mean | sd |
| Conformal seeds, early-stopped hurdle | Static split | 92.08 | 0.04 | 533.8 | 0.2 |
| Conformal seeds, early-stopped hurdle | Transport+ACI | 89.80 | 0.00 | 547.0 | 0.4 |
| Conformal seeds, reference quantile GBM | Static split | 90.20 | 0.00 | 432.7 | 0.0 |
| Conformal seeds, reference quantile GBM | Transport+ACI | 89.90 | 0.00 | 454.5 | 0.1 |
| | | Trees | CRPS | Static (cov./IS) | T+ACI (cov./IS) |
| Early-stopped hurdle | depth = 4 | 126 | 83.6 | 92.2 / 526 | 89.8 / 534 |
| Early-stopped hurdle | depth = 8 | 92 | 86.1 | 91.9 / 558 | 89.8 / 580 |
| Early-stopped hurdle | learning rate = 0.03 | 190 | 84.2 | 92.2 / 533 | 89.7 / 549 |
| Early-stopped hurdle | learning rate = 0.1 | 56 | 84.5 | 92.3 / 546 | 89.8 / 554 |
| Early-stopped hurdle | training seed = 1 | 112 | 84.2 | 92.1 / 541 | 89.7 / 556 |
| Early-stopped hurdle | training seed = 2 | 115 | 84.2 | 92.2 / 526 | 89.7 / 547 |
| Early-stopped hurdle | training seed = 3 | 99 | 84.2 | 92.0 / 523 | 89.7 / 541 |
| Early-stopped hurdle | training seed = 4 | 111 | 84.0 | 92.2 / 541 | 89.7 / 546 |
| Early-stopped hurdle | training seed = 5 | 119 | 84.2 | 92.3 / 528 | 89.7 / 546 |
| Early-stopped hurdle | training seed = 6 | 97 | 84.0 | 92.2 / 526 | 89.7 / 535 |
| Early-stopped hurdle | training seed = 7 | 108 | 84.5 | 91.9 / 532 | 89.7 / 553 |
| Early-stopped hurdle | training seed = 8 | 129 | 84.5 | 92.2 / 557 | 89.7 / 570 |
| Early-stopped hurdle | training seed = 9 | 92 | 84.7 | 91.8 / 517 | 89.7 / 536 |
| Early-stopped hurdle | training seed = 10 | 96 | 84.5 | 92.2 / 525 | 89.7 / 539 |

Table G.16: Coverage and mean interval width of the conformal methods by period, on real data with the hurdle base model, with the 7-day horizon embargo. Each cell shows coverage in percent (clustered standard error by date in parentheses) over mean interval width in MWh. Nominal coverage is 90%. The base-model CRPS is a property of the predictive, not of the calibration layer, and is reported once per period. Superscripts give the fraction of infinite intervals where nonzero; Table 3 reports the interval score, which does not let those cases improve a method's score.

| Method | Sep 2024 | Oct–Dec 2024 (transition) | 2025-S1 | 2025-S2 | 2026-S1 |
|---|---|---|---|---|---|
| Static split (PIT) | 93.5 (1.03) / 799 | 95.3 (0.45) / 1203 | 89.4 (1.04) / 455 | 94.0 (0.45) / 750 | 92.5 (0.58) / 535 |
| Sliding window 60d | 93.0 (1.04) / 742 | 92.0 (0.73) / 834 | 85.4 (1.16) / 323 | 92.9 (0.51) / 650 | 89.5 (0.76) / 419 |
| ACI ($\gamma$=0.02) | 93.0 (1.06) / 759 | 93.5 (0.55) / 915 | 85.7 (1.16) / 345 | 93.5 (0.49) / 669 | 89.1 (0.76) / 405 |
| ACI ($\gamma$=0.05) | 92.4 (1.16) / 718 | 92.1 (0.67) / 822 | 86.6 (1.14) / $405^{2.8}$ | 92.5 (0.58) / $609^{11.8}$ | 89.8 (0.75) / 446 |
| Transport+ACI ($\gamma$=0.02) | 92.6 (1.07) / 713 | 91.1 (0.8) / 787 | 86.5 (1.17) / $437^{0.6}$ | 92.4 (0.6) / $986^{1.1}$ | 90.3 (0.73) / 459 |
| Transport+ACI ($\gamma$=0.05) | 92.0 (1.19) / 681 | 90.9 (0.81) / 806 | 87.4 (1.13) / $446^{9.9}$ | 91.6 (0.63) / $718^{10.7}$ | 90.5 (0.72) / 489 |
| Base-model CRPS (MWh) | 96.1 | 133.5 | 70.5 | 94.5 | 79.0 |

$\gamma\times$window grid on the test period, a display produced after the selection was closed; and Table G.20 the calibration schemes designed for panel dependence.

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Table G.17: Effect of bounding the online update of $\alpha$ from below, whole test period. A bound of $\alpha_{\min} = 0.005$ removes infinite intervals entirely at negligible cost in coverage, and makes the unrestricted interval score finite and therefore comparable across methods.

| Bound on $\alpha_t$ | Coverage (se) | Width | % infinite | IS finite | IS all |
|---|---|---|---|---|---|
| *ACI ($\gamma$=0.05)* | | | | | |
| unbounded | 90.1 (0.43) | 544 | 4.2 | 753 | $\infty$ |
| $\alpha_{\min} = 0.005$ | 89.8 (0.43) | 550 | 0.0 | 753 | 753 |
| $\alpha_{\min} = 0.01$ | 89.8 (0.43) | 536 | 0.0 | 739 | 739 |
| $\alpha_{\min} = 0.02$ | 89.7 (0.43) | 525 | 0.0 | 729 | 729 |
| *Transport+ACI ($\gamma$=0.05)* | | | | | |
| unbounded | 90.1 (0.43) | 596 | 5.9 | 813 | $\infty$ |
| $\alpha_{\min} = 0.005$ | 89.6 (0.43) | 641 | 0.0 | 850 | 850 |
| $\alpha_{\min} = 0.01$ | 89.6 (0.43) | 598 | 0.0 | 807 | 807 |
| $\alpha_{\min} = 0.02$ | 89.5 (0.43) | 559 | 0.0 | 770 | 770 |


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Table G.18: Sensitivity to alternative definitions of the transition window, hurdle base model. $W_1$ is the Wasserstein-1 distance between the positive log-magnitudes of the window and those of the calibration set. Width and interval score are means over finite intervals, with the fraction of infinite intervals beside them.

| Method | Coverage (se) | Width | % inf. | IS finite |
|---|---|---|---|---|
| *Oct–Dec 2024 (the one used)*, $W_1$ = 1.253 | | | | |
| Static split | 95.3 (0.45) | 1203 | 0.0 | 1395 |
| ACI ($\gamma$=0.02) | 93.5 (0.55) | 915 | 0.0 | 1158 |
| Transport+ACI ($\gamma$=0.05) | 90.9 (0.81) | 806 | 0.0 | 1107 |
| *Sep–Dec 2024*, $W_1$ = 1.179 | | | | |
| Static split | 94.9 (0.43) | 1105 | 0.0 | 1279 |
| ACI ($\gamma$=0.02) | 93.4 (0.49) | 877 | 0.0 | 1091 |
| Transport+ACI ($\gamma$=0.05) | 91.1 (0.68) | 775 | 0.0 | 1038 |
| *Nov 2024–Jan 2025*, $W_1$ = 1.109 | | | | |
| Static split | 94.8 (0.79) | 1105 | 0.0 | 1256 |
| ACI ($\gamma$=0.02) | 91.2 (1.17) | 788 | 0.0 | 1019 |
| Transport+ACI ($\gamma$=0.05) | 88.6 (1.43) | 795 | 5.5 | 1083 |
| *Oct 2024–Feb 2025*, $W_1$ = 1.002 | | | | |
| Static split | 94.5 (0.57) | 1000 | 0.0 | 1164 |
| ACI ($\gamma$=0.02) | 91.2 (0.83) | 732 | 0.0 | 965 |
| Transport+ACI ($\gamma$=0.05) | 89.5 (1.01) | 787 | 8.1 | 1086 |
| *Oct–Nov 2024*, $W_1$ = 1.342 | | | | |
| Static split | 94.0 (0.58) | 1108 | 0.0 | 1373 |
| ACI ($\gamma$=0.02) | 92.4 (0.73) | 896 | 0.0 | 1219 |
| Transport+ACI ($\gamma$=0.05) | 89.9 (1.08) | 842 | 0.0 | 1223 |
| *Sep 2024–Mar 2025*, $W_1$ = 0.992 | | | | |
| Static split | 94.3 (0.5) | 968 | 0.0 | 1124 |
| ACI ($\gamma$=0.02) | 91.5 (0.72) | 736 | 0.0 | 951 |
| Transport+ACI ($\gamma$=0.05) | 89.9 (0.87) | 769 | 6.8 | 1040 |


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Table G.19: Sensitivity of the adaptive schemes to the step $\gamma$ and to the length of the recent window, whole test period, September 2024 to May 2026. This table is a display produced after the selection of Section 4.7 was closed, never a criterion: the selection used only the inner validation set. Pure ACI has no window. Bold marks the configuration selected by rolling origin; the 60-day cells at $\gamma = 0.02$ and 0.05 are the configurations of Algorithm 1 reported in the main text. †The configuration produces infinite limits, so its unrestricted interval score is infinite.

| | ACI | Transport+ACI, recent window (days) | | | | |
|---|---|---|---|---|---|---|
| $\gamma$ | | 30 | 45 | 60 | 90 | 120 |
| *Coverage (%)* | | | | | | |
| 0.005 | **90.9** | 89.9 | 90.0 | 90.0 | 90.0 | **90.0** |
| 0.01 | 90.5 | 90.0 | 90.0 | 90.0 | 90.0 | 90.0 |
| 0.02 | 90.2 | 90.0 | 90.1 | 90.1 | 90.0 | 90.0 |
| 0.05 | 90.1 | 90.1 | 90.1 | 90.1 | 90.0 | 90.0 |
| 0.10 | 90.1 | 90.0 | 90.0 | 90.0 | 90.0 | 90.0 |
| 0.20 | 90.0 | 89.9 | 90.0 | 90.0 | 90.0 | 90.0 |
| *Infinite intervals (%)* | | | | | | |
| 0.005 | **0.0** | 0.0 | 0.0 | 0.0 | 0.0 | **0.0** |
| 0.01 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 0.02 | 0.0 | 0.0 | 0.3 | 0.5 | 0.6 | 1.1 |
| 0.05 | 4.2 | 5.3 | 6.4 | 5.9 | 5.2 | 4.4 |
| 0.10 | 1.6 | 6.2 | 4.2 | 4.4 | 3.6 | 4.3 |
| 0.20 | 15.1 | 22.5 | 23.0 | 21.9 | 22.4 | 22.9 |
| *Interval score over finite intervals (MWh)* | | | | | | |
| 0.005 | **763** | 775 | 759 | 753 | 756 | **761** |
| 0.01 | 748 | 803 | 789 | 776 | 769 | 770 |
| 0.02 | 743 | 898 | 905† | 871† | 846† | 822† |
| 0.05 | 753† | 826† | 806† | 813† | 831† | 895† |
| 0.10 | 784† | 893† | 804† | 821† | 817† | 784† |
| 0.20 | 936† | 872† | 877† | 769† | 774† | 769† |


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Table G.20: Calibration schemes designed to address panel dependence, whole test period. The last four columns give the worst absolute deviation from the 90% nominal level over the groups of each axis, in percentage points, which is the measure of conditional coverage; smaller is better. Width and interval score are means over finite intervals, with the fraction of infinite intervals beside them.

| | Marginal | | | | Worst deviation from nominal (pp) | | | |
|---|---|---|---|---|---|---|---|---|
| Scheme | Coverage (se) | Width | % inf. | IS finite | Technology | Size | Region | High-curt. |
| Pooled split | 92.5 (0.37) | 678 | 0.0 | 826 | 2.5 | 3.6 | 4.8 | 2.9 |
| Block conformal, one plant per date | 92.7 (0.36) | 697 | 0.0 | 840 | 2.8 | 3.9 | 5.1 | 3.1 |
| Mondrian by technology | 92.3 (0.37) | 651 | 0.0 | 809 | 4.1 | 3.7 | 5.1 | 2.7 |
| Mondrian by region | 92.1 (0.38) | 596 | 0.0 | 764 | 4.0 | 3.7 | 7.2 | 2.5 |
| Mondrian by capacity tercile | 92.5 (0.37) | 674 | 0.0 | 823 | 2.6 | 3.9 | 4.8 | 2.9 |
| Plant-level conformal | 92.5 (0.39) | 636 | 0.0 | 773 | 4.1 | 3.0 | 7.2 | 2.8 |
| ACI ($\gamma$=0.02), for reference | 90.2 (0.43) | 551 | 0.0 | 743 | 0.3 | 1.5 | 4.7 | 1.7 |
| Transport+ACI ($\gamma$=0.05), for reference | 90.1 (0.43) | 596 | 5.9 | 813 | 0.3 | 1.6 | 4.8 | 3.3 |


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